

SOLIDWORKS API (.NET VBA C#) — Examples (40/50):

Insert_Sheet_Metal_Gusset_Feature_Example_VB

Insert_Wrap_Feature_Example_VB

SOLIDWORKS API Documentation (converted from CHM)

Insert Sheet Metal Gusset Feature Example (VBA)

This example shows how to insert a sheet metal gusset feature and modify its data.

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\SMGussetAPI.sldprt.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Inserts four sheet metal gussets.
' 2. Press F5 repeatedly and observe the gusset modifications.
' 3. Inspect the Immediate window for the flatten settings of all gussets.
' 4. Expand Flat-Pattern in the FeatureManager design tree, right-click
'    Flat-Pattern(1), and click Unsuppress.
' 5. Observe the center marks and profiles of all the gussets in their
'    flattened states.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim myFeature As SldWorks.Feature
Dim myFeature1 As SldWorks.Feature
Dim myFeature2 As SldWorks.Feature
Dim myFeature3 As SldWorks.Feature
Dim myFeature4 As SldWorks.Feature
Dim swFeat As SldWorks.Feature
Dim swFeatData As SldWorks.SMGussetFeatureData
Dim boolstatus As Boolean
Option Explicit

Sub main()
    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc

    boolstatus = Part.Extension.SelectByID2("", "FACE", -5.38403893476698E-02, 3.6701308754914E-03, 0.05530817474488, False, 0,
    Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("", "FACE", -1.77780871801474E-02, -3.07393226379986E-02, 3.41128529187245E-02, True,
    0, Nothing, 0)

    ' Gusset #1 insertion parameters
    '1. bOffset = True
    '2. dOffset = 50 mm
    '3. bFlipOffsetSide = False
    '4. profDimType = 0 (indent depth dimensioning scheme)
    '5. dIndentDepth = 10 mm
    '6. dLength = 0
    '7. bUseAngle = False
    '8. dHeight = 0
    '9. dAngle = 0
    '10. bFlipSides = False
    '11. dWidth = 10 mm
    '12. dThickness = 3 mm
    '13. bDraft = True
    '14. dDraftAngle = 3 degrees
    '15. bInnerCornerFillet = True
    '16. dInnerCornerFilletRadius = 2 mm
    '17. bOuterCornerFillet = True
    '18. dOuterCornerFilletRadius = 1 mm
    '19. gussetType = 0 (rounded back)
    '20. bEdgeFillet = False
    '21. dEdgeFilletRadius = 0 mm
    '22. bOverrideDoc = True
    '23. bShowProfile = True
    '24. bShowCenter = True

    Set myFeature = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.05, False,
    swSheetMetalGussetProfileDimType_IndentDepth, 0.01, 0, False, 0, 0, True, 0.01, 0.003, True, 3 * 0.0175, True, 0.002, True, 0.001,
    swSheetMetalRibGussetType_Rounded, False, 0, True, True, True)
    Part.ClearSelection2 True

    ' Gusset #2 insertion parameters (same as for Gusset #1 with the following exceptions)
    '2. dOffset = 30 mm
    '19. gussetType = 1 (flat back)
    '20. bEdgeFillet = True
    '21. dEdgeFilletRadius = 1 mm

    ' Select faces
    boolstatus = Part.Extension.SelectByID2("", "FACE", -5.38403893476698E-02, 3.6701308754914E-03, 0.05530817474488, False, 0,
    Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("", "FACE", -1.77780871801474E-02, -3.07393226379986E-02, 3.41128529187245E-02, True,
    0, Nothing, 0)

```

```

Set myFeature1 = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.03, False,
swSheetMetalGussetProfileDimType_IndentDepth, 0.01, 0, False, 0, 0, False, 0.01, 0.003, True, 3 * 0.0175, True, 0.002, True, 0.001,
swSheetMetalRibGussetType_Flat, True, 0.001, True, True, True)
Part.ClearSelection2 True

' Gusset #3 insertion parameters (same as for Gusset #1 with the following exceptions)
'2. dOffset = 30 mm
'4. profDimType = 1 (length + height dimensioning scheme)
'5. dIndentDepth = 0 mm
'6. dLength = 25 mm
'7. bUseAngle = False
'8. dHeight = 15 mm
'9. dAngle = 0
'10. bFlipSides = False
'19. gussetType = 1 (flat back)
'20. bEdgeFillet = True
'21. dEdgeFilletRadius = 1 mm

boolstatus = Part.Extension.SelectByID2("", "FACE", -5.38403893476698E-02, 3.6701308754914E-03, 0.05530817474488, False, 0,
Nothing, 0)
boolstatus = Part.Extension.SelectByID2("", "FACE", -1.77780871801474E-02, -3.07393226379986E-02, 3.41128529187245E-02, True,
0, Nothing, 0)

Set myFeature2 = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.03, False,
swSheetMetalGussetProfileDimType_ProfileDimensions, 0, 0.025, False, 0.015, 0, False, 0.02, 0.003, True, 3 * 0.0175, True, 0.002,
True, 0.001, swSheetMetalRibGussetType_Flat, True, 0.001, True, True, True)
Part.ClearSelection2 True

' Gusset #4 insertion parameters (same as for Gusset #1 with the following exceptions)
'2. dOffset = 30 mm
'20. bEdgeFillet = True
'21. dEdgeFilletRadius = 1 mm

' Select orientation and position references
boolstatus = Part.Extension.SelectByID2("", "FACE", -5.38403893476129E-02, -2.24553153327633E-03, 0.087801420904043, True, 0,
Nothing, 0)
boolstatus = Part.Extension.SelectByID2("", "FACE", -2.35965800548001E-02, -3.07393226379986E-02, 8.97844682415894E-02, True,
0, Nothing, 0)
boolstatus = Part.Extension.SelectByID2("Line1@Sketch6", "EXTSKETCHSEGMENT", -6.09049483400968E-03, -8.95139047397037E-02, 0,
True, 0, Nothing, 0)
boolstatus = Part.Extension.SelectByID2("Point1@Sketch7", "EXTSKETCHPOINT", 1.80407947995604E-02, -7.62728416981986E-02, 0,
True, 0, Nothing, 0)

Set myFeature3 = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.03, False,
swSheetMetalGussetProfileDimType_IndentDepth, 0.01, 0, False, 0, 0, False, 0.01, 0.003, True, 3 * 0.0175, True, 0.002, True, 0.001,
swSheetMetalRibGussetType_Rounded, True, 0.001, True, True, True)
Part.ClearSelection2 True
Stop

'Five modifications to gusset feature data:

'a. Modify type, draft, and outer corner fillet options for gusset #1
boolstatus = Part.Extension.SelectByID2("Sheet Metal Gusset1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

Set swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections Part, Nothing

swFeatData.GussetType = 1 'flat back
swFeatData.DraftSideFaces = False
swFeatData.FilletOuterCorners = False 'no outer corner fillet
Debug.Print "Sheet Metal Gusset1 Flatten Settings"
Debug.Print " Override document property settings? " & swFeatData.OverrideDocSettings
Debug.Print " Show center marks? " & swFeatData.ShowCenter
Debug.Print " Show profile? " & swFeatData.ShowProfile

swFeat.ModifyDefinition swFeatData, Part, Nothing
swFeatData.ReleaseSelectionAccess

Stop

'b. Modify orientation reference of gusset #3
boolstatus = Part.Extension.SelectByID2("Sheet Metal Gusset3", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

Set swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections Part, Nothing

boolstatus = Part.Extension.SelectByID2("Line1@Sketch6", "EXTSKETCHSEGMENT", -6.09049483400968E-03, -8.95139047397037E-02,
0, True, 0, Nothing, 0)

Dim refLine As Variant
Set refLine = Part.SelectionManager.GetSelectedObject6(1, -1)
swFeatData.ReferenceLine = refLine
swFeat.ModifyDefinition swFeatData, Part, Nothing

Stop

'c. Modify legs of gusset #2: select one bend face instead of two flat faces
boolstatus = Part.Extension.SelectByID2("Sheet Metal Gusset2", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

```

```

Set swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections Part, Nothing

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.03831148650454, -3.27672470662037E-02, 0.147978181958194, False, 0,
Nothing, 0)

Dim newBendFace As Variant
Dim bendfaces(0 To 1) As Object
Dim jj As Integer
For jj = 0 To 1
    Set bendfaces(jj) = Part.SelectionManager.GetSelectedObject6(jj + 1, -1)
Next jj
newBendFace = bendfaces
swFeatData.SupportingFaces = newBendFace
Debug.Print "Sheet Metal Gusset2 Flatten Settings"
Debug.Print " Override document property settings? " & swFeatData.OverrideDocSettings
Debug.Print " Show center marks? " & swFeatData.ShowCenter
Debug.Print " Show profile? " & swFeatData.ShowProfile

swFeat.ModifyDefinition swFeatData, Part, Nothing

Stop

'd. Modify reference position of gusset #3 - 3 mm away from vertex of hexagonal cut
boolstatus = Part.Extension.SelectByID2("Sheet Metal Gusset3", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

Set swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections Part, Nothing

boolstatus = Part.Extension.SelectByID2("", "VERTEX", -5.38403893475499E-02, -1.00654290631334E-02, 0.205954465964501,
False, 0, Nothing, 0)

Dim refPoint As Variant
Set refPoint = Part.SelectionManager.GetSelectedObject6(1, -1)
swFeatData.ReferencePoint = refPoint
Debug.Print "Sheet Metal Gusset3 Flatten Settings"
Debug.Print " Override document property settings? " & swFeatData.OverrideDocSettings
Debug.Print " Show center marks? " & swFeatData.ShowCenter
Debug.Print " Show profile? " & swFeatData.ShowProfile

swFeat.ModifyDefinition swFeatData, Part, Nothing

Stop

'e. Modify type and inner corner fillet options for gusset #4
boolstatus = Part.Extension.SelectByID2("Sheet Metal Gusset4", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

Set swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections Part, Nothing

swFeatData.GussetType = 0 'rounded back
swFeatData.FilletInnerCorners = False 'no inner corner fillet
Debug.Print "Sheet Metal Gusset4 Flatten Settings"
Debug.Print " Override document property settings? " & swFeatData.OverrideDocSettings
Debug.Print " Show center marks? " & swFeatData.ShowCenter
Debug.Print " Show profile? " & swFeatData.ShowProfile

swFeat.ModifyDefinition swFeatData, Part, Nothing
swFeatData.ReleaseSelectionAccess

End Sub

```

Insert Sheet Metal Gusset Feature Example (VB.NET)

This example shows how to insert a sheet metal gusset feature and modify its data.

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\SMGussetAPI.sldprt.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Inserts four sheet metal gussets.
' 2. Press F5 repeatedly and observe the gusset modifications.
' 3. Inspect the Immediate window for the flatten settings of all gussets.
' 4. Expand Flat-Pattern in the FeatureManager design tree, right-click
'    Flat-Pattern(1), and click Unsuppress.
' 5. Observe the center marks and profiles of all the gussets in their
'    flattened states.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----

```

```

Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

```

```
Partial Class SolidWorksMacro
```

```

    Dim Part As ModelDoc2
    Dim myFeature As Feature
    Dim myFeature1 As Feature
    Dim myFeature2 As Feature
    Dim myFeature3 As Feature
    Dim swFeat As Feature
    Dim swFeatData As SMGussetFeatureData
    Dim boolstatus As Boolean

```

```
Sub main()
```

```
    Part = swApp.ActiveDoc
```

```

    boolstatus = Part.Extension.SelectByID2("", "FACE", -0.0538403893476698, 0.0036701308754914, 0.05530817474488, False, 0, Not
    boolstatus = Part.Extension.SelectByID2("", "FACE", -0.0177780871801474, -0.0307393226379986, 0.0341128529187245, True, 0, 1

```

```

' Gusset #1 insertion parameters
'1.  bOffset                = True
'2.  dOffset                = 50 mm
'3.  bFlipOffsetSide       = False
'4.  profDimType            = 0 (indent depth dimensioning scheme)
'5.  dIndentDepth          = 10 mm
'6.  dLength               = 0
'7.  bUseAngle              = False
'8.  dHeight               = 0
'9.  dAngle                = 0
'10. bFlipSides             = False
'11. dWidth                = 10 mm
'12. dThickness            = 3 mm
'13. bDraft                = True
'14. dDraftAngle           = 3 degrees
'15. bInnerCornerFillet    = True
'16. dInnerCornerFilletRadius = 2 mm
'17. bOuterCornerFillet    = True
'18. dOuterCornerFilletRadius = 1 mm
'19. gussetType             = 0 (rounded back)
'20. bEdgeFillet           = False
'21. dEdgeFilletRadius     = 0 mm
'22. bOverrideDoc          = True
'23. bShowProfile          = True
'24. bShowCenter           = True

```

```

myFeature = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.05, False, swSheetMetalGussetProfileDimType_e.swSheet
Part.ClearSelection2(True)

```

```

' Gusset #2 insertion parameters (same as for Gusset #1 with the following exceptions)
'2.  dOffset                = 30 mm
'19. gussetType             = 1 (flat back)
'20. bEdgeFillet           = True
'21. dEdgeFilletRadius     = 1 mm

```

```
'Select faces
```

```

boolstatus = Part.Extension.SelectByID2("", "FACE", -0.0538403893476698, 0.0036701308754914, 0.05530817474488, False, 0, Not
boolstatus = Part.Extension.SelectByID2("", "FACE", -0.0177780871801474, -0.0307393226379986, 0.0341128529187245, True, 0, 1

```

```

myFeature1 = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.03, False, swSheetMetalGussetProfileDimType_e.swShee
Part.ClearSelection2(True)

```

```

' Gusset #3 insertion parameters (same as for Gusset #1 with the following exceptions)
'2.  dOffset                = 30 mm

```

```

'4. profDimType           = 1 (length + height dimensioning scheme)
'5. dIndentDepth         = 0 mm
'6. dLength              = 25 mm
'7. bUseAngle            = False
'8. dHeight              = 15 mm
'9. dAngle               = 0
'10. bFlipSides           = False
'19. gussetType           = 1 (flat back)
'20. bEdgeFillet         = True
'21. dEdgeFilletRadius    = 1 mm

boolstatus = Part.Extension.SelectByID("", "FACE", -0.0538403893476698, 0.0036701308754914, 0.05530817474488, False, 0, Not
boolstatus = Part.Extension.SelectByID("", "FACE", -0.0177780871801474, -0.0307393226379986, 0.0341128529187245, True, 0, N

myFeature2 = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.03, False, swSheetMetalGussetProfileDimType_e.swShee
Part.ClearSelection2(True)

' Gusset #4 insertion parameters (same as for Gusset #1 with the following exceptions)
'2. dOffset              = 30 mm
'20. bEdgeFillet         = True
'21. dEdgeFilletRadius    = 1 mm

'Select orientation and position references
boolstatus = Part.Extension.SelectByID("", "FACE", -0.0538403893476129, -0.00224553153327633, 0.087801420904043, True, 0, N
boolstatus = Part.Extension.SelectByID("", "FACE", -0.0235965800548001, -0.0307393226379986, 0.0897844682415894, True, 0, N
boolstatus = Part.Extension.SelectByID("Line1@Sketch6", "EXTSKETCHSEGMENT", -0.00609049483400968, -0.0895139047397037, 0, 1
boolstatus = Part.Extension.SelectByID("Point1@Sketch7", "EXTSKETCHPOINT", 0.0180407947995604, -0.0762728416981986, 0, True

myFeature3 = Part.FeatureManager.InsertSheetMetalGussetFeature3(True, 0.03, False, swSheetMetalGussetProfileDimType_e.swShee
Part.ClearSelection2(True)

Stop

'Five modifications to gusset feature data:

'a. Modify type, draft, and outer corner fillet options for gusset #1
boolstatus = Part.Extension.SelectByID("Sheet Metal Gusset1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections(Part, Nothing)

swFeatData.GussetType = 1 'flat back
swFeatData.DraftSideFaces = False
swFeatData.FilletOuterCorners = False 'no outer corner fillet
Debug.Print("Sheet Metal Gusset1 Flatten Settings")
Debug.Print(" Override document property settings? " & swFeatData.OverrideDocSettings)
Debug.Print(" Show center marks? " & swFeatData.ShowCenter)
Debug.Print(" Show profile? " & swFeatData.ShowProfile)

swFeat.ModifyDefinition(swFeatData, Part, Nothing)
swFeatData.ReleaseSelectionAccess()

Stop

'b. Modify orientation reference of gusset #3
boolstatus = Part.Extension.SelectByID("Sheet Metal Gusset3", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections(Part, Nothing)

boolstatus = Part.Extension.SelectByID("Line1@Sketch6", "EXTSKETCHSEGMENT", -0.00609049483400968, -0.0895139047397037, 0, 1

Dim refLine As Object
refLine = Part.SelectionManager.GetSelectedObject6(1, -1)
swFeatData.ReferenceLine = refLine

swFeat.ModifyDefinition(swFeatData, Part, Nothing)

Stop

'c. Modify legs of gusset #2: select one bend face instead of two flat faces
boolstatus = Part.Extension.SelectByID("Sheet Metal Gusset2", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections(Part, Nothing)

boolstatus = Part.Extension.SelectByID("", "FACE", 0.03831148650454, -0.0327672470662037, 0.147978181958194, False, 0, Notf

Dim newBendFace As Object
Dim bendfaces(0) As Face2
bendfaces(0) = Part.SelectionManager.GetSelectedObject6(1, -1)

newBendFace = bendfaces
swFeatData.SupportingFaces = newBendFace
Debug.Print("Sheet Metal Gusset2 Flatten Settings")
Debug.Print(" Override document property settings? " & swFeatData.OverrideDocSettings)
Debug.Print(" Show center marks? " & swFeatData.ShowCenter)
Debug.Print(" Show profile? " & swFeatData.ShowProfile)

swFeat.ModifyDefinition(swFeatData, Part, Nothing)

Stop

```

```

'd. Modify reference position of gusset #3 - 3 mm away from vertex of hexagonal cut
boolstatus = Part.Extension.SelectByID2("Sheet Metal Gusset3", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections(Part, Nothing)

boolstatus = Part.Extension.SelectByID2("", "VERTEX", -0.0538403893475499, -0.0100654290631334, 0.205954465964501, False, 0,

Dim refPoint As Object
refPoint = Part.SelectionManager.GetSelectedObject6(1, -1)
swFeatData.ReferencePoint = refPoint
Debug.Print("Sheet Metal Gusset3 Flatten Settings")
Debug.Print(" Override document property settings? " & swFeatData.OverrideDocSettings)
Debug.Print(" Show center marks? " & swFeatData.ShowCenter)
Debug.Print(" Show profile? " & swFeatData.ShowProfile)

swFeat.ModifyDefinition(swFeatData, Part, Nothing)

Stop

'e. Modify type and inner corner fillet options for gusset #4
boolstatus = Part.Extension.SelectByID2("Sheet Metal Gusset4", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swFeat = Part.SelectionManager.GetSelectedObject6(1, -1)

swFeatData = swFeat.GetDefinition
swFeatData.AccessSelections(Part, Nothing)

swFeatData.GussetType = 0 'rounded back
swFeatData.FilletInnerCorners = False 'no inner corner fillet
Debug.Print("Sheet Metal Gusset4 Flatten Settings")
Debug.Print(" Override document property settings? " & swFeatData.OverrideDocSettings)
Debug.Print(" Show center marks? " & swFeatData.ShowCenter)
Debug.Print(" Show profile? " & swFeatData.ShowProfile)

swFeat.ModifyDefinition(swFeatData, Part, Nothing)
swFeatData.ReleaseSelectionAccess()

```

End Sub

```

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

```

End Class

Insert Sheet Metal Hem Example (C#)

This example shows how to insert a hem into a sheet metal part.

```
//-----
// Preconditions:
// 1. Open a sheet metal part.
// 2. Select the edge to which to you can add a hem.
// 3. Open the Immediate window.
//
// Postconditions:
// 1. Adds an open hem with a custom relief of type Obround and
//    a relief ratio of 1.0.
// 2. Gets the hem type.
// 3. Examine the Immediate window.
// -----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace SheetMetalHem_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;
        CustomBendAllowance CBAObject;
        Feature myFeature;
        HemFeatureData myHem;

        public void Main()
        {
            Part = (ModelDoc2)swApp.ActiveDoc;

            CBAObject = Part.FeatureManager.CreateCustomBendAllowance();
            CBAObject.Type = 2;
            CBAObject.KFactor = 0.5;

            // Insert an open hem of custom relief type Obround and relief ratio 1.0
            myFeature = Part.FeatureManager.InsertSheetMetalHem2((int)swHemTypes_e.swHemTypeOpen,
            (int)swHemPositionTypes_e.swHemPositionTypeOutside, false, 0.01, 0.01, 0, 0.005, 0.0011, CBAObject, false,
            (int)swSheetMetalReliefTypes_e.swSheetMetalReliefObround, 0, true, 1.0, 0, 0);
            Part.ClearSelection2(true);

            myHem = (HemFeatureData)myFeature.GetDefinition();
            Debug.Print("Hem type as defined in swHemTypes_e: " + myHem.Type);
        }

        public SldWorks swApp;
    }
}
```

Insert Sheet Metal Hem Example (VBA)

This example shows how to insert a hem into a sheet metal part.

```
'-----
' Preconditions:
' 1. Open a sheet metal part.
' 2. Select the edge to which you can add a hem.
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Adds an open hem with a custom relief of type Obround and
'    a relief ratio of 1.0.
' 2. Gets the hem type.
' 3. Examine the Immediate window and graphics area.
'-----

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim CBAObject As SldWorks.CustomBendAllowance
Dim myFeature As SldWorks.Feature
Dim myHem As SldWorks.HemFeatureData

Option Explicit
Sub main()
    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc

    Set CBAObject = Part.FeatureManager.CreateCustomBendAllowance()
    CBAObject.Type = 2
    CBAObject.KFactor = 0.5

    ' Insert an open hem of custom relief type Obround and relief ratio 1.0
    Set myFeature = Part.FeatureManager.InsertSheetMetalHem2(swHemTypes_e.swHemTypeOpen,
swHemPositionTypes_e.swHemPositionTypeOutside, False, 0.01, 0.01, 0, 0.005, 0.0011, CBAObject, False,
swSheetMetalReliefTypes_e.swSheetMetalReliefObround, 0, True, 1#, 0, 0)
    Part.ClearSelection2 True

    Set myHem = myFeature.GetDefinition
    Debug.Print "Hem type as defined in swHemTypes_e: " & myHem.Type
End Sub
```

Insert Sheet Metal Hem Example (VB.NET)

This example shows how to insert a hem into a sheet metal part.

```
'-----
' Preconditions:
' 1. Open a sheet metal part.
' 2. Select the edge to which you can add a hem.
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Adds an open hem with a custom relief of type Obround and
'    a relief ratio of 1.0.
' 2. Gets the hem type.
' 3. Examine the Immediate window and graphics area.
'-----

Imports SolidWorks.Interop.sldworks

Imports SolidWorks.Interop.swconst

Imports System.Runtime.InteropServices

Imports System

Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim Part As ModelDoc2

    Dim CBAObject As CustomBendAllowance

    Dim myFeature As Feature

    Dim myHem As HemFeatureData

    Sub main()

        Part = swApp.ActiveDoc

        CBAObject = Part.FeatureManager.CreateCustomBendAllowance()

        CBAObject.Type = 2

        CBAObject.KFactor = 0.5

        ' Insert an open hem of custom
        relief type Obround and relief ratio 1.0

        myFeature =
Part.FeatureManager.InsertSheetMetalHem2(swHemTypes_e.swHemTypeOpen, swHemPositionTypes_e.swHemPositionTypeOutside, False, 0.01,
0.01, 0, 0.005, 0.0011, CBAObject, False, swSheetMetalReliefTypes_e.swSheetMetalReliefObround, 0, True, 1.0#, 0,
0)

        Part.ClearSelection2(True)

        myHem = myFeature.GetDefinition

        Debug.Print("Hem type as
defined in swHemTypes_e: " & myHem.Type)

    End Sub

    Public swApp As SldWorks

End Class
```

Insert Sketch Text and Hole Example (C#)

This example shows how to insert sketch text and a hole at the selected point on a face.

```
//-----
// Preconditions: Open a model document and select a face.
//
// Postconditions:
// 1. Creates a hole and inserts the specified text on the
//    face at its point of selection.
// 2. Examine the graphics area and FeatureManager design tree.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace InsertSketchText_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 swModel;
        SelectionMgr swSelMgr;
        MathPoint swMathPt;
        Face2 selFace;
        Entity selEnt;
        double[] selPt;
        object NewPt;
        MathUtility swMathUtil;
        MathTransform swMathTrans;
        SelectData selData;
        SketchManager swSketchMgr;
        SketchText skText;
        double[] ptArr = new double[3];
        object @params;
        Feature holeFeat;
        FeatureManager swFeatMgr;

        bool boolstatus;

        public object TransformPoint(Sketch Sketch1, double X, double Y, double Z)
        {
            ptArr[0] = X;
            ptArr[1] = Y;
            ptArr[2] = Z;

            swMathUtil = (MathUtility)swApp.GetMathUtility();
            swMathPt = (MathPoint)swMathUtil.CreatePoint((ptArr));

            @params = swMathPt.ArrayData;

            swMathTrans = Sketch1.ModelToSketchTransform;
            swMathPt = (MathPoint)swMathPt.MultiplyTransform(swMathTrans);

            NewPt = swMathPt.ArrayData;

            return NewPt;
        }

        public void Main()
        {
            swModel = (ModelDoc2)swApp.ActiveDoc;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            selFace = (Face2)swSelMgr.GetSelectedObject6(1, -1);
            selEnt = (Entity)selFace;

            selPt = (double[])swSelMgr.GetSelectionPoint2(1, -1);

            selData = swSelMgr.CreateSelectData();

            selData.X = selPt[0];
            selData.Y = selPt[1];
            selData.Z = selPt[2];

            swSketchMgr = swModel.SketchManager;

            swSketchMgr.InsertSketch(true);

            selPt = (double[])TransformPoint(swModel.IGetActiveSketch2(), selPt[0], selPt[1], selPt[2]);

            skText = (SketchText)swModel.InsertSketchText(selPt[0], selPt[1], selPt[2], "Hole", 0, 0, 0, 100, 100);
        }
    }
}
```

```

        @params = skText.GetCoordinates();

        swSketchMgr.InsertSketch(true);

        boolstatus = selEnt.Select4(false, selData);

        swFeatMgr = swModel.FeatureManager;
        holeFeat = swFeatMgr.SimpleHole2(0.001, true, false, false, 0, 0, 0.001, 0.001, false, false,
        false, false, 0, 0, false, false, false, false, true, true,
        false, false, false);
    }

    /// <summary>
    /// The SldWorks swApp variable is pre-assigned for you.
    /// </summary>

    public SldWorks swApp;
}

}

```

Insert Sketch Text and Hole Example (VBA)

This example shows how to insert sketch text and a hole at the selected point on a face.

```

'-----
' Preconditions: Open a model document and select a face.
'
' Postconditions:
' 1. Creates a hole and inserts the specified text on the
'    face at its point of selection.
' 2. Examine the graphics area and FeatureManager design tree.
'-----

```

Option Explicit

```

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swSelMgr As SldWorks.SelectionMgr
Dim swMathPt As SldWorks.MathPoint
Dim selFace As Face2
Dim selEnt As Entity
Dim selPt As Variant
Dim NewPt As Variant
Dim swMathUtil As SldWorks.MathUtility
Dim swMathTrans As SldWorks.MathTransform
Dim selData As SldWorks.SelectData
Dim swSketchMgr As SldWorks.SketchManager
Dim skText As SketchText
Dim ptArr(2) As Double
Dim params As Variant
Dim holeFeat As SldWorks.Feature
Dim swFeatMgr As SldWorks.FeatureManager
Dim boolstatus As Boolean

```

```

Function TransformPoint(ByVal Sketch1 As SldWorks.Sketch, ByVal X As Double, ByVal Y As Double, ByVal Z As Double) As Variant

```

```

    ptArr(0) = X
    ptArr(1) = Y
    ptArr(2) = Z

```

```

    Set swMathUtil = swApp.GetMathUtility
    Set swMathPt = swMathUtil.CreatePoint((ptArr))
    params = swMathPt.ArrayData
    Set swMathTrans = Sketch1.ModelToSketchTransform
    Set swMathPt = swMathPt.MultiplyTransform(swMathTrans)
    NewPt = swMathPt.ArrayData
    TransformPoint = NewPt

```

```

End Function

```

```

Sub main()

```

```

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager
    Set selFace = swSelMgr.GetSelectedObject6(1, -1)
    Set selEnt = selFace

```

```

    selPt = swSelMgr.GetSelectionPoint2(1, -1)

```

```

    Set selData = swSelMgr.CreateSelectData

```

```

    selData.X = selPt(0)
    selData.Y = selPt(1)
    selData.Z = selPt(2)

```

```

    Set swSketchMgr = swModel.SketchManager

```

```

    swSketchMgr.InsertSketch True

```

```

    selPt = TransformPoint(swModel.IGetActiveSketch2, selPt(0), selPt(1), selPt(2))

```

```

    Set skText = swModel.InsertSketchText(selPt(0), selPt(1), selPt(2), "Hole", 0, 0, 0, 100, 100)

```

```

    params = skText.GetCoordinates

```

```

    swSketchMgr.InsertSketch True

```

```

    boolstatus = selEnt.Select4(False, selData)

```

```

    Set swFeatMgr = swModel.FeatureManager
    Set holeFeat = swFeatMgr.SimpleHole2(0.001, True, False, False, 0, 0, 0.001, 0.001, False, False, False, False, 0, 0, False,
False, False, False, True, True, False, False, False)

```

End Sub

Insert Sketch Text and Hole Example (VB.NET)

This example shows how to insert sketch text and a hole at the selected point on a face.

```

'-----
' Preconditions: Open a model document and select a face.
'
' Postconditions:
' 1. Creates a hole and inserts the specified text on the
'    face at its point of selection.
' 2. Examine the graphics area and FeatureManager design tree.
'-----

```

```

Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System

```

```
Partial Class SolidWorksMacro
```

```

    Dim swModel As ModelDoc2
    Dim swSelMgr As SelectionMgr
    Dim swMathPt As MathPoint
    Dim selFace As Face2
    Dim selEnt As Entity
    Dim selPt As Object
    Dim NewPt As Object
    Dim swMathUtil As MathUtility
    Dim swMathTrans As MathTransform
    Dim selData As SelectData
    Dim swSketchMgr As SketchManager
    Dim skText As SketchText
    Dim ptArr(2) As Double
    Dim params As Object
    Dim holeFeat As Feature
    Dim swFeatMgr As FeatureManager
    Dim boolstatus As Boolean

```

```
Function TransformPoint(ByVal Sketch1 As Sketch, ByVal X As Double, ByVal Y As Double, ByVal Z As Double) As Object
```

```

    ptArr(0) = X
    ptArr(1) = Y
    ptArr(2) = Z

```

```

    swMathUtil = swApp.GetMathUtility
    swMathPt = swMathUtil.CreatePoint((ptArr))

```

```
    params = swMathPt.ArrayData
```

```

    swMathTrans = Sketch1.ModelToSketchTransform
    swMathPt = swMathPt.MultiplyTransform(swMathTrans)

```

```
    NewPt = swMathPt.ArrayData
```

```
    TransformPoint = NewPt
```

```
End Function
```

```
Sub main()
```

```

    swModel = swApp.ActiveDoc
    swSelMgr = swModel.SelectionManager
    selFace = swSelMgr.GetSelectedObject6(1, -1)
    selEnt = selFace

```

```
    selPt = swSelMgr.GetSelectionPoint2(1, -1)
```

```
    selData = swSelMgr.CreateSelectData
```

```

    selData.X = selPt(0)
    selData.Y = selPt(1)
    selData.Z = selPt(2)

```

```
    swSketchMgr = swModel.SketchManager
```

```
    swSketchMgr.InsertSketch(True)
```

```
    selPt = TransformPoint(swModel.IGetActiveSketch2, selPt(0), selPt(1), selPt(2))
```

```
    skText = swModel.InsertSketchText(selPt(0), selPt(1), selPt(2), "Hole", 0, 0, 0, 100, 100)
```

```
    params = skText.GetCoordinates
```

```
    swSketchMgr.InsertSketch(True)
```

```
    boolstatus = selEnt.Select4(False, selData)
```

```
    swFeatMgr = swModel.FeatureManager
```

```
holeFeat = swFeatMgr.SimpleHole2(0.001, True, False, False, 0, 0, 0.001, 0.001, False, False, False, False, 0, 0, False, Fal
```

```
End Sub
```

```
''' <summary>  
''' The SldWorks swApp variable is pre-assigned for you.  
''' </summary>  
Public swApp As SldWorks
```

```
End Class
```

Insert Solid Body Boundary Surface Feature Example (C#)

This example shows how to insert a solid body boundary surface feature.

```
//-----
// Preconditions: Verify that the specified part template
// exists.
//
// Postconditions:
// 1. Opens a new part.
// 2. Inserts a sketch of a rectangle, Sketch1, on Front Plane.
// 3. Creates Surface-Plane1 using Sketch1.
// 4. Creates Plane1.
// 5. Creates a sketch of a circle, Sketch2, on Plane1.
// 6. Creates Surface-Plane2 using Sketch2.
// 7. Inserts a solid body boundary surface feature, Boundary-Surface1,
//    using Surface-Plane1 and Surface-Plane2.
// 8. Examine the graphics area and expand Solid Bodies(1) in the
//    FeatureManager design tree to verify step 7.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            FeatureManager swFeatureMgr = default(FeatureManager);
            SketchManager swSketchMgr = default(SketchManager);
            RefPlane swRefPlane = default(RefPlane);
            SketchSegment swSketchSegment = default(SketchSegment);
            Feature swFeature = default(Feature);
            object[] sketchSegments = null;
            bool status = false;

            swModel = (ModelDoc2)swApp.NewDocument("C:\\ProgramData\\SolidWorks\\SOLIDWORKS 2016\\templates\\part.prtdot", 0, 0, 0);
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            swSketchMgr = (SketchManager)swModel.SketchManager;
            swFeatureMgr = (FeatureManager)swModel.FeatureManager;

            //Create Surface-Plane1
            status = swModelDocExt.SelectByID2("Front Plane", "PLANE", -0.0687189668956523, 0.0593633502290038, 0.00936526409663904, false, 0, null, 0);
            status = swModelDocExt.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstToRectEntity, (int)swUserPreferenceOption_e.swDetailingNoOptionSpecified, false);
            status = swModelDocExt.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, (int)swUserPreferenceOption_e.swDetailingNoOptionSpecified, true);
            sketchSegments = (object[])swSketchMgr.CreateCornerRectangle(-0.039991197344551, 0.02969400507229, 0, 0.0502882343966202, -0.0299334728551311, 0);

            swSketchMgr.InsertSketch(true);
            status = swModelDocExt.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, true, 0, null, 0);
            status = swModel.InsertPlanarRefSurface();
            swModel.ClearSelection2(true);

            //Create Plane1
            status = swModelDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, true, 0, null, 0);
            swRefPlane = (RefPlane)swFeatureMgr.InsertRefPlane((int)swRefPlaneReferenceConstraints_e.swRefPlaneReferenceConstraint_Distance, 0.15, 0, 0, 0);

            swModel.ClearSelection2(true);

            //Create Surface-Plane2
            status = swModelDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, true, 0, null, 0);
            swSketchSegment = (SketchSegment)swSketchMgr.CreateCircle(0.003592, 0.003353, 0.0, 0.035202, -0.057233, 0.0);
            swSketchMgr.InsertSketch(true);
            status = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, false, 1, null, 0);
            status = swModel.InsertPlanarRefSurface();
            swModel.ClearSelection2(true);
            swModel.ViewZoomtofit();

            //Insert a solid body boundary surface feature
            status = swModelDocExt.SelectByID2("Surface-Plane1", "SURFACEBODY", -0.039991197344551, 0.02969400507229, 0, false, 8193, null, 0);
            status = swModelDocExt.SelectByID2("Surface-Plane2", "SURFACEBODY", -0.0463651179854531, -0.0432741101197696, 0.15, true, 16385, null, 0);
            swFeature = (Feature)swFeatureMgr.SetNetBlendCurveData(0, 0, (int)swTangencyType_e.swTangencyNone, 0, 1, true);
            swFeature = (Feature)swFeatureMgr.SetNetBlendCurveData(0, 1, (int)swTangencyType_e.swTangencyNone, 0, 1, true);
            swFeature = (Feature)swFeatureMgr.SetNetBlendDirectionData(0, 32, 0, false, false);
            swFeature = (Feature)swFeatureMgr.SetNetBlendDirectionData(1, 32, 0, false, false);
            swFeature = (Feature)swFeatureMgr.InsertNetBlend2(2, 2, 0, false, 0.0001, false, true, true, true, false, -1, -1, false, -1, false, false, -1, false, -1, true, true);

        }
    }
}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>
public SldWorks swApp;
```

Insert Solid Body Boundary Surface Feature Example (VBA)

This example shows how to insert a solid body boundary surface feature.

```

'-----
' Preconditions: Verify that the specified part template
' exists.
'
' Postconditions:
' 1. Opens a new part.
' 2. Inserts a sketch of a rectangle, Sketch1, on Front Plane.
' 3. Creates Surface-Plane1 using Sketch1.
' 4. Creates Plane1.
' 5. Creates a sketch of a circle, Sketch2, on Plane1.
' 6. Creates Surface-Plane2 using Sketch2.
' 7. Inserts a solid body boundary surface feature, Boundary-Surface1,
'    using Surface-Plane1 and Surface-Plane2.
' 8. Examine the graphics area and expand Solid Bodies(1) in the
'    FeatureManager design tree to verify step 7.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swSketchMgr As SldWorks.SketchManager
Dim swRefPlane As SldWorks.RefPlane
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim sketchSegments As Variant
Dim status As Boolean

Sub main()

    Set swApp = Application.SldWorks

    Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2016\templates\part.prtdot", 0, 0, 0)
    Set swModelDocExt = swModel.Extension
    Set swSketchMgr = swModel.SketchManager
    Set swFeatureMgr = swModel.FeatureManager

    'Create Surface-Plane1
    status = swModelDocExt.SelectByID2("Front Plane", "PLANE", -6.87189668956523E-02, 5.93633502290038E-02, 9.36526409663904E-03, False, 0, Nothing, 0)
    status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
    status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
    sketchSegments = swSketchMgr.CreateCornerRectangle(-3.99911197344551E-02, 0.02969400507229, 0, 5.02882343966202E-02, -2.99334728551311E-02, 0)
    swSketchMgr.InsertSketch True
    status = swModelDocExt.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
    status = swModel.InsertPlanarRefSurface()

    swModel.ClearSelection2 True

    'Create Plane1
    status = swModelDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Set swRefPlane = swFeatureMgr.InsertRefPlane(swRefPlaneReferenceConstraints_e.swRefPlaneReferenceConstraint_Distance, 0.15, 0, 0, 0, 0)

    swModel.ClearSelection2 True

    'Create Surface-Plane2
    status = swModelDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Set swSketchSegment = swSketchMgr.CreateCircle(0.003592, 0.003353, 0#, 0.035202, -0.057233, 0#)
    swSketchMgr.InsertSketch True
    status = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 1, Nothing, 0)
    status = swModel.InsertPlanarRefSurface()

    swModel.ClearSelection2 True
    swModel.ViewZoomtofit2

    'Insert a solid body boundary surface feature
    status = swModelDocExt.SelectByID2("Surface-Plane1", "SURFACEBODY", -3.99911197344551E-02, 0.02969400507229, 0, False, 8193, Nothing, 0)
    status = swModelDocExt.SelectByID2("Surface-Plane2", "SURFACEBODY", -4.63651179854531E-02, -4.32741101197696E-02, 0.15, True, 16385, Nothing, 0)
    Set swFeature = swFeatureMgr.SetNetBlendCurveData(0, 0, swTangencyType_e.swTangencyNone, 0, 1, True)
    Set swFeature = swFeatureMgr.SetNetBlendCurveData(0, 1, swTangencyType_e.swTangencyNone, 0, 1, True)
    Set swFeature = swFeatureMgr.SetNetBlendDirectionData(0, 32, 0, False, False)
    Set swFeature = swFeatureMgr.SetNetBlendDirectionData(1, 32, 0, False, False)
    Set swFeature = swFeatureMgr.InsertNetBlend2(2, 2, 0, False, 0.0001, False, True, True, True, False, -1, -1, False, -1, False, False, -1, False, -1,
True, True)

End Sub

```

Insert Solid Body Boundary Surface Feature Example (VB.NET)

This example shows how to insert a solid body boundary surface feature.

```

'-----
' Preconditions: Verify that the specified part template
' exists.
'
' Postconditions:
' 1. Opens a new part.
' 2. Inserts a sketch of a rectangle, Sketch1, on Front Plane.
' 3. Creates Surface-Plane1 using Sketch1.
' 4. Creates Plane1.
' 5. Creates a sketch of a circle, Sketch2, on Plane1.
' 6. Creates Surface-Plane2 using Sketch2.
' 7. Inserts a solid body boundary surface feature, Boundary-Surface1,
'    using Surface-Plane1 and Surface-Plane2.
' 8. Examine the graphics area and expand Solid Bodies(1) in the
'    FeatureManager design tree to verify step 7.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System

Partial Class SolidWorksMacro

    Public Sub main()
        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swFeatureMgr As FeatureManager
        Dim swSketchMgr As SketchManager
        Dim swRefPlane As RefPlane
        Dim swSketchSegment As SketchSegment
        Dim swFeature As Feature
        Dim sketchSegments As Object
        Dim status As Boolean

        swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2016\templates\part.prtdot", 0, 0, 0)
        swModelDocExt = swModel.Extension
        swSketchMgr = swModel.SketchManager
        swFeatureMgr = swModel.FeatureManager

        'Create Surface-Plane1
        status = swModelDocExt.SelectByID2("Front Plane", "PLANE", -0.0687189668956523, 0.0593633502290038, 0.00936526409663904, False, 0, Nothing, 0)
        status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity, swUserPreferenceOption_e.swDetailingNoOption
Specified, False)
        status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType, swUserPreferenceOption_e.swDetailingNoOp
tionSpecified, True)
        sketchSegments = swSketchMgr.CreateCornerRectangle(-0.0399911197344551, 0.02969400507229, 0, 0.0502882343966202, -0.0299334728551311, 0)
        swSketchMgr.InsertSketch(True)
        status = swModelDocExt.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
        status = swModel.InsertPlanarRefSurface()
        swModel.ClearSelection2(True)

        'Create Plane1
        status = swModelDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
        swRefPlane = swFeatureMgr.InsertRefPlane(swRefPlaneReferenceConstraints_e.swRefPlaneReferenceConstraint_Distance, 0.15, 0, 0, 0, 0)
        swModel.ClearSelection2(True)

        'Create Surface-Plane2
        status = swModelDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
        swSketchSegment = swSketchMgr.CreateCircle(0.003592, 0.003353, 0.0#, 0.035202, -0.057233, 0.0#)
        swSketchMgr.InsertSketch(True)
        status = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 1, Nothing, 0)
        status = swModel.InsertPlanarRefSurface()
        swModel.ClearSelection2(True)
        swModel.ViewZoomtofit2()

        'Insert a solid body boundary surface feature
        status = swModelDocExt.SelectByID2("Surface-Plane1", "SURFACEBODY", -0.0399911197344551, 0.02969400507229, 0, False, 8193, Nothing, 0)
        status = swModelDocExt.SelectByID2("Surface-Plane2", "SURFACEBODY", -0.0463651179854531, -0.0432741101197696, 0.15, True, 16385, Nothing, 0)
        swFeature = swFeatureMgr.SetNetBlendCurveData(0, 0, swTangencyType_e.swTangencyNone, 0, 1, True)
        swFeature = swFeatureMgr.SetNetBlendCurveData(0, 1, swTangencyType_e.swTangencyNone, 0, 1, True)
        swFeature = swFeatureMgr.SetNetBlendDirectionData(0, 32, 0, False, False)
        swFeature = swFeatureMgr.SetNetBlendDirectionData(1, 32, 0, False, False)
        swFeature = swFeatureMgr.InsertNetBlend(2, 2, 0, False, 0.0001, False, True, True, True, False, -1, -1, False, -1, False, False, -1, False, -1, T
rue, True)

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

End Class

```

End Class

Insert Spline in Drawing Example (C++ COM)

This example shows how to insert a spline in a drawing.

```
STDMETHODIMP CDgtlPen::demo3()
```

```
{
    HRESULT                hr = S_OK;
    CComPtr <IModelDoc2>    pModel;
    CComQIPtr <IDrawingDoc> pDraw;
    CComPtr <IView>         pView;
    CComPtr <IView>         pFocusLockedView;
    CComPtr <ISheet>        pSheet;
    int                    pPropArray[4];
    double                 pKnotsArray[11];
    double                 pCntrlPntCoordArray[21];
    VARIANT_BOOL           updateEditRebuild = TRUE;
    long                   retval = NULL;
    VARIANT_BOOL           focusLocked = NULL;
    ZeroMemory(pPropArray, 4 * sizeof(int));
    ZeroMemory(pKnotsArray, 11 * sizeof(double));
    ZeroMemory(pCntrlPntCoordArray, 21 * sizeof(double));
    pPropArray[0] = 3;    //Dimension
    pPropArray[1] = 4;    //Order
    pPropArray[2] = 7;    //Number of control Points
    pPropArray[3] = 0;    //Periodicity
    pKnotsArray[0] = 0;
    pKnotsArray[1] = 0;
    pKnotsArray[2] = 0;
    pKnotsArray[3] = 0;
    pKnotsArray[4] = 0.3;
    pKnotsArray[5] = 0.3;
    pKnotsArray[6] = 0.6;
    pKnotsArray[7] = 1;
    pKnotsArray[8] = 1;
    pKnotsArray[9] = 1;
    pKnotsArray[10] = 1;
    pCntrlPntCoordArray[0] = -0.1;  pCntrlPntCoordArray[1] = -0.009;  pCntrlPntCoordArray[2] = 0;
    pCntrlPntCoordArray[3] = -0.11; pCntrlPntCoordArray[4] = 0.013;  pCntrlPntCoordArray[5] = 0;
    pCntrlPntCoordArray[6] = -0.14; pCntrlPntCoordArray[7] = 0.035;  pCntrlPntCoordArray[8] = 0;
    pCntrlPntCoordArray[9] = 0.015; pCntrlPntCoordArray[10] = 0.019;  pCntrlPntCoordArray[11] = 0;
    pCntrlPntCoordArray[12] = -0.059; pCntrlPntCoordArray[13] = -0.024; pCntrlPntCoordArray[14] = 0;
    pCntrlPntCoordArray[15] = 0.012; pCntrlPntCoordArray[16] = -0.018; pCntrlPntCoordArray[17] = 0;
    pCntrlPntCoordArray[18] = 0.054; pCntrlPntCoordArray[19] = -0.013; pCntrlPntCoordArray[20] = 0;
    hr = m_IsIdWorks->get_IActiveDoc2(&pModel);
    _ASSERT(pModel);
    pDraw = pModel;
    _ASSERT(pDraw);
    hr = pDraw->IGetFirstView(&pView);
    _ASSERT(pView);
    while(pView != NULL)
    {
        CComPtr <IView>         pNextView;
        hr = pView->get_FocusLocked(&focusLocked);

        if (focusLocked)
        {
            pFocusLockedView = pView;
            _ASSERT(pView);
            pView = NULL;
        }
        else
        {
            hr = pView->IGetNextView(&pNextView);
            pView = NULL;
            pView = pNextView;
        }
    }
}
```

```
hr = pDraw->IGetCurrentSheet(&pSheet);
_ASSERT(pSheet);
hr = pSheet->get_FocusLocked(&focusLocked);
hr = pSheet->put_FocusLocked (TRUE);

hr = pModel->InsertSketch2(updateEditRebuild);
hr = pModel->ISketchSplineByEqnParams(pPropArray, pKnotsArray, pCntrlPntCoordArray, &retval);
_ASSERT(retval);
hr = pSheet->put_FocusLocked(focusLocked);
if(pFocusLockedView != NULL)
{
    hr = pFocusLockedView->put_FocusLocked(TRUE);
}
hr = pModel->GraphicsRedraw2();
return S_OK;
}
```

Insert Spline Point Example (C#)

This example shows how to insert a spline point in a spline sketch.

```
//-----
// Preconditions: Verify that the specified template exists.
//
// Postconditions:
// 1. Sketches a spline containing four spline points.
// 2. Inserts a fifth spline point.
// 3. Examine the graphics area.
//-----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace InsertSplinePoint_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;
        SketchSegment skSegment;

        public void Main()
        {
            Part = (ModelDoc2)swApp.NewDocument("C:\\ProgramData\\SOLIDWORKS\\SOLIDWORKS 2017\\templates\\Part.prtdot", 0, 0, 0);

            object pointArray = null;
            double[] points = new double[12];
            points[0] = -0.0625070182577474;
            points[1] = 0.00739156869269664;
            points[2] = 0;
            points[3] = -0.0420233044773113;
            points[4] = 0.0350529989729012;
            points[5] = 0;
            points[6] = 0.0278754181857153;
            points[7] = -0.0165377796333246;
            points[8] = 0;
            points[9] = 0.0403878396197683;
            points[10] = 0.0406222157061507;
            points[11] = 0;
            pointArray = points;

            skSegment = Part.SketchManager.CreateSpline((pointArray));
            Part.InsertSplinePoint(0.0382447809668287, 0.00781095184375528, 0);
            Part.SketchManager.InsertSketch(true);
        }

        public SldWorks swApp;
    }
}
```

Insert Spline Point Example (VBA)

This example shows how to insert a spline point in a spline sketch.

```
'-----
' Preconditions: Verify that the specified template exists.
'
' Postconditions:
' 1. Sketches a spline containing four spline points.
' 2. Inserts a fifth spline point.
' 3. Examine the graphics area.
'-----

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim skSegment As SldWorks.SketchSegment

Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.NewDocument("C:\ProgramData\SOLIDWORKS\SOLIDWORKS 2017\templates\Part.prtdot", 0, 0, 0)

    Dim pointArray As Variant
    Dim points() As Double
    ReDim points(0 To 11) As Double
    points(0) = -6.25070182577474E-02
    points(1) = 7.39156869269664E-03
    points(2) = 0
    points(3) = -4.20233044773113E-02
    points(4) = 3.50529989729012E-02
    points(5) = 0
    points(6) = 2.78754181857153E-02
    points(7) = -1.65377796333246E-02
    points(8) = 0
    points(9) = 4.03878396197683E-02
    points(10) = 4.06222157061507E-02
    points(11) = 0
    pointArray = points

    Set skSegment = Part.SketchManager.CreateSpline((pointArray))
    Part.InsertSplinePoint 3.82447809668287E-02, 7.81095184375528E-03, 0
    Part.SketchManager.InsertSketch True

End Sub

-

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim skSegment As SldWorks.SketchSegment

Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.NewDocument("C:\ProgramData\SOLIDWORKS\SOLIDWORKS 2012\templates\Part.prtdot", 0, 0, 0)

    Dim pointArray As Variant
    Dim points() As Double
    ReDim points(0 To 11) As Double
    points(0) = -6.25070182577474E-02
    points(1) = 7.39156869269664E-03
    points(2) = 0
    points(3) = -4.20233044773113E-02
    points(4) = 3.50529989729012E-02
    points(5) = 0
    points(6) = 2.78754181857153E-02
    points(7) = -1.65377796333246E-02
    points(8) = 0
    points(9) = 4.03878396197683E-02
    points(10) = 4.06222157061507E-02
    points(11) = 0
    pointArray = points

    Set skSegment = Part.SketchManager.CreateSpline((pointArray))
    Part.InsertSplinePoint 3.82447809668287E-02, 7.81095184375528E-03, 0
    Part.SketchManager.InsertSketch True

End Sub
```

Insert Spline Point Example (VB.NET)

This example shows how to insert a spline point in a spline sketch.

```
'-----  
' Preconditions: Verify that the specified template exists.  
'-----  
' Postconditions:  
' 1. Sketches a spline containing four spline points.  
' 2. Inserts a fifth spline point.  
' 3. Examine the graphics area.  
'-----
```

```
Imports SolidWorks.Interop.sldworks
```

```
Imports SolidWorks.Interop.swconst
```

```
Imports System.Runtime.InteropServices
```

```
Imports System
```

```
Partial Class SolidWorksMacro
```

```
    Dim Part As ModelDoc2
```

```
    Dim skSegment As SketchSegment
```

```
    Sub main()
```

```
        Part = swApp.NewDocument("C:\ProgramData\SOLIDWORKS\SOLIDWORKS  
2017\templates\Part.prtdot", 0, 0, 0)
```

```
        Dim pointArray As Object
```

```
        Dim points(11) As Double
```

```
        points(0) = -0.0625070182577474
```

```
        points(1) = 0.00739156869269664
```

```
        points(2) = 0
```

```
        points(3) = -0.0420233044773113
```

```
        points(4) = 0.0350529989729012
```

```
        points(5) = 0
```

```
        points(6) = 0.0278754181857153
```

```
        points(7) = -0.0165377796333246
```

```
        points(8) = 0
```

```
        points(9) = 0.0403878396197683
```

```
        points(10) = 0.0406222157061507
```

```
        points(11) = 0
```

```
        pointArray = points
```

```
        skSegment = Part.SketchManager.CreateSpline((pointArray))
```

```
        Part.InsertSplinePoint(0.0382447809668287,  
0.00781095184375528, 0)
```

```
        Part.SketchManager.InsertSketch(True)
```

```
    End Sub
```

```
Public swApp As SldWorks
```

```
End Class
```

Insert Structural Weldment Example (C#)

This example shows how to create structural weldment features using structural member groups.

```
//-----
// Preconditions:
// 1. Verify the existence of the weldment profile pathname
// as specified in both calls to IFeatureManager::InsertStructuralWeldment4.
// 2. Open an Immediate window.
//
// Postconditions:
// 1. Creates a new part document containing a weldment and structural members.
// 2. Rotates c channel 3 x 5(1).
// 3. Inspect the FeatureManager design tree, graphics area, and
// Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;
using System.Diagnostics;
using System.Runtime.InteropServices;

namespace Macro1CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;
        bool boolstatus;
        FeatureManager FeatMgr;
        SelectionMgr SelMgr;
        Feature swWeldFeat;
        StructuralMemberFeatureData swWeldFeatData;

        public void Main()
        {
            string MacroFolder = null;
            MacroFolder = swApp.GetCurrentMacroPathFolder();
            swApp.SetCurrentWorkingDirectory(MacroFolder);

            string Template = null;
            Template = swApp.GetUserPreferenceStringValue((int)swUserPreferenceStringValue_e.swDefaultTemplatePart);
            Part = (ModelDoc2)swApp.NewDocument(Template, 0, 0, 0);

            FeatMgr = Part.FeatureManager;
            SelMgr = (SelectionMgr)Part.SelectionManager;

            Part.ClearSelection2(true);

            object vSkLines = null;
            vSkLines = Part.SketchManager.CreateCornerRectangle(-0.1872393706766, 0.1133237194389, 0, -0.07003610048208, 0.009188409684237, 0);

            Part.ClearSelection2(true);

            vSkLines = Part.SketchManager.CreateCornerRectangle(0.06513561531715, 0.03369083550887, 0, 0.1807053904567, -0.08106219210316, 0);
            Part.SketchManager.InsertSketch(true);

            Part.ViewZoomToFit2();

            object myFeature = null;
            myFeature = Part.FeatureManager.InsertWeldmentFeature();

            StructuralMemberGroup Group1 = default(StructuralMemberGroup);
            Group1 = FeatMgr.CreateStructuralMemberGroup();
            SketchSegment[] segments1 = new SketchSegment[2];
            object[] GroupArray1 = new object[1];

            boolstatus = Part.Extension.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.1495427140733, 0.1133237194389, 0, false, 0, null, 0);
            boolstatus = Part.Extension.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", -0.1872393706766, 0.08238014634844, 0, true, 0, null, 0);

            segments1[0] = (SketchSegment)SelMgr.GetSelectedObject6(1, 0);
            segments1[1] = (SketchSegment)SelMgr.GetSelectedObject6(2, 0);

            Group1.Segments = segments1;
            Group1.Angle = 0.785714285714286; //radians
            Group1.ApplyCornerTreatment = true;
            Group1.CornerTreatmentType = (int)swSolidworksWeldmentEndCondOptions_e.swEndConditionMiter;
            Group1.MirrorProfile = true;
            Group1.MirrorProfileAxis = (int)swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical;
            Group1.GapWithinGroup = 0.0127; //meters

            GroupArray1[0] = (object)Group1;
            DispatcherWrapper[] groups1 = new DispatcherWrapper[1];
            groups1[0] = new DispatcherWrapper(GroupArray1[0]);

            myFeature = Part.FeatureManager.InsertStructuralWeldment4("C:\\Program Files\\SOLIDWORKS Corp\\SOLIDWORKS\\lang\\english\\weldment profiles\\ansi inch\\c channel\\3 x 5.sldlfp", 1, false, groups1);

            Part.ClearSelection2(true);

            StructuralMemberGroup Group2 = default(StructuralMemberGroup);
            Group2 = FeatMgr.CreateStructuralMemberGroup();
            SketchSegment[] segments2 = new SketchSegment[2];
            object[] GroupArray2 = new object[1];

            boolstatus = Part.Extension.SelectByID2("Line5@Sketch1", "EXTSKETCHSEGMENT", 0.1185825251083, 0.03369083550887, 0, false, 0, null, 0);
            boolstatus = Part.Extension.SelectByID2("Line6@Sketch1", "EXTSKETCHSEGMENT", 0.06513561531715, -0.02774616865332, 0, true, 0, null, 0);

            segments2[0] = (SketchSegment)SelMgr.GetSelectedObject6(1, 0);
            segments2[1] = (SketchSegment)SelMgr.GetSelectedObject6(2, 0);

            Group2.Segments = segments2;
            Group2.AlignAxis = (int)swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical;

            GroupArray2[0] = (object)Group2;
            DispatcherWrapper[] groups2 = new DispatcherWrapper[1];
            groups2[0] = new DispatcherWrapper(GroupArray2[0]);

            myFeature = Part.FeatureManager.InsertStructuralWeldment4("C:\\Program Files\\SOLIDWORKS Corp\\SOLIDWORKS\\lang\\english\\weldment profiles\\ansi inch\\c channel\\3 x 5.sldlfp", 1, false, groups2);
        }
    }
}
```

```

nsi inch\\c channel\\3 x 5.sldlfp", 1, false, groups2);

Part.ClearSelection2(true);

// Get Group Information
StructuralMemberGroup Group = default(StructuralMemberGroup);
object[] vGroups = null;
object[] vSegments = null;

boolstatus = Part.Extension.SelectByID2("c channel 3 x 5(1)", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);
swWeldFeat = (Feature)SelMgr.GetSelectedObject6(1, 0);

swWeldFeatData = (StructuralMemberFeatureData)swWeldFeat.GetDefinition();
swWeldFeatData.AccessSelections(Part, null);

Debug.Print("");
Debug.Print("Groups Count : " + swWeldFeatData.GetGroupsCount());
Debug.Print(" Feature Name : " + swWeldFeat.Name);

vGroups = (object[])swWeldFeatData.Groups;

int i = 0;
int j = 0;
for (i = vGroups.GetLowerBound(0); i <= vGroups.GetUpperBound(0); i++)
{
    Group = (StructuralMemberGroup)vGroups[i];
    Debug.Print(" Segment Count in Group " + i + 1 + " : " + Group.GetSegmentsCount());
    Debug.Print(" Rotational angle of group: " + Group.Angle);
    Debug.Print(" ApplyCornerTreatment: " + Group.ApplyCornerTreatment);
    Debug.Print(" CornerTreatmentType: " + Group.CornerTreatmentType);
    Debug.Print(" MirrorProfile: " + Group.MirrorProfile);
    Debug.Print(" MirrorProfileAxis: " + Group.MirrorProfileAxis);
    Debug.Print(" GapWithinGroup: " + Group.GapWithinGroup);
    vSegments = (object[])Group.Segments;
    for (j = vSegments.GetLowerBound(0); j <= vSegments.GetUpperBound(0); j++)
    {
        ((SketchSegment)vSegments[j]).Select(false);
    }
}

swWeldFeatData.ReleaseSelectionAccess();

boolstatus = Part.Extension.SelectByID2("c channel 3 x 5(2)", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);
swWeldFeat = (Feature)SelMgr.GetSelectedObject6(1, 0);
swWeldFeatData = (StructuralMemberFeatureData)swWeldFeat.GetDefinition();
swWeldFeatData.AccessSelections(Part, null);

Debug.Print("");
Debug.Print("Groups Count : " + swWeldFeatData.GetGroupsCount());
Debug.Print(" Feature Name : " + swWeldFeat.Name);

vGroups = (object[])swWeldFeatData.Groups;
for (i = vGroups.GetLowerBound(0); i <= vGroups.GetUpperBound(0); i++)
{
    Group = (StructuralMemberGroup)vGroups[i];
    Debug.Print(" Segment Count in Group " + i + 1 + " : " + Group.GetSegmentsCount());
    Debug.Print(" Rotational angle of group: " + Group.Angle);
    Debug.Print(" ApplyCornerTreatment: " + Group.ApplyCornerTreatment);
    Debug.Print(" CornerTreatmentType: " + Group.CornerTreatmentType);
    Debug.Print(" MirrorProfile: " + Group.MirrorProfile);
    Debug.Print(" MirrorProfileAxis: " + Group.MirrorProfileAxis);
    Debug.Print(" GapWithinGroup: " + Group.GapWithinGroup);
    vSegments = (object[])Group.Segments;
    for (j = vSegments.GetLowerBound(0); j <= vSegments.GetUpperBound(0); j++)
    {
        ((SketchSegment)vSegments[j]).Select(false);
    }
}

swWeldFeatData.ReleaseSelectionAccess();

Part.ClearSelection2(true);
}

public SldWorks swApp;
}
}

```

Insert Structural Weldment Example (VBA)

This example shows how to create structural weldment features using structural member groups.

```

'-----
' Preconditions:
' 1. Verify the existence of the weldment profile pathname
'    as specified in both calls to IFeatureManager::InsertStructuralWeldment4.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates a new part document containing a weldment and structural members.
' 2. Rotates c channel 3 x 5(1).
' 3. Inspect the FeatureManager design tree, graphics area, and
'    Immediate window.
'-----
Dim swApp As SldWorks.SldWorks

Dim Part As ModelDoc2

Dim boolstatus As Boolean

Dim FeatMgr As FeatureManager

Dim SelMgr As SelectionMgr

Dim swWeldFeat As SldWorks.Feature

Dim swWeldFeatData As SldWorks.StructuralMemberFeatureData

Option Explicit

Public Sub Main()
    Set swApp = Application.SldWorks
    Dim MacroFolder As String
    MacroFolder = swApp.GetCurrentMacroPathFolder()
    swApp.SetCurrentWorkingDirectory MacroFolder

    Dim Template As String
    Template = swApp.GetUserPreferenceStringValue(swDefaultTemplatePart)
    Set Part = swApp.NewDocument(Template, 0, 0, 0)

    Set FeatMgr = Part.FeatureManager
    Set SelMgr = Part.SelectionManager
    Part.ClearSelection2 True

    Dim vSkLines As Variant
    vSkLines = Part.SketchManager.CreateCornerRectangle(-0.1872393706766, 0.1133237194389, 0, -0.07003610048208, 0.009188409684237,
0)

    Part.ClearSelection2 True

    vSkLines = Part.SketchManager.CreateCornerRectangle(0.06513561531715, 0.03369083550887, 0, 0.1807053904567, -0.08106219210316,
0)

    Part.SketchManager.InsertSketch True
    Part.ViewZoomtofit2

    Dim myFeature As Object
    Set myFeature = Part.FeatureManager.InsertWeldmentFeature()

    Dim Group1 As StructuralMemberGroup
    Set Group1 = FeatMgr.CreateStructuralMemberGroup

    Dim vSeg1 As Variant
    Dim segments1(1) As Object
    Dim vGroups1 As Variant
    Dim GroupArray1(0) As Object

    boolstatus = Part.Extension.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.1495427140733, 0.1133237194389, 0, False, 0,
Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", -0.1872393706766, 0.08238014634844, 0, True, 0,
Nothing, 0)

    Set segments1(0) = SelMgr.GetSelectedObject6(1, 0)
    Set segments1(1) = SelMgr.GetSelectedObject6(2, 0)

    vSeg1 = segments1
    Group1.Segments = (vSeg1)
    Group1.Angle = 0.785714285714286 'radians
    Group1.ApplyCornerTreatment = True
    Group1.CornerTreatmentType = swEndConditionMiter
    Group1.MirrorProfile = True
    Group1.MirrorProfileAxis = swMirrorProfileOrAlignmentAxis_Vertical
    Group1.GapWithinGroup = 0.0127 'meters

    Set GroupArray1(0) = Group1
    vGroups1 = GroupArray1

    Set myFeature = Part.FeatureManager.InsertStructuralWeldment4("C:\Program Files\SOLIDWORKS
Corp\SOLIDWORKS\lang\english\weldment profiles\ansi inch\c channel\3 x 5.sldlfp", 1, False, (vGroups1))

    Part.ClearSelection2 True

    Dim Group2 As StructuralMemberGroup
    Set Group2 = FeatMgr.CreateStructuralMemberGroup
    Dim vSeg2 As Variant

```

```

Dim segments2(1) As Object
Dim vGroups2 As Variant
Dim GroupArray2(0) As Object

boolstatus = Part.Extension.SelectByID2("Line5@Sketch1", "EXTSKETCHSEGMENT", 0.1185825251083, 0.03369083550887, 0, False, 0,
Nothing, 0)
boolstatus = Part.Extension.SelectByID2("Line6@Sketch1", "EXTSKETCHSEGMENT", 0.06513561531715, -0.02774616865332, 0, True, 0,
Nothing, 0)

Set segments2(0) = SelMgr.GetSelectedObject6(1, 0)
Set segments2(1) = SelMgr.GetSelectedObject6(2, 0)

vSeg2 = segments2
Group2.Segments = (vSeg2)
Group2.AlignAxis = swMirrorProfileOrAlignmentAxis_Verical

Set GroupArray2(0) = Group2
vGroups2 = GroupArray2

Set myFeature = Part.FeatureManager.InsertStructuralWeldment4("C:\Program Files\SOLIDWORKS
Corp\SOLIDWORKS\lang\english\weldment profiles\ansi inch\c channel\3 x 5.sldlfp", 1, False, (vGroups2))

Part.ClearSelection2 True

' Get Group Information

Dim Group As StructuralMemberGroup
Dim vGroups As Variant
Dim vSegments As Variant

boolstatus = Part.Extension.SelectByID2("c channel 3 x 5(1)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swWeldFeat = SelMgr.GetSelectedObject6(1, 0)

Set swWeldFeatData = swWeldFeat.GetDefinition
swWeldFeatData.AccessSelections Part, Nothing

Debug.Print ""
Debug.Print "Groups Count : " & swWeldFeatData.GetGroupsCount
Debug.Print " Feature Name : " & swWeldFeat.Name

vGroups = swWeldFeatData.Groups

Dim i As Long, j As Long
For i = LBound(vGroups) To UBound(vGroups)
    Set Group = vGroups(i)
    Debug.Print " Segment Count in Group " & i + 1 & " : " & Group.GetSegmentsCount; ""
    Debug.Print " Rotational angle of group: " & Group.Angle
    Debug.Print " ApplyCornerTreatment: " & Group.ApplyCornerTreatment
    Debug.Print " CornerTreatmentType: " & Group.CornerTreatmentType
    Debug.Print " MirrorProfile: " & Group.MirrorProfile
    Debug.Print " MirrorProfileAxis: " & Group.MirrorProfileAxis
    Debug.Print " GapWithinGroup: " & Group.GapWithinGroup
    vSegments = Group.Segments
    For j = LBound(vSegments) To UBound(vSegments)
        vSegments(j).Select False
    Next j
Next i

swWeldFeatData.ReleaseSelectionAccess

boolstatus = Part.Extension.SelectByID2("c channel 3 x 5(2)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swWeldFeat = SelMgr.GetSelectedObject6(1, 0)
Set swWeldFeatData = swWeldFeat.GetDefinition
swWeldFeatData.AccessSelections Part, Nothing

Debug.Print ""
Debug.Print "Groups Count : " & swWeldFeatData.GetGroupsCount
Debug.Print " Feature Name : " & swWeldFeat.Name

vGroups = swWeldFeatData.Groups
For i = LBound(vGroups) To UBound(vGroups)
    Set Group = vGroups(i)
    Debug.Print " Segment Count in Group " & i + 1 & " : " & Group.GetSegmentsCount; ""
    Debug.Print " Rotational angle of group: " & Group.Angle
    Debug.Print " ApplyCornerTreatment: " & Group.ApplyCornerTreatment
    Debug.Print " CornerTreatmentType: " & Group.CornerTreatmentType
    Debug.Print " MirrorProfile: " & Group.MirrorProfile
    Debug.Print " MirrorProfileAxis: " & Group.MirrorProfileAxis
    Debug.Print " GapWithinGroup: " & Group.GapWithinGroup
    vSegments = Group.Segments
    For j = LBound(vSegments) To UBound(vSegments)
        vSegments(j).Select False
    Next j
Next i

swWeldFeatData.ReleaseSelectionAccess
Part.ClearSelection2 True

End Sub

```

Insert Structural Weldment Example (VB.NET)

This example shows how to create structural weldment features using structural member groups.

```

-----
Preconditions:
1. Verify the existence of the weldment profile pathname
   as specified in both calls to IFeatureManager::InsertStructuralWeldment4.
2. Open an Immediate window.
Postconditions:
1. Creates a new part document containing a weldment and structural members.
2. Rotates c channel 3 x 5(1).
3. Inspect the FeatureManager design tree, graphics area, and
   Immediate window.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System
Imports System.Diagnostics
Imports System.Runtime.InteropServices

Partial Class SolidWorksMacro

    Dim Part As ModelDoc2
    Dim boolstatus As Boolean
    Dim FeatMgr As FeatureManager
    Dim SelMgr As SelectionMgr
    Dim swWeldFeat As Feature
    Dim swWeldFeatData As StructuralMemberFeatureData

    Public Sub Main()

        Dim MacroFolder As String
        MacroFolder = swApp.GetCurrentMacroPathFolder()
        swApp.SetCurrentWorkingDirectory(MacroFolder)

        Dim Template As String
        Template = swApp.GetUserPreferenceStringValue(swUserPreferenceStringValue_e.swDefaultTemplatePart)
        Part = swApp.NewDocument(Template, 0, 0, 0)

        FeatMgr = Part.FeatureManager
        SelMgr = Part.SelectionManager

        Part.ClearSelection2(True)

        Dim vSkLines As Object
        vSkLines = Part.SketchManager.CreateCornerRectangle(-0.1872393706766, 0.1133237194389, 0, -0.07003610048208, 0.009188409684237, 0)

        Part.ClearSelection2(True)

        vSkLines = Part.SketchManager.CreateCornerRectangle(0.06513561531715, 0.03369083550887, 0, 0.1807053904567, -0.08106219210316, 0)
        Part.SketchManager.InsertSketch(True)

        Part.ViewZoomtofit2()

        Dim myFeature As Object
        myFeature = Part.FeatureManager.InsertWeldmentFeature()

        Dim Group1 As StructuralMemberGroup
        Group1 = FeatMgr.CreateStructuralMemberGroup
        Dim segments1(1) As SketchSegment
        Dim GroupArray1(0) As Object

        boolstatus = Part.Extension.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.1495427140733, 0.1133237194389, 0, False, 0, Nothing, 0)
        boolstatus = Part.Extension.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", -0.1872393706766, 0.08238014634844, 0, True, 0, Nothing, 0)

        segments1(0) = SelMgr.GetSelectedObject6(1, 0)
        segments1(1) = SelMgr.GetSelectedObject6(2, 0)

        Group1.Segments = segments1
        Group1.Angle = 0.785714285714286 'radians
        Group1.ApplyCornerTreatment = True
        Group1.CornerTreatmentType = swSolidWorksWeldmentEndCondOptions_e.swEndConditionMiter
        Group1.MirrorProfile = True
        Group1.MirrorProfileAxis = swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical
        Group1.GapWithinGroup = 0.0127 'meters

        GroupArray1(0) = Group1
        Dim groups1(0) As DispatcherWrapper
        groups1(0) = New DispatcherWrapper(GroupArray1(0))

        myFeature = Part.FeatureManager.InsertStructuralWeldment4("C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\lang\english\weldment profiles\ansi inch\c
channel\3 x 5.sldlfp", 1, False, groups1)

        Part.ClearSelection2(True)

        Dim Group2 As StructuralMemberGroup
        Group2 = FeatMgr.CreateStructuralMemberGroup
        Dim segments2(1) As SketchSegment
        Dim GroupArray2(0) As Object

        boolstatus = Part.Extension.SelectByID2("Line5@Sketch1", "EXTSKETCHSEGMENT", 0.1185825251083, 0.03369083550887, 0, False, 0, Nothing, 0)
        boolstatus = Part.Extension.SelectByID2("Line6@Sketch1", "EXTSKETCHSEGMENT", 0.06513561531715, -0.02774616865332, 0, True, 0, Nothing, 0)

        segments2(0) = SelMgr.GetSelectedObject6(1, 0)
        segments2(1) = SelMgr.GetSelectedObject6(2, 0)

        Group2.Segments = segments2
        Group2.AlignAxis = swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical

        GroupArray2(0) = Group2
        Dim groups2(0) As DispatcherWrapper
        groups2(0) = New DispatcherWrapper(GroupArray2(0))

        myFeature = Part.FeatureManager.InsertStructuralWeldment4("C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\lang\english\weldment profiles\ansi inch\c
channel\3 x 5.sldlfp", 1, False, groups2)

        Part.ClearSelection2(True)
    
```

```

' Get Group Information
Dim Group As StructuralMemberGroup
Dim vGroups As Object
Dim vSegments As Object

boolstatus = Part.Extension.SelectByID2("c channel 3 x 5(1)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swWeldFeat = SelMgr.GetSelectedObject6(1, 0)

swWeldFeatData = swWeldFeat.GetDefinition
swWeldFeatData.AccessSelections(Part, Nothing)

Debug.Print("")
Debug.Print("Groups Count : " & swWeldFeatData.GetGroupsCount)
Debug.Print(" Feature Name : " & swWeldFeat.Name)

vGroups = swWeldFeatData.Groups

Dim i As Long, j As Long
For i = LBound(vGroups) To UBound(vGroups)
    Group = vGroups(i)
    Debug.Print(" Segment Count in Group " & i + 1 & " : " & Group.GetSegmentsCount)
    Debug.Print(" Rotational angle of group: " & Group.Angle)
    Debug.Print(" ApplyCornerTreatment: " & Group.ApplyCornerTreatment)
    Debug.Print(" CornerTreatmentType: " & Group.CornerTreatmentType)
    Debug.Print(" MirrorProfile: " & Group.MirrorProfile)
    Debug.Print(" MirrorProfileAxis: " & Group.MirrorProfileAxis)
    Debug.Print(" GapWithinGroup: " & Group.GapWithinGroup)
    vSegments = Group.Segments
    For j = LBound(vSegments) To UBound(vSegments)
        vSegments(j).Select(False)
    Next j
Next i

swWeldFeatData.ReleaseSelectionAccess()

boolstatus = Part.Extension.SelectByID2("c channel 3 x 5(2)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swWeldFeat = SelMgr.GetSelectedObject6(1, 0)
swWeldFeatData = swWeldFeat.GetDefinition
swWeldFeatData.AccessSelections(Part, Nothing)

Debug.Print("")
Debug.Print("Groups Count : " & swWeldFeatData.GetGroupsCount)
Debug.Print(" Feature Name : " & swWeldFeat.Name)

vGroups = swWeldFeatData.Groups
For i = LBound(vGroups) To UBound(vGroups)
    Group = vGroups(i)
    Debug.Print(" Segment Count in Group " & i + 1 & " : " & Group.GetSegmentsCount)
    Debug.Print(" Rotational angle of group: " & Group.Angle)
    Debug.Print(" ApplyCornerTreatment: " & Group.ApplyCornerTreatment)
    Debug.Print(" CornerTreatmentType: " & Group.CornerTreatmentType)
    Debug.Print(" MirrorProfile: " & Group.MirrorProfile)
    Debug.Print(" MirrorProfileAxis: " & Group.MirrorProfileAxis)
    Debug.Print(" GapWithinGroup: " & Group.GapWithinGroup)
    vSegments = Group.Segments
    For j = LBound(vSegments) To UBound(vSegments)
        vSegments(j).Select(False)
    Next j
Next i

swWeldFeatData.ReleaseSelectionAccess()
Part.ClearSelection2(True)

End Sub

Public swApp As SldWorks

End Class

```

Insert Structural Weldments Using Custom Weldment Profile Example (C#)

This example shows how to insert a structural weldment feature using a custom weldment profile configuration and structural member groups.

```
//-----
// Preconditions:
// 1. Verify that the specified library feature and part document template
// exist.
// 2. Verify that a valid pathname exists in Parts in Tools > Options >
// System Options > Default Templates.
// 3. Create C:\Test\Pipes.
// 4. Open the Immediate window.
//
// Postconditions:
// 1. Opens the specified library feature, adds the nxn configuration,
// saves the library feature as nxn.sldlfp, and closes the new library
// feature, which is called a custom weldment profile when used to create
// structural weldment features.
// 2. Creates a new part document that contains a sketch of two
// rectangles.
// 3. Creates a weldment and two structural member features using the
// sketch and the nxn configuration of the custom weldment profile created in
// step 1.
// 4. Rotates Pipes nxn(1).
// 5. Examine the FeatureManager design tree, graphics area, and
// the Immediate window.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            FeatureManager swFeatMgr = default(FeatureManager);
            SelectionMgr swSelMgr = default(SelectionMgr);
            SketchManager swSketchMgr = default(SketchManager);
            FeatureManager swFeatureMgr = default(FeatureManager);
            ConfigurationManager swConfigMgr = default(ConfigurationManager);
            Configuration swConfig = default(Configuration);
            Feature swFeature = default(Feature);
            StructuralMemberGroup swStructuralMemberGroup1 = default(StructuralMemberGroup);
            StructuralMemberGroup swStructuralMemberGroup2 = default(StructuralMemberGroup);
            Feature swWeldFeat = default(Feature);
            StructuralMemberFeatureData swStructuralMemberFeatData = default(StructuralMemberFeatureData);
            bool status = false;
            int errors = 0;
            int warnings = 0;
            string libFeature = null;
            string newLibFeature = null;
            string macroFolder = null;
            string template = null;
            object sketchLines = null;
            SketchSegment[] segs1 = new SketchSegment[2];
            object[] groupArray1 = new object[1];
            DispatcherWrapper[] groups1 = new DispatcherWrapper[1];
            SketchSegment[] segs2 = new SketchSegment[2];
            DispatcherWrapper[] groups2 = new DispatcherWrapper[1];
            object[] groupArray2 = new object[1];
            StructuralMemberGroup group = default(StructuralMemberGroup);
            object[] groups = new object[2];
            object[] segs = new object[2];
            string weldmentProfile = null;
            string weldmentConfigurationName = null;

            //Open existing library feature, add nxn configuration,
            //and save library feature as nxn.sldlfp
            libFeature = "C:\\Program Files\\SolidWorks Corp\\SOLIDWORKS\\lang\\english\\weldment profiles\\ansi inch\\pipe\\0.5 sch 40.sldlfp";
            swModel = (ModelDoc2)swApp.OpenDoc6(libFeature, (int)swDocumentTypes_e.swDocPART, (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref errors, ref warnings);
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            status = swModelDocExt.SelectByID2("0.5 sch 40.SLDPRT", "COMPONENT", 0, 0, 0, false, 0, null, 0);
            swModel.ClearSelection2(true);
            swConfigMgr = (ConfigurationManager)swModel.ConfigurationManager;
            swConfig = (Configuration)swConfigMgr.AddConfiguration("nxn", "", "", (int)swConfigurationOptions2_e.swConfigOption_DontActivate, "", "");
            newLibFeature = "C:\\Test\\Pipes\\nxn.sldlfp";
            status = swModelDocExt.SaveAs(newLibFeature, (int)swSaveAsVersion_e.swSaveAsCurrentVersion, (int)swSaveAsOptions_e.swSaveAsOptions_Silent, null, ref errors, ref warnings);
            swModel = null;
            swApp.CloseDoc(newLibFeature);

            //Open new part document and
            //create weldment and structural members
            macroFolder = swApp.GetCurrentMacroPathFolder();
            swApp.SetCurrentWorkingDirectory(macroFolder);
            template = swApp.GetUserPreferenceStringValue((int)swUserPreferenceStringValue_e.swDefaultTemplatePart);
            swModel = (ModelDoc2)swApp.NewDocument(template, 0, 0, 0);
            swFeatMgr = (FeatureManager)swModel.FeatureManager;
            swSelMgr = (SelectionMgr)swModel.SelectionManager;
            swModel.ClearSelection2(true);
            swSketchMgr = (SketchManager)swModel.SketchManager;
            sketchLines = (object)swSketchMgr.CreateCornerRectangle(-0.1872393706766, 0.1133237194389, 0, -0.07003610048208, 0.009188409684237, 0);
            swModel.ClearSelection2(true);
            sketchLines = (object)swSketchMgr.CreateCornerRectangle(0.06513561531715, 0.03369083550887, 0, 0.1807053904567, -0.08106219210316, 0);
            swSketchMgr.InsertSketch(true);
            swModel.ViewZoomToFit2();
            swFeatureMgr = (FeatureManager)swModel.FeatureManager;
            swFeature = (Feature)swFeatureMgr.InsertWeldmentFeature();
            swStructuralMemberGroup1 = (StructuralMemberGroup)swFeatMgr.CreateStructuralMemberGroup();
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            status = swModelDocExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.1495427140733, 0.1133237194389, 0, false, 0, null, 0);
            status = swModelDocExt.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", -0.1872393706766, 0.08238014634844, 0, true, 0, null, 0);
        }
    }
}
```

```

segs1[0] = (SketchSegment)swSelMgr.GetSelectedObject6(1, 0);
segs1[1] = (SketchSegment)swSelMgr.GetSelectedObject6(2, 0);
swStructuralMemberGroup1.Segments = segs1;
swStructuralMemberGroup1.Angle = 0.785714285714286; //radians
swStructuralMemberGroup1.ApplyCornerTreatment = true;
swStructuralMemberGroup1.CornerTreatmentType = (int)swSolidworksWeldmentEndCondOptions_e.swEndConditionMiter;
swStructuralMemberGroup1.MirrorProfile = true;
swStructuralMemberGroup1.MirrorProfileAxis = (int)swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical;

groupArray1[0] = (object)swStructuralMemberGroup1;
groups1[0] = new DispatcherWrapper(groupArray1[0]);

weldmentProfile = "C:\\Test\\Pipes\\nxn.SLDLFP";
weldmentConfigurationName = "nxn";
swFeature = (Feature)swFeatureMgr.InsertStructuralWeldment5(weldmentProfile, (int)swConnectedSegmentsOption_e.swConnectedSegments_SimpleCut, f
else, (groups1), weldmentConfigurationName);
swModel.ClearSelection2(true);

swStructuralMemberGroup2 = (StructuralMemberGroup)swFeatMgr.CreateStructuralMemberGroup();
status = swModelDocExt.SelectByID2("Line5@Sketch1", "EXTSKETCHSEGMENT", 0.1185825251083, 0.03369083550887, 0, false, 0, null, 0);
status = swModelDocExt.SelectByID2("Line6@Sketch1", "EXTSKETCHSEGMENT", 0.06513561531715, -0.02774616865332, 0, true, 0, null, 0);
segs2[0] = (SketchSegment)swSelMgr.GetSelectedObject6(1, 0);
segs2[1] = (SketchSegment)swSelMgr.GetSelectedObject6(2, 0);

swStructuralMemberGroup2.Segments = segs2;
swStructuralMemberGroup2.AlignAxis = (int)swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical;

groupArray2[0] = (object)swStructuralMemberGroup2;
groups2[0] = new DispatcherWrapper(groupArray2[0]);

swFeature = (Feature)swFeatureMgr.InsertStructuralWeldment5(weldmentProfile, (int)swConnectedSegmentsOption_e.swConnectedSegments_SimpleCut, f
else, (groups2), weldmentConfigurationName);
swModel.ClearSelection2(true);

//Get feature data for each structural member group
status = swModelDocExt.SelectByID2("Pipes nxn(1)", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);
swWeldFeat = (Feature)swSelMgr.GetSelectedObject6(1, 0);
swStructuralMemberFeatData = (StructuralMemberFeatureData)swWeldFeat.GetDefinition();
swStructuralMemberFeatData.AccessSelections(swModel, null);
Debug.Print("");
Debug.Print("Groups count: " + swStructuralMemberFeatData.GetGroupsCount());
Debug.Print(" Feature name: " + swWeldFeat.Name);
Debug.Print(" Custom weldment profile configuration name: " + swStructuralMemberFeatData.ConfigurationName);
Debug.Print(" Transfer material? " + swStructuralMemberFeatData.TransferMaterial);
Debug.Print(" Library material profile: " + swStructuralMemberFeatData.LibraryProfileMaterial);

groups = (object[])swStructuralMemberFeatData.Groups;
long i = 0;
long j = 0;
Debug.Print(" Group:");
for (i = 0; i < groups.Length; i++)
{
    group = (StructuralMemberGroup)groups[i];
    Debug.Print(" Segment count: " + group.GetSegmentsCount());
    Debug.Print(" Rotational angle: " + group.Angle);
    Debug.Print(" Apply corner treatment: " + group.ApplyCornerTreatment);
    Debug.Print(" Corner treatment type: " + group.CornerTreatmentType);
    Debug.Print(" Mirror profile: " + group.MirrorProfile);
    Debug.Print(" Mirror profile axis: " + group.MirrorProfileAxis);
    Debug.Print(" Gap within: " + group.GapWithinGroup);
    segs = (object[])group.Segments;
    for (j = 0; j < segs.Length; j++)
    {
        ((SketchSegment)segs[j]).Select(false);
    }
}
swStructuralMemberFeatData.ReleaseSelectionAccess();

status = swModelDocExt.SelectByID2("Pipes nxn(2)", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);
swWeldFeat = (Feature)swSelMgr.GetSelectedObject6(1, 0);
swStructuralMemberFeatData = (StructuralMemberFeatureData)swWeldFeat.GetDefinition();
swStructuralMemberFeatData.AccessSelections(swModel, null);
Debug.Print("");
Debug.Print("Groups count: " + swStructuralMemberFeatData.GetGroupsCount());
Debug.Print(" Feature name: " + swWeldFeat.Name);
Debug.Print(" Custom weldment profile configuration name: " + swStructuralMemberFeatData.ConfigurationName);
Debug.Print(" Transfer material? " + swStructuralMemberFeatData.TransferMaterial);
Debug.Print(" Library material profile: " + swStructuralMemberFeatData.LibraryProfileMaterial);

groups = (object[])swStructuralMemberFeatData.Groups;
Debug.Print(" Group:");
for (i = 0; i < groups.Length; i++)
{
    group = (StructuralMemberGroup)groups[i];
    Debug.Print(" Segment count: " + group.GetSegmentsCount());
    Debug.Print(" Rotational angle: " + group.Angle);
    Debug.Print(" Apply corner treatment: " + group.ApplyCornerTreatment);
    Debug.Print(" Corner treatment type: " + group.CornerTreatmentType);
    Debug.Print(" Mirror profile: " + group.MirrorProfile);
    Debug.Print(" Mirror profile axis: " + group.MirrorProfileAxis);
    Debug.Print(" Gap within: " + group.GapWithinGroup);
    segs = (object[])group.Segments;
    for (j = 0; j < segs.Length; j++)
    {
        ((SketchSegment)segs[j]).Select(false);
    }
}
swStructuralMemberFeatData.ReleaseSelectionAccess();
swModel.ClearSelection2(true);
}

/// <summary>
/// The SldWorks swApp variable is pre-assigned for you.
/// </summary>
public SldWorks swApp;
}
}

```

Insert Structural Weldments Using Custom Weldment Profile Example (VBA)

This example shows how to insert a structural weldment feature using a custom weldment profile configuration and structural member groups.

```

'-----
' Preconditions:
' 1. Verify that the specified library feature exists.
' 2. Verify that a valid pathname exists in Parts in Tools > Options >
'   System Options > Default Templates.
' 3. Create C:\Test\Pipes.
' 4. Open the Immediate window.
'
' Postconditions:
' 1. Opens the specified library feature, adds the nxn configuration,
'   saves the library feature as nxn.sldlfp, and closes the new library
'   feature, which is called a custom weldment profile when used to create
'   structural weldment features.
' 2. Creates a new part document that contains a sketch of two
'   rectangles.
' 3. Creates a weldment and two structural member features using the
'   sketch and the nxn configuration of the custom weldment profile created in
'   step 1.
' 4. Rotates Pipes nxn(1).
' 5. Examines the FeatureManager design tree, graphics area, and
'   the Immediate window.
'-----

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swFeatMgr As SldWorks.FeatureManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim swSketchMgr As SldWorks.SketchManager
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swConfigMgr As SldWorks.ConfigurationManager
Dim swConfig As SldWorks.Configuration
Dim swFeature As SldWorks.Feature
Dim swStructuralMemberGroup1 As SldWorks.StructuralMemberGroup
Dim swStructuralMemberGroup2 As SldWorks.StructuralMemberGroup
Dim swWeldFeat As SldWorks.Feature
Dim swStructuralMemberFeatData As SldWorks.StructuralMemberFeatureData
Dim status As Boolean
Dim errors As Long
Dim warnings As Long
Dim libFeature As String
Dim newLibFeature As String
Dim macroFolder As String
Dim template As String
Dim sketchLines As Variant
Dim segs1(1) As Object
Dim groups1 As Variant
Dim groupArray1(0) As Object
Dim segs2(1) As Object
Dim groups2 As Variant
Dim groupArray2(0) As Object
Dim group As SldWorks.StructuralMemberGroup
Dim groups As Variant
Dim segs As Variant
Dim weldmentProfile As String
Dim weldmentConfigurationName As String

Option Explicit

Sub main()

    Set swApp = Application.SldWorks

    'Open existing library feature, add nxn configuration,
    'and save library feature as nxn.sldlfp
    libFeature = "C:\Program Files\SolidWorks Corp\SolidWorks\lang\english\weldment profiles\ansi inch\pipe\0.5 sch 40.sldlfp"
    Set swModel = swApp.OpenDoc6(libFeature, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
    Set swModelDocExt = swModel.Extension
    status = swModelDocExt.SelectByID2("0.5 sch 40.SLDPRT", "COMPONENT", 0, 0, 0, False, 0, Nothing, 0)
    swModel.ClearSelection2 True
    Set swConfigMgr = swModel.ConfigurationManager
    Set swConfig = swConfigMgr.AddConfiguration("nxn", "", "", swConfigurationOptions2_e.swConfigOption_DontActivate, "", "")
    newLibFeature = "C:\Test\Pipes\nxn.sldlfp"
    status = swModelDocExt.SaveAs(newLibFeature, swSaveAsVersion_e.swSaveAsCurrentVersion, swSaveAsOptions_e.swSaveAsOptions_Silent, Nothing, errors,
warnings)
    Set swModel = Nothing
    swApp.CloseDoc newLibFeature

    'Open new part document and
    'create weldment and structural members
    macroFolder = swApp.GetCurrentMacroPathFolder()
    swApp.SetCurrentWorkingDirectory macroFolder
    template = swApp.GetUserPreferenceStringValue(swUserPreferenceStringValue_e.swDefaultTemplatePart)
    Set swModel = swApp.NewDocument(template, 0, 0, 0)
    Set swFeatMgr = swModel.FeatureManager
    Set swSelMgr = swModel.SelectionManager

    swModel.ClearSelection2 True

    Set swSketchMgr = swModel.SketchManager
    sketchLines = swSketchMgr.CreateCornerRectangle(-0.1872393706766, 0.1133237194389, 0, -0.07003610048208, 0.009188409684237, 0)

```

```

swModel.ClearSelection2 True

sketchLines = swSketchMgr.CreateCornerRectangle(0.06513561531715, 0.03369083550887, 0, 0.1807053904567, -0.08106219210316, 0)
swSketchMgr.InsertSketch True

swModel.ViewZoomtofit2

Set swFeatureMgr = swModel.FeatureManager
Set swFeature = swFeatureMgr.InsertWeldmentFeature()

Set swStructuralMemberGroup1 = swFeatMgr.CreateStructuralMemberGroup

Set swModelDocExt = swModel.Extension
status = swModelDocExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.1495427140733, 0.1133237194389, 0, False, 0, Nothing, 0)
status = swModelDocExt.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", -0.1872393706766, 0.08238014634844, 0, True, 0, Nothing, 0)

Set segs1(0) = swSelMgr.GetSelectedObject6(1, 0)
Set segs1(1) = swSelMgr.GetSelectedObject6(2, 0)

swStructuralMemberGroup1.segments = segs1
swStructuralMemberGroup1.Angle = 0.785714285714286 'radians
swStructuralMemberGroup1.ApplyCornerTreatment = True
swStructuralMemberGroup1.CornerTreatmentType = swSolidworksWeldmentEndCondOptions_e.swEndConditionMiter
swStructuralMemberGroup1.MirrorProfile = True
swStructuralMemberGroup1.MirrorProfileAxis = swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical

Set groupArray1(0) = swStructuralMemberGroup1
groups1 = groupArray1

weldmentProfile = "C:\Test\Pipes\nxn.SLDLFP"
weldmentConfigurationName = "nxn"
Set swFeature = swFeatureMgr.InsertStructuralWeldment5(weldmentProfile, swConnectedSegmentsOption_e.swConnectedSegments_SimpleCut, False, (groups1),
weldmentConfigurationName)

swModel.ClearSelection2 True

Set swStructuralMemberGroup2 = swFeatMgr.CreateStructuralMemberGroup

status = swModelDocExt.SelectByID2("Line5@Sketch1", "EXTSKETCHSEGMENT", 0.1185825251083, 0.03369083550887, 0, False, 0, Nothing, 0)
status = swModelDocExt.SelectByID2("Line6@Sketch1", "EXTSKETCHSEGMENT", 0.06513561531715, -0.02774616865332, 0, True, 0, Nothing, 0)
Set segs2(0) = swSelMgr.GetSelectedObject6(1, 0)
Set segs2(1) = swSelMgr.GetSelectedObject6(2, 0)

swStructuralMemberGroup2.segments = segs2
swStructuralMemberGroup2.AlignAxis = swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical

Set groupArray2(0) = swStructuralMemberGroup2
groups2 = groupArray2

Set swFeature = swFeatureMgr.InsertStructuralWeldment5(weldmentProfile, swConnectedSegmentsOption_e.swConnectedSegments_SimpleCut, False, (groups2),
weldmentConfigurationName)

swModel.ClearSelection2 True

'Get feature data for each structural member group
status = swModelDocExt.SelectByID2("Pipes nxn(1)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swWeldFeat = swSelMgr.GetSelectedObject6(1, 0)
Set swStructuralMemberFeatData = swWeldFeat.GetDefinition
swStructuralMemberFeatData.AccessSelections swModel, Nothing

Debug.Print ""
Debug.Print "Groups count: " & swStructuralMemberFeatData.GetGroupsCount
Debug.Print " Feature name: " & swWeldFeat.Name
Debug.Print " Custom weldment profile configuration name: " & swStructuralMemberFeatData.ConfigurationName
Debug.Print " Transfer material? " & swStructuralMemberFeatData.TransferMaterial
Debug.Print " Library material profile: " & swStructuralMemberFeatData.LibraryProfileMaterial

groups = swStructuralMemberFeatData.groups

Dim i As Long, j As Long
Debug.Print " Group:"
For i = LBound(groups) To UBound(groups)
    Set group = groups(i)
    Debug.Print " Segment count: " & group.GetSegmentsCount
    Debug.Print " Rotational angle: " & group.Angle
    Debug.Print " Apply corner treatment: " & group.ApplyCornerTreatment
    Debug.Print " Corner treatment type: " & group.CornerTreatmentType
    Debug.Print " Mirror profile: " & group.MirrorProfile
    Debug.Print " Mirror profile axis: " & group.MirrorProfileAxis
    Debug.Print " Gap within: " & group.GapWithinGroup
    segs = group.segments
    For j = LBound(segs) To UBound(segs)
        segs(j).Select False
    Next j
Next i

swStructuralMemberFeatData.ReleaseSelectionAccess

status = swModelDocExt.SelectByID2("Pipes nxn(2)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swWeldFeat = swSelMgr.GetSelectedObject6(1, 0)
Set swStructuralMemberFeatData = swWeldFeat.GetDefinition
swStructuralMemberFeatData.AccessSelections swModel, Nothing

Debug.Print ""
Debug.Print "Groups count: " & swStructuralMemberFeatData.GetGroupsCount
Debug.Print " Feature name: " & swWeldFeat.Name
Debug.Print " Custom weldment profile configuration name: " & swStructuralMemberFeatData.ConfigurationName
Debug.Print " Transfer material? " & swStructuralMemberFeatData.TransferMaterial
Debug.Print " Library material profile: " & swStructuralMemberFeatData.LibraryProfileMaterial

groups = swStructuralMemberFeatData.groups
Debug.Print " Group:"
For i = LBound(groups) To UBound(groups)
    Set group = groups(i)
    Debug.Print " Segment count: " & group.GetSegmentsCount
    Debug.Print " Rotational angle: " & group.Angle
    Debug.Print " Apply corner treatment: " & group.ApplyCornerTreatment

```

```
    Debug.Print "        Corner treatment type: " & group.CornerTreatmentType  
    Debug.Print "        Mirror profile: " & group.MirrorProfile  
    Debug.Print "        Mirror profile axis: " & group.MirrorProfileAxis  
    Debug.Print "        Gap within: " & group.GapWithinGroup  
    segs = group.segments  
    For j = LBound(segs) To UBound(segs)  
        segs(j).Select False  
    Next j  
Next i
```

```
swStructuralMemberFeatData.ReleaseSelectionAccess  
swModel.ClearSelection2 True
```

```
End Sub
```

Insert Structural Weldments Using Custom Weldment Profile Example (VB.NET)

This example shows how to insert a structural weldment feature using a custom weldment profile configuration and structural member groups.

```

-----
Preconditions:
1. Verify that the specified library feature and part document template exist.
2. Verify that a valid pathname exists in Parts in Tools > Options > System Options > Default Templates.
3. Create C:\Test\Pipes.
4. Open the Immediate window.
Postconditions:
1. Opens the specified library feature, adds the nxn configuration, saves the library feature as nxn.sldlfp, and closes the new library feature, which is called a custom weldment profile when used to create structural weldment features.
2. Creates a new part document that contains a sketch of two rectangles.
3. Creates a weldment and two structural member features using the sketch and the nxn configuration of the custom weldment profile created in step 1.
4. Rotates Pipes nxn(1).
5. Examine the FeatureManager design tree, graphics area, and the Immediate window.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swFeatMgr As FeatureManager
        Dim swSelMgr As SelectionMgr
        Dim swSketchMgr As SketchManager
        Dim swFeatureMgr As FeatureManager
        Dim swConfigMgr As ConfigurationManager
        Dim swConfig As Configuration
        Dim swFeature As Feature
        Dim swStructuralMemberGroup1 As StructuralMemberGroup
        Dim swStructuralMemberGroup2 As StructuralMemberGroup
        Dim swWeldFeat As Feature
        Dim swStructuralMemberFeatData As StructuralMemberFeatureData
        Dim status As Boolean
        Dim errors As Integer
        Dim warnings As Integer
        Dim libFeature As String
        Dim newLibFeature As String
        Dim macroFolder As String
        Dim template As String
        Dim sketchLines As Object
        Dim segs1(1) As SketchSegment
        Dim groupArray1(0) As Object
        Dim groups1(0) As DispatchWrapper
        Dim segs2(1) As SketchSegment
        Dim groups2(0) As DispatchWrapper
        Dim groupArray2(0) As Object
        Dim group As StructuralMemberGroup
        Dim groups(1) As Object
        Dim segs(1) As Object
        Dim weldmentProfile As String
        Dim weldmentConfigurationName As String

        'Open existing library feature, add nxn configuration,
        'and save library feature as nxn.sldlfp
        libFeature = "C:\Program Files\SolidWorks Corp\SolidWorks\lang\english\weldment_profiles\ansi_inch\pipe\0.5 sch 40.sldlfp"
        swModel = swApp.OpenDoc6(libFeature, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
        swModelDocExt = swModel.Extension
        status = swModelDocExt.SelectByID2("0.5 sch 40.SLDPRT", "COMPONENT", 0, 0, 0, False, 0, Nothing, 0)
        swModel.ClearSelection2(True)
        swConfigMgr = swModel.ConfigurationManager
        swConfig = swConfigMgr.AddConfiguration("nxn", "", "", swConfigurationOptions2_e.swConfigOption_DontActivate, "", "")
        newLibFeature = "C:\Test\Pipes\nxn.sldlfp"
        status = swModelDocExt.SaveAs(newLibFeature, swSaveAsVersion_e.swSaveAsCurrentVersion, swSaveAsOptions_e.swSaveAsOptions_Silent, Nothing, errors,
warnings)
        swModel = Nothing
        swApp.CloseDoc(newLibFeature)

        'Open new part document and
        'create weldment and structural members
        macroFolder = swApp.GetCurrentMacroPathFolder()
        swApp.SetCurrentWorkingDirectory(macroFolder)
        template = swApp.GetUserPreferenceStringValue(swUserPreferenceStringValue_e.swDefaultTemplatePart)
        swModel = swApp.NewDocument(template, 0, 0, 0)
        swFeatMgr = swModel.FeatureManager
        swSelMgr = swModel.SelectionManager
        swModel.ClearSelection2(True)
        swSketchMgr = swModel.SketchManager
        sketchLines = swSketchMgr.CreateCornerRectangle(-0.1872393706766, 0.1133237194389, 0, -0.07003610048208, 0.009188409684237, 0)
        swModel.ClearSelection2(True)
        sketchLines = swSketchMgr.CreateCornerRectangle(0.06513561531715, 0.03369083550887, 0, 0.1807053904567, -0.08106219210316, 0)
        swSketchMgr.InsertSketch(True)
        swModel.ViewZoomtofit2()
        swFeatureMgr = swModel.FeatureManager
        swFeature = swFeatureMgr.InsertWeldmentFeature()
        swStructuralMemberGroup1 = swFeatMgr.CreateStructuralMemberGroup
        swModelDocExt = swModel.Extension
        status = swModelDocExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.1495427140733, 0.1133237194389, 0, False, 0, Nothing, 0)
        status = swModelDocExt.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", -0.1872393706766, 0.08238014634844, 0, True, 0, Nothing, 0)
        segs1(0) = swSelMgr.GetSelectedObject6(1, 0)
        segs1(1) = swSelMgr.GetSelectedObject6(2, 0)
        swStructuralMemberGroup1.Segments = segs1
        swStructuralMemberGroup1.Angle = 0.785714285714286 'radians
    
```

```

swStructuralMemberGroup1.ApplyCornerTreatment = True
swStructuralMemberGroup1.CornerTreatmentType = swSolidworksWeldmentEndCondOptions_e.swEndConditionMiter
swStructuralMemberGroup1.MirrorProfile = True
swStructuralMemberGroup1.MirrorProfileAxis = swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical

groupArray1(0) = swStructuralMemberGroup1
groups1(0) = New DispatchWrapper(groupArray1(0))

weldmentProfile = "C:\Test\Pipes\nxn.SLDLFP"
weldmentConfigurationName = "nxn"
swFeature = swFeatureMgr.InsertStructuralWeldment5(weldmentProfile, swConnectedSegmentsOption_e.swConnectedSegments_SimpleCut, False, (groups1), weldmentConfigurationName)
swModel.ClearSelection2(True)

swStructuralMemberGroup2 = swFeatMgr.CreateStructuralMemberGroup
status = swModelDocExt.SelectByID2("Line5@Sketch1", "EXTSKETCHSEGMENT", 0.1185825251083, 0.03369083550887, 0, False, 0, Nothing, 0)
status = swModelDocExt.SelectByID2("Line6@Sketch1", "EXTSKETCHSEGMENT", 0.06513561531715, -0.02774616865332, 0, True, 0, Nothing, 0)
segs2(0) = swSelMgr.GetSelectedObject6(1, 0)
segs2(1) = swSelMgr.GetSelectedObject6(2, 0)

swStructuralMemberGroup2.Segments = segs2
swStructuralMemberGroup2.AlignAxis = swMirrorProfileOrAlignmentAxis_e.swMirrorProfileOrAlignmentAxis_Vertical

groupArray2(0) = swStructuralMemberGroup2
groups2(0) = New DispatchWrapper(groupArray2(0))

swFeature = swFeatureMgr.InsertStructuralWeldment5(weldmentProfile, swConnectedSegmentsOption_e.swConnectedSegments_SimpleCut, False, (groups2), weldmentConfigurationName)
swModel.ClearSelection2(True)

'Get feature data for each structural member group
status = swModelDocExt.SelectByID2("Pipes nxn(1)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swWeldFeat = swSelMgr.GetSelectedObject6(1, 0)
swStructuralMemberFeatData = swWeldFeat.GetDefinition
swStructuralMemberFeatData.AccessSelections(swModel, Nothing)
Debug.Print("")
Debug.Print("Groups count: " & swStructuralMemberFeatData.GetGroupsCount)
Debug.Print("  Feature name: " & swWeldFeat.Name)
Debug.Print("  Custom weldment profile configuration name: " & swStructuralMemberFeatData.ConfigurationName)
Debug.Print("  Transfer material? " & swStructuralMemberFeatData.TransferMaterial)
Debug.Print("  Library material profile: " & swStructuralMemberFeatData.LibraryProfileMaterial)

groups = swStructuralMemberFeatData.groups
Dim i As Long, j As Long
Debug.Print("  Group:")
For i = LBound(groups) To UBound(groups)
    group = groups(i)
    Debug.Print("    Segment count: " & group.GetSegmentsCount)
    Debug.Print("    Rotational angle: " & group.Angle)
    Debug.Print("    Apply corner treatment: " & group.ApplyCornerTreatment)
    Debug.Print("    Corner treatment type: " & group.CornerTreatmentType)
    Debug.Print("    Mirror profile: " & group.MirrorProfile)
    Debug.Print("    Mirror profile axis: " & group.MirrorProfileAxis)
    Debug.Print("    Gap within: " & group.GapWithinGroup)
    segs = group.segments
    For j = LBound(segs) To UBound(segs)
        segs(j).Select(False)
    Next j
Next i
swStructuralMemberFeatData.ReleaseSelectionAccess()

status = swModelDocExt.SelectByID2("Pipes nxn(2)", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
swWeldFeat = swSelMgr.GetSelectedObject6(1, 0)
swStructuralMemberFeatData = swWeldFeat.GetDefinition
swStructuralMemberFeatData.AccessSelections(swModel, Nothing)
Debug.Print("")
Debug.Print("Groups count: " & swStructuralMemberFeatData.GetGroupsCount)
Debug.Print("  Feature name: " & swWeldFeat.Name)
Debug.Print("  Custom weldment profile configuration name: " & swStructuralMemberFeatData.ConfigurationName)
Debug.Print("  Transfer material? " & swStructuralMemberFeatData.TransferMaterial)
Debug.Print("  Library material profile: " & swStructuralMemberFeatData.LibraryProfileMaterial)

groups = swStructuralMemberFeatData.groups
Debug.Print("  Group:")
For i = LBound(groups) To UBound(groups)
    group = groups(i)
    Debug.Print("    Segment count: " & group.GetSegmentsCount)
    Debug.Print("    Rotational angle: " & group.Angle)
    Debug.Print("    Apply corner treatment: " & group.ApplyCornerTreatment)
    Debug.Print("    Corner treatment type: " & group.CornerTreatmentType)
    Debug.Print("    Mirror profile: " & group.MirrorProfile)
    Debug.Print("    Mirror profile axis: " & group.MirrorProfileAxis)
    Debug.Print("    Gap within: " & group.GapWithinGroup)
    segs = group.segments
    For j = LBound(segs) To UBound(segs)
        segs(j).Select(False)
    Next j
Next i
swStructuralMemberFeatData.ReleaseSelectionAccess()
swModel.ClearSelection2(True)

End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class

```

Insert Surface-cut Feature Example (C#)

This example shows how to insert a surface-cut feature.

```
//-----
// Preconditions:
// 1. Verify that the specified file to open exists.
// 2. Open the Immediate window.
//
// Postconditions:
// 1. Opens the part whose intersecting solid bodies to cut with a plane.
// 2. Creates a plane named Plane1.
// 3. Selects Plane1 to cut all intersecting solid bodies.
// 4. Inserts the surface-cut feature, which cuts all intersecting
//    solid bodies by the plane.
// 5. Examine the graphics area and Immediate window to verify.
//
// NOTE: Because this part document is used elsewhere, do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;
using System.Diagnostics;

namespace InsertCutSurface2CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            PartDoc swPart = default(PartDoc);
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            Feature swFeature = default(Feature);
            FeatureManager swFeatureManager = default(FeatureManager);
            RefPlane swRefPlane = default(RefPlane);
            SurfCutFeatureData swSurfaceCutFeature = default(SurfCutFeatureData);
            bool status = false;
            string fileName = null;
            int errors = 0;
            int warnings = 0;

            // Open part to cut with a plane
            fileName = "C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS 2018\\samples\\tutorial\\multibody\\multi_inter.sldprt";
            swPart = (PartDoc)swApp.OpenDoc6(fileName, (int)swDocumentTypes_e.swDocPART, (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref errors,
            ref warnings);
            swModel = (ModelDoc2)swPart;
            swModelDocExt = (ModelDocExtension)swModel.Extension;

            // Create and select the plane to cut the
            // intersecting solid bodies in the part
            status = swModelDocExt.SelectByID2("Front", "PLANE", 0, 0, 0, true, 0, null, 0);
            swFeatureManager = (FeatureManager)swModel.FeatureManager;
            swRefPlane = (RefPlane)swFeatureManager.InsertRefPlane((int)swRefPlaneReferenceConstraints_e.swRefPlaneReferenceConstraint_Distance, 0.045, 0,
            0, 0, 0);
            status = swModelDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, true, 0, null, 0);

            // Insert surface-cut feature to cut
            // all intersecting solid bodies
            swFeature = (Feature)swFeatureManager.InsertCutSurface(false, 0, false, true, null, out errors);
            Debug.Print("Were any errors generated by the surface cut (0 = no errors)? " + errors);

            // Get surface-cut feature and some properties
            swSurfaceCutFeature = (SurfCutFeatureData)swFeature.GetDefinition();
            Debug.Print("Name of surface-cut feature: " + swFeature.Name);
            Debug.Print(" Is feature scope on? " + swSurfaceCutFeature.FeatureScope);
            Debug.Print(" Number of bodies cut by the plane: " + swSurfaceCutFeature.GetFeatureScopeBodiesCount());
            Debug.Print(" Is auto-select on? " + swSurfaceCutFeature.AutoSelect);
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Insert Surface-cut Feature Example (VBA)

This example shows how to insert a surface-cut feature.

```
'-----
' Preconditions:
' 1. Verify that specified part to open exists.
' 2. Open the Immediate window.
'
' Postconditions:
' 1. Opens the part whose intersecting solid bodies to cut with a plane.
' 2. Creates a plane named Plane1.
' 3. Selects Plane1 to cut all intersecting solid bodies.
' 4. Inserts the surface-cut feature, which cuts all intersecting
'    solid bodies by the plane.
' 5. Examine the graphics area and Immediate window.
'
' NOTE: Because this part document is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swFeature As SldWorks.Feature
Dim swFeatureManager As SldWorks.FeatureManager
Dim swRefPlane As SldWorks.RefPlane
Dim swSurfaceCutFeature As SldWorks.SurfCutFeatureData
Dim fileName As String
Dim status As Boolean
Dim errors As Long, warnings As Long

Sub main()

    Set swApp = Application.SldWorks

    ' Open part to cut with a plane
    fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\multibody\multi_inter.sldprt"
    Set swModel = swApp.OpenDoc6(fileName, swDocPart, swOpenDocOptions_Silent, "", errors, warnings)
    Set swModelDocExt = swModel.Extension

    ' Create and select the plane to cut
    ' all intersecting solid bodies in the part
    status = swModelDocExt.SelectByID2("Front", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Set swFeatureManager = swModel.FeatureManager
    Set swRefPlane = swFeatureManager.InsertRefPlane(swRefPlaneReferenceConstraint_Distance, 0.045, 0, 0, 0, 0)
    status = swModelDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, True, 0, Nothing, 0)

    ' Insert a surface-cut feature that cuts
    ' all intersecting solid bodies
    Set swFeature = swFeatureManager.InsertCutSurface(False, 0, False, True, Nothing, errors)
    Debug.Print ("Were any errors generated by the surface cut (0 = no errors)? " & errors)
    Set swSurfaceCutFeature = swFeature.GetDefinition

    ' Get surface-cut feature and some properties
    Debug.Print ("Name of surface-cut feature: " & swFeature.Name)
    Debug.Print (" Is feature scope on? " & swSurfaceCutFeature.FeatureScope)
    Debug.Print (" Number of bodies cut by the plane: " & swSurfaceCutFeature.GetFeatureScopeBodiesCount)
    Debug.Print (" Is auto-select on? " & swSurfaceCutFeature.AutoSelect)

End Sub
```

Insert Surface-cut Feature Example (VB.NET)

This example shows how to insert a surface-cut feature.

```

-----
Preconditions:
1. Verify that the specified part to open exists.
2. Open the Immediate window.

Postconditions:
1. Opens the part whose intersecting solid bodies
   to cut with a plane.
2. Creates a plane named Plane1.
3. Selects Plane1 to cut all intersecting solid bodies.
4. Inserts the surface-cut feature, which cuts all intersecting
   solid bodies by the plane.
5. Examine the Immediate window and graphics area to verify.

NOTE: Because this part document is used elsewhere, do not save changes.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Public Sub Main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swFeature As Feature
        Dim swFeatureManager As FeatureManager
        Dim swRefPlane As RefPlane
        Dim swSurfaceCutFeature As SurfCutFeatureData
        Dim status As Boolean
        Dim fileName As String
        Dim errors As Integer, warnings As Integer

        ' Open part to cut with a plane
        fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS_2018\samples\tutorial\multibody\multi_inter.sldprt"
        swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)
        swModelDocExt = swModel.Extension

        ' Create and select the plane to cut
        ' all intersecting solid bodies in the part
        status = swModelDocExt.SelectByID2("Front", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
        swFeatureManager = swModel.FeatureManager
        swRefPlane = swFeatureManager.InsertRefPlane(swRefPlaneReferenceConstraints_e.swRefPlaneReferenceConstraint_Distance, 0.045, 0, 0, 0, 0)
        status = swModelDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, True, 0, Nothing, 0)

        ' Insert a surface-cut feature that cuts all
        ' intersecting solid bodies
        swFeature = swFeatureManager.InsertCutSurface(False, 0, False, True, Nothing, errors)
        Debug.Print("Were any errors generated by the surface cut (0 = no errors)? " & errors)

        ' Get surface-cut feature and some properties
        swSurfaceCutFeature = swFeature.GetDefinition
        Debug.Print("Name of surface-cut feature: " & swFeature.Name)
        Debug.Print(" Is feature scope on? " & swSurfaceCutFeature.FeatureScope)
        Debug.Print(" Number of bodies cut by the plane: " & swSurfaceCutFeature.GetFeatureScopeBodiesCount)
        Debug.Print(" Is auto-select on? " & swSurfaceCutFeature.AutoSelect)
    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

```

End Class

Insert Sweep Cut Feature Example (C#)

This example shows how to create a swept-cut feature and get its properties.

```
//-----
// Preconditions:
// 1. Verify that the part to open exists.
// 2. Open an Immediate window.
//
// Postconditions:
// 1. Creates Cut-Sweep1.
// 2. Inspect the FeatureManager design tree, graphics area,
//    and Immediate window.
//
// NOTE: Because this part document is used elsewhere,
// do not save changes.
//-----

using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;
using System.Diagnostics;
namespace InsertSweepCut_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        ModelDoc2 Part;
        bool boolstatus;
        int longstatus;
        int longwarnings;
        SweepFeatureData swSweep;
        Feature swProfFeat;
        Sketch swProfSketch;
        Feature swPathFeat;
        Sketch swPathSketch;
        bool bRet;

        public void Main()
        {
            Part = swApp.OpenDoc6("C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS
2018\\samples\\tutorial\\api\\sweepcutextrude.SLDPRT",
1, 0, "", ref longstatus, ref longwarnings);

            swApp.ActivateDoc2("sweepcutextrude.SLDPRT", false, ref longstatus);

            Part = (ModelDoc2)swApp.ActiveDoc;
            ModelView myModelView = null;
            myModelView = (ModelView)Part.ActiveView;
            myModelView.FrameLeft = 0;
            myModelView.FrameTop = 0;

            myModelView.FrameState = (int)swWindowState_e.swWindowMaximized;

            Part.ShowNamedView2("Isometric",
7);

            boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH",
0.01948983274156, -0.02564816935317, 0, false, 1, null, 0);

            // profile has Mark = 1

            boolstatus =
Part.Extension.SelectByID2("Sketch3", "SKETCH",
-0.03797488317814, -0.02133214444164, 0, true, 4, null, 0);

            // path sweep has Mark = 4
```

```

        Feature myFeature = default(Feature);

myFeature = Part.FeatureManager.InsertCutSwept4(false, true, 0, false, false, 0,
0, false,
0, 0,

0, 0, true, true, 0, true, true, true, false);

swSweep = (SweepFeatureData)myFeature.GetDefinition();
swProfFeat = (Feature)swSweep.Profile;
Debug.Assert((null != swProfFeat));
swProfSketch = (Sketch)swProfFeat.GetSpecificFeature();
Debug.Assert((null != swProfSketch));

bRet = swSweep.AccessSelections(Part, null);
Debug.Assert(bRet);

swPathFeat = (Feature)swSweep.Path;
Debug.Assert((null != swPathFeat));
swPathSketch = (Sketch)swPathFeat.GetSpecificFeature();
Debug.Assert((null != swPathSketch));

Debug.Print("File = " + Part.GetPathName());
Debug.Print("
" + myFeature.Name);
Debug.Print("
Path = " + swPathFeat.Name);

//swTangencyType_e
Debug.Print("
Path alignment type = " + swSweep.PathAlignmentType);

Profile = "
Debug.Print("
" + swProfFeat.Name);
Debug.Print(" AdvancedSmoothing = " + swSweep.AdvancedSmoothing);
Debug.Print(" AlignWithEndFaces = " + swSweep.AlignWithEndFaces);
Debug.Print(" AutoSelect = " + swSweep.AutoSelect);
Debug.Print(" AutoSelectComponents = " + swSweep.AutoSelectComponents);

Debug.Print(" EndTangencyType = " + swSweep.EndTangencyType);

Debug.Print(" AssemblyFeatureScope = " + swSweep.AssemblyFeatureScope);
Debug.Print(" FeatureScope = " + swSweep.FeatureScope);
Debug.Print(" FeatureScopeBodiesCnt = " + swSweep.GetFeatureScopeBodiesCount());

//swSelectType_e
Debug.Print(" GetPathType = " + swSweep.GetPathType());

Debug.Print("
Wall thickness foward = " + swSweep.GetWallThickness(true)
* 1000.0 + " mm");
Debug.Print("
Wall thickness reverse = " + swSweep.GetWallThickness(false)
* 1000.0 + " mm");

Debug.Print(" IsBossFeature = " + swSweep.IsBossFeature());
Debug.Print(" IsThinFeature = " + swSweep.IsThinFeature());
Debug.Print(" MaintainTangency = " + swSweep.MaintainTangency);

Merge = "
Debug.Print("
" + swSweep.Merge);
Debug.Print(" MergeSmoothFaces = " + swSweep.MergeSmoothFaces);
Debug.Print(" PropagateFeatureToParts = " + swSweep.PropagateFeatureToParts);

Debug.Print(" StartTangencyType = " + swSweep.StartTangencyType);

```

```
Debug.Print(" TangentPropagation = " + swSweep.TangentPropagation);  
Debug.Print(" ThinWallType = " + swSweep.ThinWallType);  
  
//swTwistControlType_e  
Debug.Print(" TwistControlType = " + swSweep.TwistControlType);  
  
//swCutSweepOption_e  
Debug.Print(" CutSweepOption = " + swSweep.GetCutSweepOption());  
  
swSweep.ReleaseSelectionAccess();  
}  
  
public SldWorks swApp;  
  
}  
}
```

Insert Sweep Cut Feature Example (VBA)

This example shows how to create a swept-cut feature and get its properties.

```

'-----
' Preconditions:
' 1. Verify that the part to open exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates Cut-Sweep1.
' 2. Inspect the FeatureManager design tree, graphics area,
'    and Immediate window.
'
' NOTE: Because this part document is used elsewhere,
' do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Dim swSweep As SldWorks.SweepFeatureData
Dim swProfFeat As SldWorks.Feature
Dim swProfSketch As SldWorks.Sketch
Dim swPathFeat As SldWorks.Feature
Dim swPathSketch As SldWorks.Sketch
Dim bRet As Boolean
Dim myModelView As Object
Dim myFeature As SldWorks.Feature

Sub main()
    Set swApp = Application.SldWorks

    Set Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS
2018\samples\tutorial\api\sweepcutextrude.SLDPRT", 1, 0, "", longstatus, longwarnings)
    swApp.ActivateDoc2 "sweepcutextrude.SLDPRT", False, longstatus
    Set Part = swApp.ActiveDoc

    Set myModelView = Part.ActiveView
    myModelView.FrameLeft = 0
    myModelView.FrameTop = 0

    myModelView.FrameState = swWindowState_e.swWindowMaximized
    Part.ShowNamedView2 "Isometric", 7

    boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 0.01948983274156, -0.02564816935317, 0, False, 1, Nothing,
0) ' profile has Mark = 1
    boolstatus = Part.Extension.SelectByID2("Sketch3", "SKETCH", -0.03797488317814, -0.02133214444164, 0, True, 4, Nothing,
0) ' path sweep has Mark = 4

    Set myFeature = Part.FeatureManager.InsertCutSwept4(False, True, 0, False, False, 0, 0, False, 0, 0, 0, 0, True, True, 0,
True, True, True, False)

    Set swSweep = myFeature.GetDefinition
    Set swProfFeat = swSweep.Profile
    Set swProfSketch = swProfFeat.GetSpecificFeature

    bRet = swSweep.AccessSelections(Part, Nothing)

    Set swPathFeat = swSweep.Path
    Set swPathSketch = swPathFeat.GetSpecificFeature

    Debug.Print "File = " & Part.GetPathName
    Debug.Print " " & myFeature.Name
    Debug.Print " Sweep path = " & swPathFeat.Name
    Debug.Print " Path type as defined in swSelectType_e = " & swSweep.GetPathType
    Debug.Print " Path alignment type as defined in swTangencyType_e = " & swSweep.PathAlignmentType
    Debug.Print " Sweep Profile = " & swProfFeat.Name
    Debug.Print " Profile type as defined in swSelectType_e = " & swSweep.GetProfileType
    Debug.Print " Profile orientation/twist type as defined in swTwistControlType_e = " & swSweep.TwistControlType
    Debug.Print " Cut sweep type as defined in swCutSweepOption_e = " & swSweep.GetCutSweepOption
    Debug.Print " Align sweep with end faces? " & swSweep.AlignWithEndFaces
    Debug.Print " Automatically select all bodies in the multibody part to be affected? " & swSweep.AutoSelect
    Debug.Print " Start of sweep tangency type as defined in swTangencyType_e = " & swSweep.StartTangencyType
    Debug.Print " End of sweep tangency type as defined in swTangencyType_e = " & swSweep.EndTangencyType
    Debug.Print " Only specific bodies affected by this sweep feature? " & swSweep.FeatureScope
    Debug.Print " Feature scope bodies count = " & swSweep.GetFeatureScopeBodiesCount
    Debug.Print " Is a boss feature? " & swSweep.IsBossFeature
    Debug.Print " Is a thin-walled feature? " & swSweep.IsThinFeature
    Debug.Print " Thin-walled type (0=1D, 1=1DReverse, 2=MidPlane, 3=2D) = " & swSweep.ThinWallType
    Debug.Print " Wall thickness Direction 1 = " & swSweep.GetWallThickness(True) * 1000# & " mm"
    Debug.Print " Wall thickness Direction 2 = " & swSweep.GetWallThickness(False) * 1000# & " mm"
    Debug.Print " Merge tangent faces? " & swSweep.MaintainTangency
    Debug.Print " Merge results for a multibody part? " & swSweep.Merge
    Debug.Print " Merge smooth faces if using guide curves? " & swSweep.MergeSmoothFaces

    swSweep.ReleaseSelectionAccess
End Sub

```

Insert Sweep Cut Feature Example (VB.NET)

This example shows how to create a swept-cut feature and get its properties.

```

'-----
' Preconditions:
' 1. Verify that the part to open exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates Cut-Sweep1.
' 2. Inspect the FeatureManager design tree, graphics area,
'    and Immediate window.
'
' NOTE: Because this part document is used elsewhere,
' do not save changes.
'-----

Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim Part As ModelDoc2

    Dim boolstatus As Boolean

    Dim longstatus As Long, longwarnings As Long

    Dim swSweep As SweepFeatureData

    Dim swProfFeat As Feature

    Dim swProfSketch As Sketch

    Dim swPathFeat As Feature

    Dim swPathSketch As Sketch

    Dim bRet As Boolean

    Sub main()

        Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS
2018\samples\tutorial\api\sweepcutextrude.SLDPRT",
1, 0, "", longstatus, longwarnings)

        swApp.ActivateDoc2("sweepcutextrude.SLDPRT", False, longstatus)

        Part = swApp.ActiveDoc

        Dim myModelView As Object

        myModelView = Part.ActiveView

        myModelView.FrameLeft = 0

        myModelView.FrameTop = 0

        myModelView.FrameState = swWindowState_e.swWindowMaximized

        Part.ShowNamedView2("Isometric",
7)

        boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH",
0.01948983274156, -0.02564816935317, 0, False, 1, Nothing, 0) ' profile has Mark = 1

        boolstatus =
Part.Extension.SelectByID2("Sketch3", "SKETCH",
-0.03797488317814, -0.02133214444164, 0, True, 4, Nothing, 0) ' path sweep has Mark = 4

        Dim myFeature As Feature

        myFeature = Part.FeatureManager.InsertCutSwept4(False, True, 0, False, False, 0,
0, False,
0, 0, 0, 0, True, True, 0, True, True, True, False)

        swSweep = myFeature.GetDefinition

        swProfFeat = swSweep.Profile : Debug.Assert(Not Nothing Is swProfFeat)

        swProfSketch = swProfFeat.GetSpecificFeature : Debug.Assert(Not Nothing Is swProfSketch)

```

```

bRet = swSweep.AccessSelections(Part, Nothing) : Debug.Assert(bRet)

swPathFeat = swSweep.Path : Debug.Assert(Not Nothing Is swPathFeat)

swPathSketch = swPathFeat.GetSpecificFeature : Debug.Assert(Not Nothing Is swPathSketch)

Debug.Print("File = " & Part.GetPathName)

Debug.Print(" " & myFeature.Name)

Debug.Print("      Path                      =
" & swPathFeat.Name)

Debug.Print("      Path alignment
type      = " & swSweep.PathAlignmentType) 'swTangencyType_e

Debug.Print("      Profile
= " & swProfFeat.Name)

Debug.Print("      AdvancedSmoothing
= " & swSweep.AdvancedSmoothing)

Debug.Print("      AlignWithEndFaces
= " & swSweep.AlignWithEndFaces)

Debug.Print("      AutoSelect                      =
" & swSweep.AutoSelect)

Debug.Print("      AutoSelectComponents          =
" & swSweep.AutoSelectComponents)

Debug.Print("      EndTangencyType
= " & swSweep.EndTangencyType)

Debug.Print("      AssemblyFeatureScope          =
" & swSweep.AssemblyFeatureScope)

Debug.Print("      FeatureScope                      =
" & swSweep.FeatureScope)

Debug.Print("      FeatureScopeBodiesCnt
= " & swSweep.GetFeatureScopeBodiesCount)

Debug.Print("      GetPathType
= " & swSweep.GetPathType) 'swSelectType_e

Debug.Print("      Wall thickness foward      = " & swSweep.GetWallThickness(True)
* 1000.0# & " mm")

Debug.Print("      Wall thickness
reverse   = " & swSweep.GetWallThickness(False)
* 1000.0# & " mm")

Debug.Print("      IsBossFeature
= " & swSweep.IsBossFeature)

Debug.Print("      IsThinFeature
= " & swSweep.IsThinFeature)

Debug.Print("      MaintainTangency                  =
" & swSweep.MaintainTangency)

Debug.Print("      Merge
= " & swSweep.Merge)

Debug.Print("      MergeSmoothFaces                  =
" & swSweep.MergeSmoothFaces)

Debug.Print("      PropagateFeatureToParts
= " & swSweep.PropagateFeatureToParts)

Debug.Print("      StartTangencyType
= " & swSweep.StartTangencyType)

Debug.Print("      TangentPropagation                  =
" & swSweep.TangentPropagation)

Debug.Print("      ThinWallType                      =
" & swSweep.ThinWallType)

Debug.Print("      TwistControlType                  =
" & swSweep.TwistControlType) 'swTwistControlType_e

Debug.Print("      CutSweepOption
= " & swSweep.GetCutSweepOption) 'swCutSweepOption_e

swSweep.ReleaseSelectionAccess()

End Sub

Public swApp As SldWorks

```

End Class

Insert Swept-boss Feature Example (VBA)

This example shows how to create a swept-boss feature and get its guide curves.

```
'-----
' Preconditions:
' 1. Verify that the part template exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates a new part.
' 2. Creates a swept-boss feature.
' 3. Gets the number of guide curves in the feature.
' 4. Accesses the guide curves in the feature.
' 5. Gets the feature types of the guide curves.
' 6. Releases access to the sweep feature.
' 7. Examine the Immediate window.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchMgr As SldWorks.SketchManager
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim swPathFeat As SldWorks.Feature
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swSweepFeatureData As SldWorks.SweepFeatureData
Dim swSweep As SldWorks.SweepFeatureData
Dim pointArray As Variant
Dim points() As Double
Dim guideCurves As Variant
Dim guideCurve As Object
Dim nbrGuideCurves As Long
Dim status As Boolean
Dim i As Long

Sub main()

    Set swApp = Application.SldWorks

    'Create new model document
    Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SolidWorks 2018\templates\Part.prtdot", 0, 0, 0)
    Set swModelDocExt = swModel.Extension

    'Sketch an ellipse for sweep profile
    swModel.ClearSelection2 True
    Set swSketchMgr = swModel.SketchManager
    status = swModelDocExt.SelectByID("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    swSketchMgr.InsertSketch True
    Set swSketchSegment = swSketchMgr.CreateEllipse(0, 0, 0, -0.064925207354862, 0, 0, 0, -3.60377802938881E-02, 0)
    swModel.ClearSelection2 True
    swSketchMgr.InsertSketch True

    'Sketch a line for sweep path
    status = swModelDocExt.SelectByID("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    swSketchMgr.InsertSketch True
    Set swSketchSegment = swSketchMgr.CreateLine(0#, 0#, 0#, 0#, 0.059816, 0#)
    swModel.ClearSelection2 True
    swSketchMgr.InsertSketch True

    'Sketch a spline for sweep guide curve
    status = swModelDocExt.SelectByID("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    swSketchMgr.InsertSketch True

    ReDim points(0 To 5) As Double
    points(0) = -0.064925207354862
    points(1) = 0
    points(2) = 0
    points(3) = -5.76005360247873E-03
    points(4) = 5.95205538922803E-02
    points(5) = 0
    pointArray = points
    Set swSketchSegment = swSketchMgr.CreateSpline((pointArray))
    swModel.ClearSelection2 True

    status = swModelDocExt.SelectByID("Unknown", "MANIPULATOR", -4.81685228359519E-02, 1.68573405240843E-02, 0, False, 0, Nothing, 0)
    swModel.ClearSelection2 True
    swSketchMgr.InsertSketch True
    swModel.ViewZoomtofit2

    'Select the profile, path, and guide curve
    status = swModelDocExt.SelectByID("Sketch1", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID("Sketch2", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID("Sketch3", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
    swModel.ClearSelection2 True
    status = swModelDocExt.SelectByID("Sketch1", "SKETCH", 0, 0, 0, False, 1, Nothing, 0)
    status = swModelDocExt.SelectByID("Sketch2", "SKETCH", 0, 0, 0, True, 4, Nothing, 0)
    status = swModelDocExt.SelectByID("Sketch3", "SKETCH", 0, 0, 0, True, 2, Nothing, 0)

    'Create the sweep feature
    Set swFeatureMgr = swModel.FeatureManager

    Set swSweep = swFeatureMgr.CreateDefinition(swFmSweep)

    swSweep.TangentPropagation = False
```

```

swSweep.AlignWithEndFaces = False
swSweep.TwistControlType = 0
swSweep.MaintainTangency = False
swSweep.AdvancedSmoothing = False
swSweep.StartTangencyType = 0
swSweep.EndTangencyType = 0
swSweep.ThinFeature = False
swSweep.SetWallThickness True, 0#
swSweep.SetWallThickness False, 0#
swSweep.ThinWallType = 0
swSweep.PathAlignmentType = 0
swSweep.Merge = True
swSweep.FeatureScope = True
swSweep.AutoSelect = True
swSweep.SetTwistAngle 0#
swSweep.MergeSmoothFaces = True
swSweep.CircularProfile = False
swSweep.CircularProfileDiameter = 0#
swSweep.Direction = 0

Set swPathFeat = swFeatureMgr.CreateFeature(swSweep)

Debug.Print "Feature type: " & swPathFeat.GetTypeName2

'Change the orientation of the view
swModel.ShowNamedView2 "Isometric", 7

'Access sweep feature data, get guide curves,
'get feature types of guide curves, and release
'access to sweep feature
Set swSweepFeatureData = swPathFeat.GetDefinition
nbrGuideCurves = swSweepFeatureData.GetGuideCurvesCount
Debug.Print (" Number of guide curves: " & nbrGuideCurves)
status = swSweepFeatureData.AccessSelections(swModel, Nothing)
Debug.Print (" Guide curve: ")
guideCurves = swSweepFeatureData.guideCurves
For i = 0 To (nbrGuideCurves - 1)
    Set guideCurve = guideCurves(i)
    Debug.Print (" Type of feature as defined in swSelectType_e: " & swSweepFeatureData.GetGuideCurvesType(i))
Next i
swSweepFeatureData.ReleaseSelectionAccess

End Sub

```

Insert Swept-cut Feature Using Circular Profile Example (VBA)

This example shows how to create a swept-cut feature using a circular profile and get its properties.

```
'-----
' Preconditions:
' 1. Verify that the part exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Opens the part.
' 2. Selects an edge on the part.
' 3. Creates a cut-sweep feature using a circular profile.
' 4. Accesses the cut-sweep feature.
' 5. Changes the diameter of the circular profile.
' 6. Examine the Immediate window, FeatureManager
'    design tree, and graphics area.
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swPathFeat As SldWorks.Feature
Dim swFeatureManager As SldWorks.FeatureManager
Dim swSweep As SldWorks.SweepFeatureData
Dim status As Boolean
Dim errors As Long
Dim warnings As Long

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block20.sldprt", 1, 0, "", errors, warnings)
    Set swModelDocExt = swModel.Extension

    'Select an edge for the circular profile
    status = swModelDocExt.SelectByID("", "EDGE", -3.72983839416747E-02, 3.93904284381961E-02, 4.95042874504179E-02, True, 4, Nothing, 0)
    Set swFeatureManager = swModel.FeatureManager

    Set swSweep = swModel.FeatureManager.CreateDefinition(swFmSweepCut)

    swSweep.TangentPropagation = False
    swSweep.AlignWithEndFaces = True
    swSweep.TwistControlType = 0
    swSweep.MaintainTangency = False
    swSweep.AdvancedSmoothing = False
    swSweep.StartTangencyType = 0
    swSweep.EndTangencyType = 0
    swSweep.PathAlignmentType = 0
    swSweep.FeatureScope = True
    swSweep.AutoSelect = True
    swSweep.MergeSmoothFaces = True
    swSweep.CircularProfile = True
    swSweep.CircularProfileDiameter = 0.008

    Set swPathFeat = swModel.FeatureManager.CreateFeature(swSweep)

    Set swSweep = swPathFeat.GetDefinition
    status = swSweep.AccessSelections(swModel, Nothing)
    Debug.Print "Using a circular profile? " & swSweep.CircularProfile
    Debug.Print "Original size of circular profile = " & swSweep.CircularProfileDiameter
    swSweep.CircularProfileDiameter = 0.003
    Debug.Print "Modified size of circular profile = " & swSweep.CircularProfileDiameter
    swPathFeat.ModifyDefinition swSweep, swModel, Nothing

End Sub
```

Insert Swept-cut Feature Using Tool Body Example (VBA)

This example shows how to create a swept-cut feature using a tool body.

```

'-----
' Preconditions:
' 1. Verify that the specified part template exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates a boss-extrude feature.
' 2. Creates a sketch.
' 3. Creates a revolve feature.
' 4. Selects the revolve feature, sketch, and extrude feature and
'    creates a swept-cut feature.
' 5. Accesses the swept-cut feature.
' 6. Gets the names of the swept-cut feature's tool body and path.
' 7. Releases access to the swept-cut feature.
' 8. Examine the Immediate window, FeatureManager design tree,
'    and graphics area.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchMgr As SldWorks.SketchManager
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim swPathFeat As SldWorks.Feature
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swSelectionMgr As SldWorks.SelectionMgr
Dim swSweepFeatureData As SldWorks.SweepFeatureData
Dim swSweep As SldWorks.SweepFeatureData
Dim swProfileObj As Object
Dim swProfileBody As SldWorks.Body2
Dim swPathFeature As SldWorks.Feature
Dim sketchLines As Variant
Dim status As Boolean

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2018\templates\Part.prtdot", 0, 0, 0)
    Set swModelDocExt = swModel.Extension

    'Create extrude feature
    status = swModelDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    Set swSketchMgr = swModel.SketchManager
    Set swSketchSegment = swSketchMgr.CreateCircle(-0.000361, 0.001416, 0#, 0.024462, -0.045092, 0#)
    Set swFeatureMgr = swModel.FeatureManager
    Set swFeature = swFeatureMgr.FeatureExtrusion3(True, False, True, 0, 0, 0.09, 0.01, False, False, False, False, 1.74532925199433E-02,
    1.74532925199433E-02, False, False, False, False, True, True, True, 0, 0, False)

    'Create sketch
    status = swModelDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    Set swSelectionMgr = swModel.SelectionManager
    swSelectionMgr.EnableContourSelection = False
    Set swSketchSegment = swSketchMgr.CreateCircle(-0.000019, 0.00051, 0#, 0.026716, -0.0401, 0#)
    swSketchMgr.InsertSketch True

    swModel.ClearSelection2 True

    'Create revolve feature
    status = swModelDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity,
    swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
    status = swModelDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType,
    swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
    sketchLines = swSketchMgr.CreateCornerRectangle(-2.66210577384013E-02, -2.48555003438298E-02, 0, -3.78465609175683E-02, -4.75106067599669E-02, 0)
    swModel.ClearSelection2 True
    status = swModelDocExt.SelectByID2("Line2", "SKETCHSEGMENT", -2.64169576805983E-02, -4.49999999999998E-02, 2.93457016154969E-02, False, 16, Nothing,
    0)
    Set swFeature = swFeatureMgr.FeatureRevolve2(True, True, False, False, False, False, 0, 0, 6.2831853071796, 0, False, False, 0.01, 0.01, 0, 0, 0,
    False, True, True)
    swSelectionMgr.EnableContourSelection = False

    swModel.ClearSelection2 True

    'Create swept-cut feature
    status = swModelDocExt.SelectByID2("Revolve1", "SOLIDBODY", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, True, 0, Nothing, 0)
    status = swModelDocExt.SelectByID2("Boss-Extrude1", "SOLIDBODY", 0, 0, 0, True, 0, Nothing, 0)
    swModel.ClearSelection2 True
    status = swModelDocExt.SelectByID2("Revolve1", "SOLIDBODY", 0, 0, 0, False, 1, Nothing, 0) ' Tool body used to make the cut
    status = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, True, 4, Nothing, 0) ' Sweep path
    status = swModelDocExt.SelectByID2("Boss-Extrude1", "SOLIDBODY", 0, 0, 0, True, 2048, Nothing, 0) ' Solid body to cut

    Set swSweep = swFeatureMgr.CreateDefinition(swFmSweepCut)

    swSweep.TangentPropagation = False
    swSweep.AlignWithEndFaces = True
    swSweep.TwistControlType = 0
    swSweep.MaintainTangency = False
    swSweep.AdvancedSmoothing = False
    swSweep.StartTangencyType = 0
    swSweep.EndTangencyType = 0

```

```
swSweep.ThinFeature = False
swSweep.SetWallThickness True, 0#
swSweep.SetWallThickness False, 0#
swSweep.ThinWallType = 0
swSweep.PathAlignmentType = 0
swSweep.FeatureScope = True
swSweep.AutoSelect = False
swSweep.SetTwistAngle 0#
swSweep.MergeSmoothFaces = True
swSweep.AssemblyFeatureScope = True
swSweep.AutoSelectComponents = True
swSweep.PropagateFeatureToParts = False
swSweep.CircularProfile = False
swSweep.CircularProfileDiameter = 0#
swSweep.Direction = 0
```

```
Set swPathFeat = swFeatureMgr.CreateFeature(swSweep)
```

```
Debug.Print "Feature name = " & swPathFeat.Name
Set swSweepFeatureData = swPathFeat.GetDefinition
```

```
status = swSweepFeatureData.AccessSelections(swModel, Nothing)
```

```
Set swProfileObj = swSweepFeatureData.Profile
Set swProfileBody = swProfileObj
```

```
Debug.Print " Cut sweep option as defined in swCutSweepOption_e: " & swSweepFeatureData.GetCutSweepOption
Debug.Print " Tool body = " & swProfileBody.Name
```

```
Set swPathFeature = swSweepFeatureData.Path
Debug.Print " Path = " & swPathFeature.Name
swSweepFeatureData.ReleaseSelectionAccess
```

```
End Sub
```

Insert Swept-cut Thin-walled Feature Example (VBA)

This example shows how to create a swept-cut thin-walled feature.

```

'-----
' Preconditions: Verify that the specified part exists.
'
' Postconditions:
' 1. Creates a thin-walled swept-cut feature.
' 2. Examine the FeatureManager design tree and graphics area.
'-----
Option Explicit
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchMgr As SldWorks.SketchManager
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim swPathFeat As SldWorks.Feature
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swSelectionMgr As SldWorks.SelectionMgr
Dim swSweepFeatureData As SldWorks.SweepFeatureData
Dim swSweep As SldWorks.SweepFeatureData
Dim swProfileObj As Object
Dim swProfileBody As SldWorks.Body2
Dim swPathFeature As SldWorks.Feature
Dim sketchLines As Variant
Dim status As Boolean
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS\
2018\samples\tutorial\api\sweepcutextrude.SLDPRT", 1, 0, "", longstatus, longwarnings)

    boolstatus = swModel.Extension.SelectByID2("Sketch2", "SKETCH", 0.01948983274156, -0.02564816935317, 0, False, 1, Nothing, 0)
    boolstatus = swModel.Extension.SelectByID2("Sketch3", "SKETCH", -0.03797488317814, -0.02133214444164, 0, True, 4, Nothing, 0)

    Set swSweep = swModel.FeatureManager.CreateDefinition(swFmSweepCut)

    swSweep.TwistControlType = 0
    swSweep.PathAlignmentType = 0
    swSweep.ThinFeature = True
    swSweep.SetWallThickness True, 0.001 'Meters in Direction 1
    swSweep.SetWallThickness False, 0.001 'Meters in Direction 2
    swSweep.ThinWallType = 3 '2 Directions
    swSweep.TangentPropagation = False
    swSweep.AlignWithEndFaces = True
    swSweep.MaintainTangency = False
    swSweep.AdvancedSmoothing = False
    swSweep.StartTangencyType = 0
    swSweep.EndTangencyType = 0
    swSweep.FeatureScope = True
    swSweep.AutoSelect = True
    swSweep.SetTwistAngle 0#
    swSweep.MergeSmoothFaces = True
    swSweep.AssemblyFeatureScope = True
    swSweep.AutoSelectComponents = True
    swSweep.PropagateFeatureToParts = False
    swSweep.CircularProfile = False
    swSweep.CircularProfileDiameter = 0#

    Set swPathFeat = swModel.FeatureManager.CreateFeature(swSweep)

End Sub

```

Insert Swept-surface Feature Example (VBA)

This example shows how to create a swept-surface feature.

```

'-----
' Preconditions: Verify that the part template exists.
'
' Postconditions:
' 1. Opens a new part.
' 2. Creates two sketches.
' 3. Inserts a swept-surface feature.
' 4. Examine the FeatureManager design tree and
'    graphics area.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchManager As SldWorks.SketchManager
Dim swSketchSegment As SldWorks.SketchSegment
Dim swPathFeat As SldWorks.Feature
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swSweep As SldWorks.SweepFeatureData
Dim status As Boolean

Sub main()

    Set swApp = Application.SldWorks

    'Open new part document
    Set swModel = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2018\templates\part.prtdot", 0, 0, 0)
    Set swModelDocExt = swModel.Extension

    'Create a sketch
    Set swSketchManager = swModel.SketchManager
    swSketchManager.InsertSketch True
    status = swModelDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    Set swSketchSegment = swSketchManager.CreateLine(0#, 0#, 0#, 0.068491, 0.049604, 0#)
    Set swSketchSegment = swSketchManager.CreateLine(0.068491, 0.049604, 0#, 0.10923, 0.112837, 0#)
    Set swSketchSegment = swSketchManager.CreateLine(0.10923, 0.112837, 0#, 0.194652, 0.154023, 0#)
    swSketchManager.InsertSketch True

    swModel.ViewZoomtofit2
    swModel.ShowNamedView2 "Isometric", 7
    swModel.ClearSelection2 True

    'Create another sketch
    status = swModelDocExt.SelectByID2("Right Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    swSketchManager.InsertSketch True
    Set swSketchSegment = swSketchManager.CreateLine(-0#, 0#, 0#, 0.021042, 0.091756, 0#)
    Set swSketchSegment = swSketchManager.CreateLine(0.021042, 0.091756, 0#, 0.098366, 0.085093, 0#)
    Set swSketchSegment = swSketchManager.CreateLine(0.098366, 0.085093, 0#, 0.143062, 0.122696, 0#)
    swModel.ClearSelection2 True
    swSketchManager.InsertSketch True

    'Insert surface sweep
    status = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, False, 1, Nothing, 0)
    status = swModelDocExt.SelectByID2("Sketch1", "SKETCH", 0, 0, 0, True, 4, Nothing, 0)

    Set swFeatureMgr = swModel.FeatureManager
    Set swSweep = swFeatureMgr.CreateDefinition(swFmRefSurface)

    swSweep.TangentPropagation = False
    swSweep.TwistControlType = 0
    swSweep.MaintainTangency = False
    swSweep.AdvancedSmoothing = False
    swSweep.StartTangencyType = 0
    swSweep.EndTangencyType = 0
    swSweep.PathAlignmentType = 0
    swSweep.FeatureScope = True
    swSweep.AutoSelect = True
    swSweep.SetTwistAngle 0#
    swSweep.MergeSmoothFaces = True
    swSweep.CircularProfile = False
    swSweep.CircularProfileDiameter = 0#
    swSweep.Direction = 0

    Set swPathFeat = swFeatureMgr.CreateFeature(swSweep)

End Sub

```

Insert Swept-cut Feature Using Sketch Profile Example (VBA)

This example shows how to create a swept-cut feature using a sketch profile and get its properties.

```
'-----
' Preconditions:
' 1. Verify that the part to open exists.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Creates Cut-Sweep1.
' 2. Inspect the FeatureManager design tree, graphics area,
'    and Immediate window.
'
' NOTE: Because this part document is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Dim swSweep As SldWorks.SweepFeatureData
Dim swProfFeat As SldWorks.Feature
Dim swProfSketch As SldWorks.Sketch
Dim swPathFeat As SldWorks.Feature
Dim swPathSketch As SldWorks.Sketch
Dim bRet As Boolean
Dim myModelView As Object
Dim myFeature As SldWorks.Feature

Sub main()

    Set swApp = Application.SldWorks

    Set Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS
2018\samples\tutorial\api\sweepcutextrude.SLDPRT", 1, 0, "", longstatus, longwarnings)
    swApp.ActivateDoc2 "sweepcutextrude.SLDPRT", False, longstatus
    Set Part = swApp.ActiveDoc

    Set myModelView = Part.ActiveView
    myModelView.FrameLeft = 0
    myModelView.FrameTop = 0

    myModelView.FrameState = swWindowState_e.swWindowMaximized
    Part.ShowNamedView2 "Isometric", 7

    boolstatus = Part.Extension.SelectByID2("Sketch2", "SKETCH", 0.01948983274156, -0.02564816935317, 0, False, 1, Nothing,
0)
    boolstatus = Part.Extension.SelectByID2("Sketch3", "SKETCH", -0.03797488317814, -0.02133214444164, 0, True, 4, Nothing,
0)

    Set swSweep = Part.FeatureManager.CreateDefinition(swFmSweepCut)

    swSweep.TwistControlType = 0

    swSweep.PathAlignmentType = 0

    swSweep.CircularProfile = False

    Set myFeature = Part.FeatureManager.CreateFeature(swSweep)

    Set swSweep = myFeature.GetDefinition
    Set swProfFeat = swSweep.Profile
    Set swProfSketch = swProfFeat.GetSpecificFeature

    bRet = swSweep.AccessSelections(Part, Nothing)

    Set swPathFeat = swSweep.Path
    Set swPathSketch = swPathFeat.GetSpecificFeature

    Debug.Print "File = " & Part.GetPathName
    Debug.Print " " & myFeature.Name
    Debug.Print " Sweep path = " & swPathFeat.Name
    Debug.Print " Path type as defined in swSelectType_e = " & swSweep.GetPathType
    Debug.Print " Path alignment type as defined in swTangencyType_e = " & swSweep.PathAlignmentType
    Debug.Print " Sweep Profile = " & swProfFeat.Name
    Debug.Print " Profile type as defined in swSelectType_e = " & swSweep.GetProfileType
    Debug.Print " Profile orientation/twist type as defined in swTwistControlType_e = " & swSweep.TwistControlType
    Debug.Print " Cut sweep type as defined in swCutSweepOption_e = " & swSweep.GetCutSweepOption
    Debug.Print " Align sweep with end faces? " & swSweep.AlignWithEndFaces
    Debug.Print " Automatically select all bodies to be affected if a multibody part? " & swSweep.AutoSelect
    Debug.Print " Start of sweep tangency type as defined in swTangencyType_e = " & swSweep.StartTangencyType
    Debug.Print " End of sweep tangency type as defined in swTangencyType_e = " & swSweep.EndTangencyType
    Debug.Print " Only specific bodies affected by this sweep feature? " & swSweep.FeatureScope
    Debug.Print " Feature scope bodies count = " & swSweep.GetFeatureScopeBodiesCount
    Debug.Print " Is a boss feature? " & swSweep.IsBossFeature
    Debug.Print " Is a thin-walled feature? " & swSweep.IsThinFeature
    Debug.Print " Thin-walled type (0=1D, 1=1DReverse, 2=MidPlane, 3=2D) = " & swSweep.ThinWallType
    Debug.Print " Wall thickness Direction 1 = " & swSweep.GetWallThickness(True) * 1000# & " mm"
```

```
Debug.Print "    Wall thickness Direction 2 = " & swSweep.GetWallThickness(False) * 1000# & " mm"
Debug.Print "    Merge tangent faces? " & swSweep.MaintainTangency
Debug.Print "    Merge results if a multibody part? " & swSweep.Merge
Debug.Print "    Merge smooth faces if using guide curves? " & swSweep.MergeSmoothFaces

    swSweep.ReleaseSelectionAccess
End Sub
```

Insert Table-driven Pattern Example (C#)

This example shows how to insert a table-driven pattern.

```
//-----
// Preconditions:
// 1. Verify that c:\temp exists.
// 2. Open public_documents\samples\tutorial\api\tablepattern.sldprt.
// 3. Edit TPattern1.
// 4. Click Save As, browse to c:\temp, type TestTable.sldptab in File name,
// click Save, and click OK.
// 5. Edit c:\temp\TestTable.sldptab and change:
// a. 0.01m to 0.04m.
// b. -0.03m to -0.025m.
// 6. Save c:\temp\TestTable.sldptab.
// 7. Delete TPattern1.
//
// Postconditions:
// 1. Creates TPattern1.
// 2. Examine the FeatureManager design tree and the graphics
// area.
//
// NOTE: Because the part is used elsewhere, do not save changes.
//-----

using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
using System;

namespace Macro1CSharp.csproj
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            FeatureManager swFeatureMgr = default(FeatureManager);
            Feature swPatternFeature = default(Feature);
            object swPatternPoints = null;
            bool status = false;

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            swFeatureMgr = (FeatureManager)swModel.FeatureManager;
            status = swModelDocExt.SelectByID2("Coordinate System1", "COORDSYS", 0, 0, 0, true, 16, null, 0);
            status = swModelDocExt.SelectByID2("Cut-Extrude1", "BODYFEATURE", 0, 0, 0, true, 4, null, 0);
            swPatternFeature = (Feature)swFeatureMgr.InsertTableDrivenPattern2("c:\\temp\\TestTable.sldptab", (swPatternPoints), true, false, true);
        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Insert Table-driven Pattern Example (VBA)

This example shows how to insert a table-driven pattern.

```
'-----
' Preconditions:
' 1. Verify that c:\temp exists.
' 2. Open public_documents\samples\tutorial\api\tablepattern.sldprt.
' 3. Edit TPattern1.
' 4. Click Save As, browse to c:\temp, type TestTable.sldptab in File name,
'    click Save, and click OK.
' 5. Edit c:\temp\TestTable.sldptab and change:
'    a. 0.01m to 0.04m.
'    b. -0.03m to -0.025m.
' 6. Save c:\temp\TestTable.sldptab.
' 7. Delete TPattern1.
'
' Postconditions:
' 1. Creates TPattern1.
' 2. Examine the FeatureManager design tree and the graphics
'    area.
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swPatternFeature As SldWorks.Feature
Dim swPatternPoints As Variant
Dim status As Boolean

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swModelDocExt = swModel.Extension
    Set swFeatureMgr = swModel.FeatureManager
    status = swModelDocExt.SelectByID("Coordinate System1", "COORDSYS", 0, 0, 0, True, 16, Nothing, 0)
    status = swModelDocExt.SelectByID("Cut-Extrude1", "BODYFEATURE", 0, 0, 0, True, 4, Nothing, 0)
    Set swPatternPoints = Nothing
    Set swPatternFeature = swFeatureMgr.InsertTableDrivenPattern2("c:\temp\TestTable.sldptab", (swPatternPoints), True, False, True)

End Sub
```

Insert Table-driven Pattern Example (VB.NET)

This example shows how to insert a table-driven pattern.

```

-----
Preconditions:
1. Verify that c:\temp exists.
2. Open public_documents\samples\tutorial\api\tablepattern.sldprt.
3. Edit TPattern1.
4. Click Save As, browse to c:\temp, type TestTable.sldptab in File name,
   click Save, and click OK.
5. Edit c:\temp\TestTable.sldptab and change:
   a. 0.01m to 0.04m.
   b. -0.03m to -0.025m.
6. Save c:\temp\TestTable.sldptab.
7. Delete TPattern1.

Postconditions:
1. Creates TPattern1.
2. Examine the FeatureManager design tree and the graphics
   area.

NOTE: Because the part is used elsewhere, do not save changes.
-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System

Partial Class SolidWorksMacro

    Public Sub main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swFeatureMgr As FeatureManager
        Dim swPatternFeature As Feature
        Dim swPatternPoints As Object
        Dim status As Boolean

        swModel = swApp.ActiveDoc
        swModelDocExt = swModel.Extension
        swFeatureMgr = swModel.FeatureManager
        status = swModelDocExt.SelectByID2("Coordinate System1", "COORDSYS", 0, 0, 0, True, 16, Nothing, 0)
        status = swModelDocExt.SelectByID2("Cut-Extrude1", "BODYFEATURE", 0, 0, 0, True, 4, Nothing, 0)
        swPatternPoints = Nothing
        swPatternFeature = swFeatureMgr.InsertTableDrivenPattern2("c:\temp\TestTable.sldptab", (swPatternPoints), True, False, True)

    End Sub

    ''' <summary>
    ''' The SldWorks swApp variable is pre-assigned for you.
    ''' </summary>
    Public swApp As SldWorks

End Class

```

Insert Table Example (VBA)

This example shows how to insert a table into a drawing.

```
'-----  
' Preconditions:  
' 1. Open a drawing.  
' 2. Select a drawing view.  
'  
' Postconditions: Inserts the table.  
'-----
```

Option Explicit

```
Public Enum swBOMConfigurationAnchorType_e  
    swBOMConfigurationAnchor_TopLeft = 1  
    swBOMConfigurationAnchor_TopRight = 2  
    swBOMConfigurationAnchor_BottomLeft = 3  
    swBOMConfigurationAnchor_BottomRight = 4  
End Enum
```

Sub main()

```
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSelMgr As SldWorks.SelectionMgr  
    Dim swDraw As SldWorks.DrawingDoc  
    Dim swTable As SldWorks.TableAnnotation  
    Dim vPickPt As Variant  
    Dim nNumCol As Long  
    Dim nNumRow As Long  
    Dim sRowStr As String  
    Dim i As Long  
    Dim j As Long  
    Dim bRet As Boolean  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.ActiveDoc  
    Set swDraw = swModel  
    Set swSelMgr = swModel.SelectionManager  
    vPickPt = swSelMgr.GetSelectionPoint(1)  
    Set swTable = swDraw.InsertTableAnnotation(vPickPt(0), vPickPt(1), swBOMConfigurationAnchor_TopLeft, 6, 5)  
    nNumCol = swTable.ColumnCount  
    nNumRow = swTable.RowCount  
    For i = 0 To nNumRow  
        For j = 0 To nNumCol  
            swTable.Text(i, j) = "[" & i & ", " & j & "]"  
        Next j  
    Next i
```

End Sub

Insert Text at Angle Example (VBA)

This example shows how to insert a sketch of text at an angle.

```
' -----  
' Preconditions: Verify that the specified document to open  
' exists.  
'  
' Postconditions:  
' 1. Opens the specified part document.  
' 2. Inserts the string Part2.SLDPRT as Sketch2  
'    at the origin and at an angle.  
' 3. Examine the graphics area and FeatureManager design tree.  
'  
' NOTE: Because the model is used elsewhere, do not  
' save any changes.  
' -----  
Option Explicit  
  
Sub main()  
  
    Const sSketchText As String = "<r65>Part2.SLDPRT</r>"  
    Dim swApp As SldWorks.SldWorks  
    Dim swModel As SldWorks.ModelDoc2  
    Dim swSketchText As SldWorks.SketchText  
    Dim errors As Long  
    Dim warnings As Long  
  
    Set swApp = Application.SldWorks  
    Set swModel = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\block20.sldprt", 1, 0, "", errors, warnings)  
    Set swSketchText = swModel.InsertSketchText(0#, 0#, 0#, sSketchText, 0, 0, 100, 100)  
    swModel.InsertSketch2 True  
  
End Sub
```

Insert Thin Cut Extrude Example (C#)

This example shows how to insert a thin cut extrude feature.

```
//-----
// Preconditions: Verify that the specified part exists.
//
// Postconditions:
// 1. Opens the part.
// 2. Inserts a thin cut extrude feature in the part.
// 3. Examine the FeatureManager design tree and
//    graphics area.
//
// NOTE: Because this part document is used elsewhere,
// do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;

namespace FeatureCutThin2FeatureManagerCSharp.csproj
{
    partial
    class SolidWorksMacro
    {
        public
        void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            ModelDocExtension swModelDocExt = default(ModelDocExtension);
            SketchManager swSketchManager = default(SketchManager);
            SketchSegment swSketchSegment = default(SketchSegment);
            FeatureManager swFeatureManager = default(FeatureManager);
            Feature swFeature = default(Feature);
            bool boolstatus = false;
            int errorstatus = 0;
            int warnings = 0;

            // Open part
            swApp.OpenDoc6("C:\\Users\\Public\\Documents\\SOLIDWORKS\\SOLIDWORKS 2018\\samples\\tutorial\\api\\water.sldprt", 1, 0,
            "",
            ref errorstatus, ref warnings);
            swModel = (ModelDoc2)swApp.ActiveDoc;

            // Select face on which to sketch a circle
            swModelDocExt = (ModelDocExtension)swModel.Extension;
            boolstatus = swModelDocExt.SelectByID2("", "FACE", 0.0001655362220845, -0.0477671348753, 0.072, false, 0, null, 0)

            swModel.ShowNamedView2("**Normal To", (int)swStandardViews_e.swBackView);
            swModel.ClearSelection2(true);

            // Sketch a circle
            swSketchManager = (SketchManager)swModel.SketchManager;
            swSketchSegment = (SketchSegment)swSketchManager.CreateCircle(0.0, 0.0, 0.0, 0.030255, -0.042492, 0.0);
            swModel.ClearSelection2(true);

            // Create the thin cut extrude
            boolstatus = swModelDocExt.SelectByID2("Arc1", "SKETCHSEGMENT", 0, 0, 0, false, 0, null, 0);
            swFeatureManager = (FeatureManager)swModel.FeatureManager;
            swFeature = (Feature)swFeatureManager.FeatureCutThin2(true, false, false, (int)swEndConditions_e.swEndCondBlind,
            (int)swEndConditions_e.swEndCondBlind, 0.01, 0.01, false, false, false,
            false, 0.01745329251994, 0.01745329251994, false, false, false, false, 0.01, 0.01, 0.01,
            0, 0, false, 0.005, true, true, (int)swStartConditions_e.swStartSketchPlane, 0, false);
            swModel.ShowNamedView2("**Isometric", (int)swStandardViews_e.swIsometricView);

        }

        /// <summary>
        /// The SldWorks swApp variable is pre-assigned for you.
        /// </summary>
        public SldWorks swApp;
    }
}
```

Insert Thin Cut Extrude Example (VBA)

This example shows how to insert a thin cut extrude feature.

```
'-----
' Preconditions: Verify that the specified part exists.
'
' Postconditions:
' 1. Opens the part.
' 2. Inserts a thin cut extrude feature in the part.
' 3. Examine the FeatureManager design tree and
'    graphics area.
'
' NOTE: Because this part document is used elsewhere,
' do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchManager As SldWorks.SketchManager
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeatureManager As SldWorks.FeatureManager
Dim swFeature As SldWorks.Feature
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long

Sub main()

    Set swApp = Application.SldWorks

    ' Open part
    swApp.OpenDoc6 "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\water.sldprt", 1, 0, "", longstatus, longwarnings
    Set swModel = swApp.ActiveDoc

    ' Select face on which to sketch a circle
    Set swModelDocExt = swModel.Extension
    boolstatus = swModelDocExt.SelectByID("", "FACE", 1.655362220845E-04, -0.0477671348753, 0.072, False, 0, Nothing, 0)
    swModel.ShowNamedView2 "Normal To", swBackView
    swModel.ClearSelection2 True

    ' Sketch a circle
    Set swSketchManager = swModel.SketchManager
    Set swSketchSegment = swSketchManager.CreateCircle(0#, 0#, 0#, 0.030255, -0.042492, 0#)
    swModel.ClearSelection2 True

    ' Create the thin cut extrude
    boolstatus = swModelDocExt.SelectByID("Arc1", "SKETCHSEGMENT", 0, 0, 0, False, 0, Nothing, 0)
    Set swFeatureManager = swModel.FeatureManager
    Set swFeature = swFeatureManager.FeatureCutThin2(True, False, False, swEndCondBlind, swEndCondBlind, 0.01, 0.01, False, False, False, False,
    0.01745329251994, 0.01745329251994, False, False, False, False, 0.01, 0.01, 0.01, 0, 0, False, 0.005, True, True, swStartSketchPlane, 0, False)
    swModel.ShowNamedView2 "Isometric", swIsometricView

End Sub
```

Insert Thin Cut Extrude Example (VB.NET)

This example shows how to insert a thin cut extrude feature.

```
'-----
' Preconditions: Verify that the specified part exists.
'
' Postconditions:
' 1. Opens the part.
' 2. Inserts a thin cut extrude feature in the part.
' 3. Examine the FeatureManager design tree and
'    graphics area.
'
' NOTE: Because this part document is used elsewhere,
' do not save changes.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System

Partial Class SolidWorksMacro

    Public
    Sub main()

        Dim swModel As ModelDoc2
        Dim swModelDocExt As ModelDocExtension
        Dim swSketchManager As SketchManager
        Dim swSketchSegment As SketchSegment
        Dim swFeatureManager As FeatureManager
        Dim swFeature As Feature
        Dim boolstatus As Boolean
        Dim longstatus As Integer, longwarnings As Integer

        ' Open part
        swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2018\samples\tutorial\api\water.sldprt", 1, 0, "",
longstatus,
longwarnings)
        swModel = swApp.ActiveDoc

        ' Select face on which to sketch a circle
        swModelDocExt = swModel.Extension
        boolstatus = swModelDocExt.SelectByID2("", "FACE", 0.0001655362220845, -0.0477671348753, 0.072, False, 0, Nothing, 0)
        swModel.ShowNamedView2("Normal To", swStandardViews_e.swBackView)
        swModel.ClearSelection2(True)

        ' Sketch a circle
        swSketchManager = swModel.SketchManager
        swSketchSegment = swSketchManager.CreateCircle(0.0#, 0.0#, 0.0#, 0.030255, -0.042492, 0.0#)
        swModel.ClearSelection2(True)

        ' Create the thin cut extrude
        boolstatus = swModelDocExt.SelectByID2("Arc1", "SKETCHSEGMENT", 0, 0, 0, False, 0, Nothing, 0)
        swFeatureManager = swModel.FeatureManager
        swFeature = swFeatureManager.FeatureCutThin2(True, False, False, swEndConditions_e.swEndCondBlind,
swEndConditions_e.swEndCondBlind, 0.01, 0.01, False, False, False, False, 0.01745329251994, 0.01745329251994, False, False,
False, False, 0.01, 0.01, 0.01, 0, 0, False, 0.005, True, True, swStartConditions_e.swStartSketchPlane, 0, False)
        swModel.ShowNamedView2("Isometric", swStandardViews_e.swIsometricView)

    End Sub

''' <summary>
''' The SldWorks swApp variable is pre-assigned for you.
''' </summary>
Public swApp As SldWorks

End Class
```

Insert Weld Symbol Example (C#)

This example shows how to insert a weld symbol into a model.

```
//-----
// Preconditions:
// 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
// 2. Select a face, edge, or sketch segment for Weld Symbol insertion.
// 3. Open an Immediate window.
//
// Postconditions:
// 1. Inserts the specified ISO Weld Symbol.
// 2. Inspect the Immediate window.
//
// NOTE: Because the model is used elsewhere, do not save changes.
// -----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;

namespace InsertWeldSymbol_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        public void Main()
        {
            ModelDoc2 swModel = default(ModelDoc2);
            WeldSymbol swWeldSymbol = default(WeldSymbol);

            swModel = (ModelDoc2)swApp.ActiveDoc;
            swWeldSymbol = (WeldSymbol)swModel.InsertWeldSymbol3();

            swWeldSymbol.SetFieldWeld((int)swWeldSymbolField_e.swFieldWeldNone);
            swWeldSymbol.SetPeripheral(false);
            swWeldSymbol.SetProcess(true, "Process", true);
            swWeldSymbol.SetStagger(true);
            swWeldSymbol.SetSymmetric((int)swWeldSymbolSymmetric_e.swWeldSymmetric);
            swWeldSymbol.SetText(true, "Left", "BUTT", "Right", "Stagger", (int)swWeldSymbolContourTypes_e.swWeldContourNone);

            Debug.Print("Arc count: " + swWeldSymbol.GetArcCount());
            Debug.Print("Arrow head count: " + swWeldSymbol.GetArrowHeadCount());
            Debug.Print("Contour setting as defined in swWeldSymbolContourTypes_e: " + swWeldSymbol.GetContour(true));
            Debug.Print("Field weld setting as defined in swWeldSymbolField_e: " + swWeldSymbol.GetFieldWeld());
            Debug.Print("Number of leaders on this weld symbol: " + swWeldSymbol.GetLeaderCount());
            Debug.Print("Number of line segments in this weld symbol: " + swWeldSymbol.GetLineCount());
            Debug.Print("Weld all around the contour? " + swWeldSymbol.GetPeripheral());
            Debug.Print("Welding process indicated? " + swWeldSymbol.GetProcess());
            Debug.Print("Reference box around the specification process text? " + swWeldSymbol.GetProcessReference());
            Debug.Print("Stagger symbols above and below the line? " + swWeldSymbol.GetStagger());
            Debug.Print("Weld symbol symmetry as defined in swWeldSymbolSymmetric_e: " + swWeldSymbol.GetSymmetric());
            Debug.Print("Number of text items in this weld symbol: " + swWeldSymbol.GetTextCount());
            Debug.Print("Number of triangles in this weld symbol: " + swWeldSymbol.GetTriangleCount());
            Debug.Print("Extra leader line? " + swWeldSymbol.HasExtraLeader());
            Debug.Print("Weld symbol is attached to a leaderLine? " + swWeldSymbol.IsAttached());

        }

        public SldWorks swApp;
    }
}
```

Insert Weld Symbol Example (VBA)

This example shows how to insert a weld symbol into a model.

```
'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
' 2. Select a face, edge, or sketch segment for Weld Symbol insertion.
' 3. Open an Immediate window.
'
' Postconditions:
' 1. Inserts the specified ISO Weld Symbol.
' 2. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----
```

Option Explicit

Sub main()

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swWeldSymbol As SldWorks.WeldSymbol
```

```
Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
Set swWeldSymbol = swModel.InsertWeldSymbol3
```

```
swWeldSymbol.SetFieldWeld swFieldWeldNone
swWeldSymbol.SetPeripheral False
swWeldSymbol.SetProcess True, "Process", True
swWeldSymbol.SetStagger True
swWeldSymbol.SetSymmetric swWeldSymmetric
swWeldSymbol.SetText True, "Left", "BUTT", "Right", "Stagger", swWeldContourNone
```

```
Debug.Print "Arc count: " & swWeldSymbol.GetArcCount
Debug.Print "Arrow head count: " & swWeldSymbol.GetArrowHeadCount
Debug.Print "Contour setting as defined in swWeldSymbolContourTypes_e: " & swWeldSymbol.GetContour(True)
Debug.Print "Field weld setting as defined in swWeldSymbolField_e: " & swWeldSymbol.GetFieldWeld
Debug.Print "Number of leaders on this weld symbol: " & swWeldSymbol.GetLeaderCount
Debug.Print "Number of line segments in this weld symbol: " & swWeldSymbol.GetLineCount
Debug.Print "Weld all around the contour? " & swWeldSymbol.GetPeripheral
Debug.Print "Welding process indicated? " & swWeldSymbol.GetProcess
Debug.Print "Reference box around the specification process text? " & swWeldSymbol.GetProcessReference
Debug.Print "Stagger symbols above and below the line? " & swWeldSymbol.GetStagger
Debug.Print "Weld symbol symmetry as defined in swWeldSymbolsSymmetric_e: " & swWeldSymbol.GetSymmetric
Debug.Print "Number of text items in this weld symbol: " & swWeldSymbol.GetTextCount
Debug.Print "Number of triangles in this weld symbol: " & swWeldSymbol.GetTriangleCount
Debug.Print "Extra leader line? " & swWeldSymbol.HasExtraLeader
Debug.Print "Weld symbol is attached to a leaderline? " & swWeldSymbol.IsAttached
```

End Sub

Insert Weld Symbol Example (VB.NET)

This example shows how to insert a weld symbol into a model.

```
'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
' 2. Select a face, edge, or sketch segment for Weld Symbol insertion.
' 3. Open an Immediate window.
'
' Postconditions:
' 1. Inserts the specified ISO Weld Symbol.
' 2. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----
```

```
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics
```

Partial Class SolidWorksMacro

```
Sub main()

    Dim swModel As ModelDoc2
    Dim swWeldSymbol As WeldSymbol

    swModel = swApp.ActiveDoc
    swWeldSymbol = swModel.InsertWeldSymbol3

    swWeldSymbol.SetFieldWeld(swWeldSymbolField_e.swFieldWeldNone)
    swWeldSymbol.SetPeripheral(False)
    swWeldSymbol.SetProcess(True, "Process", True)
    swWeldSymbol.SetStagger(True)
    swWeldSymbol.SetSymmetric(swWeldSymbolSymmetric_e.swWeldSymmetric)
    swWeldSymbol.SetText(True, "Left", "BUTT", "Right", "Stagger", swWeldSymbolContourTypes_e.swWeldContourNone)

    Debug.Print("Arc count: " & swWeldSymbol.GetArcCount)
    Debug.Print("Arrow head count: " & swWeldSymbol.GetArrowHeadCount)
    Debug.Print("Contour setting as defined in swWeldSymbolContourTypes_e: " & swWeldSymbol.GetContour(True))
    Debug.Print("Field weld setting as defined in swWeldSymbolField_e: " & swWeldSymbol.GetFieldWeld)
    Debug.Print("Number of leaders on this weld symbol: " & swWeldSymbol.GetLeaderCount)
    Debug.Print("Number of line segments in this weld symbol: " & swWeldSymbol.GetLineCount)
    Debug.Print("Weld all around the contour? " & swWeldSymbol.GetPeripheral)
    Debug.Print("Welding process indicated? " & swWeldSymbol.GetProcess)
    Debug.Print("Reference box around the specification process text? " & swWeldSymbol.GetProcessReference)
    Debug.Print("Stagger symbols above and below the line? " & swWeldSymbol.GetStagger)
    Debug.Print("Weld symbol symmetry as defined in swWeldSymbolSymmetric_e: " & swWeldSymbol.GetSymmetric)
    Debug.Print("Number of text items in this weld symbol: " & swWeldSymbol.GetTextCount)
    Debug.Print("Number of triangles in this weld symbol: " & swWeldSymbol.GetTriangleCount)
    Debug.Print("Extra leader line? " & swWeldSymbol.HasExtraLeader)
    Debug.Print("Weld symbol is attached to a leaderline? " & swWeldSymbol.IsAttached)


```

End Sub

Public swApp As SldWorks

End Class

Insert Weld Table Example (C#)

This example shows how to insert a weld table into a drawing view.

```
//-----
// Preconditions:
// 1. Open the drawing of a part that contains a weld bead.
// 2. Verify that install_dir\lang\english\weldtable-standard.sldwldtbt exists.
//
// Postconditions:
// 1. Inserts a weld table.
// 2. Examine the drawing and the FeatureManager design tree.
// -----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;
namespace InsertWeldTable_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        DrawingDoc Draw;
        ModelDoc2 Part;
        View swActiveView;

        bool boolstatus;
        Const WeldTableTemplate As String = "C:\\Program Files\\SOLIDWORKS Corp\\SOLIDWORKS\\lang\\english\\weldtable-
standard.sldwldtbt"

        public void Main()
        {
            Part = (ModelDoc2)swApp.ActiveDoc;
            boolstatus = Part.Extension.SelectByID2("Drawing View1", "DRAWINGVIEW", 0, 0, 0, false, 0, null, 0);
            Draw = (DrawingDoc)Part;

            boolstatus = Draw.ActivateView("Drawing View1");
            swActiveView = (View)Draw.ActiveDrawingView;
            bool boolResult = false;
            boolResult = swActiveView.InsertWeldTable(false, true, true, 0, 0,
(int)swBOMConfigurationAnchorType_e.swBOMConfigurationAnchor_BottomLeft, "Default", WeldTableTemplate);
        }

        public SldWorks swApp;
    }
}
```

Insert Weld Table Example (VBA)

This example shows how to insert a weld table into a drawing view.

```
'-----  
' Preconditions:  
' 1. Open the drawing of a part that contains a weld bead.  
' 2. Verify that install_dir\lang\english\weldtable-standard.sldwldtbt exists.  
'  
' Postconditions:  
' 1. Inserts a weld table.  
' 2. Examine the drawing and the FeatureManager design tree.  
'-----  
Dim swApp As SldWorks.SldWorks  
Dim Draw As SldWorks.DrawingDoc  
Dim swActiveView As SldWorks.View  
Dim boolstatus As Boolean  
  
Option Explicit  
Const WeldTableTemplate As String = "C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\lang\english\weldtable-standard.sldwldtbt"  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set Draw = swApp.ActiveDoc  
    boolstatus = Draw.Extension.SelectByID("Drawing View1", "DRAWINGVIEW", 0, 0, 0, False, 0, Nothing, 0)  
    boolstatus = Draw.ActivateView("Drawing View1")  
  
    Set swActiveView = Draw.ActiveDrawingView  
    Dim boolResult As Boolean  
    boolResult = swActiveView.InsertWeldTable(False, True, False, 0#, 0#, swBOMConfigurationAnchorType_e.swBOMConfigurationAnchor_BottomLeft, "Default",  
WeldTableTemplate)  
  
End Sub
```

Insert Weld Table Example (VB.NET)

This example shows how to insert a weld table into a drawing view.

```
'-----  
' Preconditions:  
' 1. Open the drawing of a part that contains a weld bead.  
' 2. Verify that install_dir\lang\english\weldtable-standard.sldwldtbt exists.  
'-----  
' Postconditions:  
' 1. Inserts a weld table.  
' 2. Examine the drawing and the FeatureManager design tree.  
'-----
```

```
Imports SolidWorks.Interop.sldworks
```

```
Imports SolidWorks.Interop.swconst
```

```
Imports System
```

```
Partial Class SolidWorksMacro
```

```
    Dim Draw As DrawingDoc
```

```
    Dim Part As ModelDoc2
```

```
    Dim swActiveView As View
```

```
    Dim boolstatus As Boolean
```

```
    Const WeldTableTemplate As String = "C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\lang\english\weldtable-standard.sldwldtbt"
```

```
    Sub main()
```

```
        Part = swApp.ActiveDoc
```

```
        boolstatus = Part.Extension.SelectByID2("Drawing View1", "DRAWINGVIEW", 0, 0, 0, False, 0, Nothing, 0)
```

```
        Draw = Part
```

```
        boolstatus = Draw.ActivateView("Drawing View1")
```

```
        swActiveView = Draw.ActiveDrawingView
```

```
        Dim boolResult As Boolean
```

```
        boolResult = swActiveView.InsertWeldTable(False, True, False, 0.0#, 0.0#,
```

```
        swBOMConfigurationAnchorType_e.swBOMConfigurationAnchor_BottomLeft, "Default", WeldTableTemplate)
```

```
    End Sub
```

```
    Public swApp As SldWorks
```

```
End Class
```

Insert Weldment Cut List Example #2 (C#)

This example shows how to insert a weldment cut list into the FeatureManager design tree.

```
//-----  
// Preconditions:  
// 1. Open public_documents\samples\tutorial\assemblymates\bracket.sldprt.  
// 2. Click Tools > Options > System Options > FeatureManager >  
// Solid Bodies > Show > OK.  
// 3. Expand the Solid Bodies(1) folder in the FeatureManager design tree  
// and note its contents.  
//  
// Postconditions:  
// 1. Inserts a cut-list-item folder feature containing the specified  
// weldment body.  
// 2. Expand the Cut-List-Item1 folder in the Solid Bodies(1) folder  
// in the the FeatureManager design tree to verify step 1.  
//  
// NOTE: Because this part is used elsewhere,  
// do not save changes.  
//-----
```

```
using SolidWorks.Interop.sldworks;  
using SolidWorks.Interop.swconst;  
using System;  
namespace InsertWeldmentCutList2_CSharp.csproj  
{  
    partial class SolidWorksMacro  
    {  
  
        ModelDoc2 Part;  
        Body2[] obj = new Body2[1];  
        Object[] v;  
        long i;  
  
        public void Main()  
        {  
            Part = (ModelDoc2)swApp.ActiveDoc;  
            v = (object[])((PartDoc)Part).GetBodies2(0, true);  
            for (i = 0; i <= v.GetUpperBound(0); i++)  
            {  
                obj[i] = (Body2)v[i];  
            }  
            Feature cutListFeature = default(Feature);  
            cutListFeature = Part.FeatureManager.InsertWeldmentCutList2(obj);  
        }  
        public SldWorks swApp;  
    }  
}
```

Insert Weldment Cut List Example #2 (VBA)

This example shows how to insert a weldment cut list into the FeatureManager design tree.

```
'-----  
' Preconditions:  
' 1. Open public_documents\samples\tutorial\assemblymates\bracket.sldprt.  
' 2. Click Tools > Options > System Options > FeatureManager >  
' Solid Bodies > Show > OK.  
' 3. Expand the Solid Bodies(1) folder in the FeatureManager design tree  
' and note its contents.  
'  
' Postconditions:  
' 1. Inserts a cut-list-item folder feature containing the specified  
' weldment body.  
' 2. Expand the Cut-List-Item1 folder in the Solid Bodies(1) folder  
' in the the FeatureManager design tree to verify step 1.  
'  
' NOTE: Because this part is used elsewhere,  
' do not save changes.  
'-----
```

Option Explicit

Dim swApp As SldWorks.SldWorks

Dim Part As ModelDoc2

Dim obj() As Object

Dim v As Variant

Sub main()

Dim PartName As String

Set swApp = Application.SldWorks

Set Part = swApp.ActiveDoc

v = Part.GetBodies2(0, True)

Dim cutListFeature As Feature

Set cutListFeature = Part.FeatureManager.InsertWeldmentCutList2(v)

End Sub

Insert Weldment Cut List Example #2 (VB.NET)

This example shows how to insert a weldment cut list into the FeatureManager design tree.

```
'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\assemblymates\bracket.sldprt.
' 2. Click Tools > Options > System Options > FeatureManager >
'    Solid Bodies > Show > OK.
' 3. Expand the Solid Bodies(1) folder in the FeatureManager design tree
'    and note its contents.
'
' Postconditions:
' 1. Inserts a cut-list-item folder feature containing the specified
'    weldment body.
' 2. Expand the Cut-List-Item1 folder in the Solid Bodies(1) folder
'    in the the FeatureManager design tree to verify step 1.
'
' NOTE: Because this part is used elsewhere,
' do not save changes.
'-----
```

Imports SolidWorks.Interop.sldworks

Imports SolidWorks.Interop.swconst

Imports System

Partial Class SolidWorksMacro

Dim Part As ModelDoc2

Dim obj(0) As Body2

Dim v As Array

Dim i As Long

Sub main()

Part = swApp.ActiveDoc

v = Part.GetBodies2(0, True)

For i = 0 To UBound(v)

obj(i) = CType(v(i), Body2)

Next

Dim cutListFeature As Feature

cutListFeature = Part.FeatureManager.InsertWeldmentCutList2(obj)

End Sub

Public swApp As SldWorks

End Class

Insert Weldment Cut List Example (VBA)

This example shows how to insert a weldment cut list in a weldment part document.

```
'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\weldments\weldment_box2.sldprt.
' 2. Expand StructuralMember1 in the FeatureManager design tree and
'    Ctrl+click StructuralMember[1] and StructuralMember[2].
' 3. Open the Immediate window.
'
' Postconditions:
' 1. Creates a new weldment cut-list item folder called Cut-List-Item1
'    in the FeatureManager design tree.
' 2. Places the selected solid bodies in Cut-List-Item1.
' 3. To verify:
'    * Examine the Immediate window.
'    * Expand Cut list(31) and expand Cut-List-Item1(2).
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
```

Option Explicit

Sub main()

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swSelMgr As SldWorks.SelectionMgr
Dim swFeatMgr As SldWorks.FeatureManager
Dim swCutListFeat As SldWorks.Feature
Dim nSelCount As Long
Dim swBody() As SldWorks.Body2
Dim i As Long
Dim bRet As Boolean
```

```
Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
Set swSelMgr = swModel.SelectionManager
Set swFeatMgr = swModel.FeatureManager
```

```
nSelCount = swSelMgr.GetSelectedObjectCount
ReDim swBody(nSelCount - 1)
For i = 1 To nSelCount
    Set swBody(i - 1) = swSelMgr.GetSelectedObject6(i, -1)
Next i
```

```
Set swCutListFeat = swFeatMgr.InsertWeldmentCutList
```

```
Debug.Print "File = " & swModel.GetPathName
Debug.Print " " & swCutListFeat.Name
For i = 0 To nSelCount - 1
    Debug.Print " " & swBody(i).GetSelectionId
Next i
```

End Sub

Insert Weldment Cut List Table Example (C#)

This example shows how to insert a weldment cut list table in a drawing.

```
//-----
// Preconditions:
// 1. Verify that the specified cut list template exists.
// 2. Open public_documents\samples\tutorial\weldments\weldment_box2.sldprt.
// 3. Click File > Make Drawing from Part > OK > drag a view onto
//    the sheet > OK.
// 4. Expand the drawing view in the FeatureManager design tree.
// 5. Right-click weldment_box2 and select Open Part(weldment_box2.sldprt).
// 6. Right-click the Cut list folder and click Update Automatically.
// 7. Click Window > weldment_box2 - Sheet1*.
//
// Postconditions:
// 1. Inserts a weldment cut list table.
// 2. Examine the FeatureManager design tree and graphics area.
//
// NOTE: Because this part is used elsewhere, do not save changes.
//-----
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;
namespace InsertWeldmentCutList_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        const string WeldmentTableTemplate = "C:\\Program Files\\SOLIDWORKS Corp\\SOLIDWORKS\\lang\\english\\cut_list.sldwldtbt";

        public void Main()
        {
            DrawingDoc oDrawing = default(DrawingDoc);
            View swView = default(View);
            WeldmentCutListAnnotation WMTTable = default(WeldmentCutListAnnotation);

            oDrawing = (DrawingDoc)swApp.ActiveDoc;
            swView = (View)oDrawing.GetFirstView();
            swView = (View)swView.GetNextView();

            // Insert the weldment cut list table

            WMTTable = swView.InsertWeldmentTable(false, 0.1996662889191, 0.1013905859662,
(int)swBOMConfigurationAnchorType_e.swBOMConfigurationAnchor_TopLeft, "Default<As Welded>", WeldmentTableTemplate);
        }

        public SldWorks swApp;
    }
}
```

Insert Weldment Cut List Table Example (VBA)

This example shows how to insert a weldment cut list table in a drawing.

```
'-----  
' Preconditions:  
' 1. Verify that the specified cut list template exists.  
' 2. Open public_documents\samples\tutorial\weldments\weldment_box2.sldprt.  
' 3. Click File > Make Drawing from Part > OK > drag a view onto  
'    the sheet > OK.  
' 4. Expand the drawing view in the FeatureManager design tree.  
' 5. Right-click weldment_box2 and select Open Part(weldment_box2.sldprt).  
' 6. Right-click the Cut list folder and click Update Automatically.  
' 7. Click Window > weldment_box2 - Sheet1*.  
'  
' Postconditions:  
' 1. Inserts a weldment cut list table.  
' 2. Examine the FeatureManager design tree and graphics area.  
'  
' NOTE: Because this part is used elsewhere, do not save changes.  
'-----
```

Option Explicit

```
Const WeldmentTableTemplate As String = "C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\lang\english\cut list.sldwldtbt"
```

```
Sub Main()
```

```
    Dim swapp As SldWorks.SldWorks  
    Dim oDrawing As DrawingDoc  
    Dim swView As View  
    Dim WMTTable As SldWorks.WeldmentCutListAnnotation
```

```
    Set swapp = Application.SldWorks  
    Set oDrawing = swapp.ActiveDoc  
    Set swView = oDrawing.GetFirstView  
    Set swView = swView.GetNextView
```

```
    ' Insert the weldment cut list table  
    Set WMTTable = swView.InsertWeldmentTable(False, 0.1996662889191, 0.1013905859662, swBOMConfigurationAnchor_TopLeft, "Default<As  
Welded>", WeldmentTableTemplate)
```

```
End Sub
```

Insert Weldment Cut List Table Example (VB.NET)

This example shows how to insert a weldment cut list table in a drawing.

```
'-----
' Preconditions:
' 1. Verify that the specified cut list template exists.
' 2. Open public_documents\samples\tutorial\weldments\weldment_box2.sldprt.
' 3. Click File > Make Drawing from Part > OK > drag a view onto
'    the sheet > OK.
' 4. Expand the drawing view in the FeatureManager design tree.
' 5. Right-click weldment_box2 and select Open Part(weldment_box2.sldprt).
' 6. Right-click the Cut list folder and click Update Automatically.
' 7. Click Window > weldment_box2 - Sheet1*.
'
' Postconditions:
' 1. Inserts a weldment cut list table.
' 2. Examine the FeatureManager design tree and graphics area.
'
' NOTE: Because this part is used elsewhere, do not save changes.
'-----
```

Imports SolidWorks.Interop.sldworks

Imports SolidWorks.Interop.swconst

Imports System

Partial Class SolidWorksMacro

Const WeldmentTableTemplate **As String** = "C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\lang\english\cut list.sldwldtbt"

Sub Main()

Dim oDrawing **As** DrawingDoc
Dim swView **As** View
Dim WMTTable **As** WeldmentCutListAnnotation

oDrawing = swapp.ActiveDoc
swView = oDrawing.GetFirstView
swView = swView.GetNextView

' Insert the weldment cut list table

WMTTable = swView.**InsertWeldmentTable**(**False**, 0.1996662889191, 0.1013905859662,
swBOMConfigurationAnchorType_e.swBOMConfigurationAnchor_TopLeft, "Default<As Welded>", WeldmentTableTemplate)

End Sub

Public swApp **As** SldWorks

End Class

Insert Weldment End Cap Example (C#)

This example shows how to create an end cap on the open face of a structural member.

```
//-----
// Preconditions:
// 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
// 2. Open an Immediate window.
//
// Postconditions:
// 1. Deletes End cap1.
// 2. Inserts End cap3 in the FeatureManager design tree.
// 3. Inspect the Immediate window.
//
// NOTE: Because the model is used elsewhere, do not save changes.
// -----
using Microsoft.VisualBasic;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Data;
using System.Diagnostics;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System.Runtime.InteropServices;
namespace InsertEndCap_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        Feature myFeature;
        ModelDoc2 Part;
        EndCapFeatureData swEndCap;
        bool boolstatus;

        public void Main()
        {
            Part = (ModelDoc2)swApp.ActiveDoc;

            boolstatus = Part.Extension.SelectByID2("End cap1", "BODYFEATURE", 0, 0, 0, false, 0, null, 0);
            Part.EditDelete();

            Part.ViewZoomTo2(0.632542197290199, 0.972121141705638, 0.0346184961022406, 1.1852319686392, 0.619681287512073,
0.0346184961022431);

            boolstatus = Part.Extension.SelectByID2("", "FACE", 0.58771345904097, 0.614999999999952, -1.01293869257864, true, 0,
null, 0);
            boolstatus = Part.Extension.SelectByID2("", "FACE", -0.0124763445314215, 0.614999999999839, -1.0014248149476, true, 0,
null, 0);

            myFeature = Part.FeatureManager.InsertEndCapFeature3(0.005, false, false, 0.003, 0.6, 0.003, true, 0.002, false, 2);
            swEndCap = (EndCapFeatureData)myFeature.GetDefinition();

            Debug.Print("File = " + Part.GetPathName());
            Debug.Print(" " + myFeature.Name);
            Debug.Print("    Chamfer distance or fillet radius                = " + swEndCap.ChamferDistance * 1000.0 + "
mm");
            Debug.Print("    Inset distance                                = " + swEndCap.DepthDistance * 1000.0 + " mm");
            Debug.Print("    Thickness direction (0=outward, 1=inward, 2=inset) = " + swEndCap.IsEndCapInward);
            Debug.Print("    Offset distance                                = " + swEndCap.OffsetDistance * 1000.0 + "
mm");
            Debug.Print("    Thickness of end cap                                = " + swEndCap.Thickness * 1000.0 + " mm");
            Debug.Print("    Thickness ratio for offset                        = " + swEndCap.ThicknessRatioForOffset);
            Debug.Print("    Chamfer corners                                    = " + swEndCap.UseChamferCorners);
            Debug.Print("    Apply corner treatment                            = " + swEndCap.UseCornerTreatment);
            Debug.Print("    Reverse offset                                    = " + swEndCap.UseReverse);
            Debug.Print("    Use thickness ratio for offset                    = " + swEndCap.UseThicknessRatioForOffset);

        }

        public SldWorks swApp;
    }
}
```

Insert Weldment End Cap Example (VBA)

This example shows how to create an end cap on the open face of a structural member.

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Deletes End cap1.
' 2. Inserts End cap3 in the FeatureManager design tree.
' 3. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----

Dim swApp As SldWorks.SldWorks
Dim myFeature As SldWorks.Feature
Dim Part As SldWorks.ModelDoc2
Dim swEndCap As SldWorks.EndCapFeatureData
Dim boolstatus As Boolean

Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc

    boolstatus = Part.Extension.SelectByID2("End cap1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
    Part.EditDelete

    Part.ViewZoomTo2 0.632542197290199, 0.972121141705638, 3.46184961022406E-02, 1.1852319686392, 0.619681287512073,
    3.46184961022431E-02

    boolstatus = Part.Extension.SelectByID2("", "FACE", 0.58771345904097, 0.614999999999952, -1.01293869257864, True, 0, Nothing,
0)
    boolstatus = Part.Extension.SelectByID2("", "FACE", -1.24763445314215E-02, 0.6149999999999839, -1.0014248149476, True, 0,
Nothing, 0)

    Set myFeature = Part.FeatureManager.InsertEndCapFeature3(0.005, False, False, 0.003, 0.6, 0.003, True, 0.002, False, 2)
    Set swEndCap = myFeature.GetDefinition

    Debug.Print "File = " & Part.GetPathName
    Debug.Print " " & myFeature.Name
    Debug.Print " Chamfer distance or fillet radius = " & swEndCap.ChamferDistance * 1000# & " mm"
    Debug.Print " Inset distance = " & swEndCap.DepthDistance * 1000# & " mm"
    Debug.Print " Thickness direction (0=outward, 1=inward, 2=inset) = " & swEndCap.IsEndCapInward
    Debug.Print " Offset distance = " & swEndCap.OffsetDistance * 1000# & " mm"
    Debug.Print " Thickness of end cap = " & swEndCap.Thickness * 1000# & " mm"
    Debug.Print " Thickness ratio for offset = " & swEndCap.ThicknessRatioForOffset
    Debug.Print " Chamfer corners = " & swEndCap.UseChamferCorners
    Debug.Print " Apply corner treatment = " & swEndCap.UseCornerTreatment
    Debug.Print " Reverse offset = " & swEndCap.UseReverse
    Debug.Print " Use thickness ratio for offset = " & swEndCap.UseThicknessRatioForOffset

End Sub

```

Insert Weldment End Cap Example (VB.NET)

This example shows how to create an end cap on the open face of a structural member.

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorial\api\weldment_box3.sldprt.
' 2. Open an Immediate window.
'
' Postconditions:
' 1. Deletes End cap1.
' 2. Inserts End cap3 in the FeatureManager design tree.
' 3. Inspect the Immediate window.
'
' NOTE: Because the model is used elsewhere, do not save changes.
'-----
Imports SolidWorks.Interop.sldworks
Imports SolidWorks.Interop.swconst
Imports System.Runtime.InteropServices
Imports System
Imports System.Diagnostics

Partial Class SolidWorksMacro

    Dim myFeature As Feature
    Dim Part As ModelDoc2
    Dim swEndCap As EndCapFeatureData
    Dim boolstatus As Boolean

    Sub main()

        Part = swApp.ActiveDoc

        boolstatus = Part.Extension.SelectByID2("End cap1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
        Part.EditDelete()

        Part.ViewZoomTo2(0.632542197290199, 0.972121141705638, 0.0346184961022406, 1.1852319686392, 0.619681287512073,
0.0346184961022431)

        boolstatus = Part.Extension.SelectByID2("", "FACE", 0.58771345904097, 0.614999999999952, -1.01293869257864, True, 0,
Nothing, 0)
        boolstatus = Part.Extension.SelectByID2("", "FACE", -0.0124763445314215, 0.614999999999839, -1.0014248149476, True, 0,
Nothing, 0)

        myFeature = Part.FeatureManager.InsertEndCapFeature3(0.005, False, False, 0.003, 0.6, 0.003, True, 0.002, False, 2)

        swEndCap = myFeature.GetDefinition

        Debug.Print("File = " & Part.GetPathName)
        Debug.Print(" " & myFeature.Name)
        Debug.Print("    Chamfer distance or fillet radius                = " & swEndCap.ChamferDistance * 1000.0# & " mm")
        Debug.Print("    Inset distance                    = " & swEndCap.DepthDistance * 1000.0# & " mm")
        Debug.Print("    Thickness direction (0=outward, 1=inward, 2=inset)        = " & swEndCap.IsEndCapInward)
        Debug.Print("    Offset distance                = " & swEndCap.OffsetDistance * 1000.0# & " mm")
        Debug.Print("    Thickness of end cap            = " & swEndCap.Thickness * 1000.0# & " mm")
        Debug.Print("    Thickness ratio for offset      = " & swEndCap.ThicknessRatioForOffset)
        Debug.Print("    Chamfer corners                  = " & swEndCap.UseChamferCorners)
        Debug.Print("    Apply corner treatment          = " & swEndCap.UseCornerTreatment)
        Debug.Print("    Reverse offset                  = " & swEndCap.UseReverse)
        Debug.Print("    Use thickness ratio for offset  = " & swEndCap.UseThicknessRatioForOffset)

    End Sub

    Public swApp As SldWorks

End Class

```

Insert Weldment Features Example (C#)

This example shows how to insert the following weldment features into the FeatureManager design tree:

- * End cap feature
- * Fillet bead feature
- * Gusset feature
- * Structural weldment
- * Sub weld folder
- * Weldment trim feature

```
//-----
// Preconditions:
// 1. Open public_documents\samples\tutorial\weldments\weldment_box2.sldprt
// 2. Expand the Cut list folder in the FeatureManager design tree
//    and observe its contents.
// 3. Delete End cap1 from the FeatureManager design tree.
// 4. Change the path specified for profilePathName, if necessary.
//
// Postconditions: Observe the following in the FeatureManager design tree:
// * Gusset1 moves to the sub weld folder, Sub Folder1, in the Cut list
//   folder
// * Trim/Extend8 feature appears at the bottom of the design tree and at
//   the end of the Cut list folder
// * Structural Member6 feature appears at the bottom of the design tree
//   and at the end of the Cut list folder
// * End cap5 feature appears at the bottom of the design tree and in the
//   Cut list folder
// * Gusset5 feature appears at the bottom of the design tree and in the
//   Cut list folder
// * Fillet Bead5 feature appears at the bottom of the design tree and
//   in the Cut list folder
//
// NOTE: Because this part is used elsewhere,
// do not save any changes when you close it.
//-----

using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
using System;

namespace InsertWeldmentFeatures_CSharp.csproj
{
    partial class SolidWorksMacro
    {
        FeatureManager fm;
        SelectionMgr selMgr;
        ModelDoc2 Part;
        bool boolstatus;
        public void Main()
        {
            object myFeature = null;
            Body2[] obj = new Body2[1];
            Array v = default(Array);
            Part = (ModelDoc2)swApp.ActiveDoc;
            object myModelView = null;
            myModelView = Part.ActiveView;
            selMgr = (SelectionMgr)Part.SelectionManager;
            //InsertSubWeldFolder2

            boolstatus = Part.Extension.SelectByID2("Gusset1", "SOLIDBODY", 0, 0, 0, false, 0, null, 0);
            obj[0] = (Body2)selMgr.GetSelectedObject6(1, -1);
            v = obj;

            fm = Part.FeatureManager;
            myFeature = fm.InsertSubWeldFolder2(v);
            Part.ClearSelection2(true);
            //InsertWeldmentTrimFeature2
            Body2[] obj1 = new Body2[1];
            Body2[] obj2 = new Body2[1];

            long Options = 0;
            Array v1 = default(Array);
            Array v2 = default(Array);

            boolstatus = Part.Extension.SelectByID2("Structural Member1[2]", "SOLIDBODY", 0, 0, 0, true, 2, null, 0);
            boolstatus = Part.Extension.SelectByID2("Structural Member1[1]", "SOLIDBODY", 0, 0, 0, true, 1, null, 0);
            long Count = 0;

            Count = selMgr.GetSelectedObjectCount();
            if (Count == 2)
            {
                obj1[0] = (Body2)selMgr.GetSelectedObject2(1);
                v1 = obj1;
                obj2[0] = (Body2)selMgr.GetSelectedObject2(2);
```

```

v2 = obj2;

Part.ClearSelection2(true);

Options = (long)swWeldmentTrimExtendOptionType_e.swWeldmentTrimExtendOption_AllowTrimmedExtensionTrim +
(long)swWeldmentTrimExtendOptionType_e.swWeldmentTrimExtendOption_AllowTrimmingExtensionTrim + (long)swWeldmentTrimExtendOptionType_e.swWeldmentTrimExtendOption_CopedCut +
(long)swWeldmentTrimExtendOptionType_e.swWeldmentTrimExtendOption_WeldGap;
myFeature = fm.InsertWeldmentTrimFeature2(1, (int)Options, 0.01, v1, v2);
}

//InsertEndCapFeature
Feature endCapFeature = default(Feature);

Face2[] face1 = new Face2[1];

Array x = default(Array);

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.6023443450227, 0.6150000000001, -1.013201139555, true, 1, null, 0);

face1[0] = (Face2)selMgr.GetSelectedObject6(1, -1);

x = face1;

endCapFeature = fm.InsertEndCapFeature2(0.005, false, true, 0.003, 0.5, 0.003, x);

Part.ClearSelection2(true);

//InsertStructuralWeldment3
Array segmentObjects = default(Array);

SketchSegment[] sketchSegments = new SketchSegment[4];

string profilePathName = null;

Feature structuralMember = default(Feature);

StructuralMemberGroup @group = default(StructuralMemberGroup);

StructuralMemberGroup[] GroupArray1 = new StructuralMemberGroup[1];

long i = 0;

@group = fm.CreateStructuralMemberGroup();

Part.ClearSelection2(true);

Part.InsertSketch2(true);

segmentObjects = (System.Array)Part.SketchManager.CreateCornerRectangle(0, 0, 0, 1.0, 2.0, 0);

for (i = 0; i <= segmentObjects.GetUpperBound(0); i++)
{
    sketchSegments[i] = (SketchSegment)segmentObjects.GetValue(i);
}

@group.Segments = sketchSegments;

GroupArray1[0] = @group;

Part.ViewZoomtofit2();

Part.InsertSketch2(true);

profilePathName = "C:\\Program Files\\SOLIDWORKS Corp\\SOLIDWORKS\\langlenglish\\weldment profiles\\ansi inch\\square tube\\2 x 2 x 0.25.SLDLFP";

structuralMember = fm.InsertStructuralWeldment3(profilePathName, 1, 0.0, false, GroupArray1);

Part.ClearSelection2(true);

//InsertGussetFeature2
Face2[] faceGFObj = new Face2[2];

object z = null;

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.025399999999988, 1.94542628561, 0.00429028534694, true, 1, null, 0);

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.07780718114736, 1.9746, -0.001856983219, true, 2, null, 0);

Count = selMgr.GetSelectedObjectCount();

if (Count == 2)

{
    faceGFObj[0] = (Face2)selMgr.GetSelectedObject6(1, 1);
    faceGFObj[1] = (Face2)selMgr.GetSelectedObject6(1, 2);
    z = faceGFObj;
    Part.ClearSelection2(true);
    myFeature = fm.InsertGussetFeature2(0.005, 0, 0, false, 0.025, 0.025, 0.015, 0.7853981633975, 0.015, true,
    0.005, 0, false, false, false, z);
}

//InsertFilletBeadFeature3
Face2[] fbFaceObj1 = new Face2[1];
Face2[] fbFaceObj2 = new Face2[2];

Part.ClearSelection2(true);

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.0412896304482, 0.02548020566445, 0, true, 1, null, 0);

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.09804264728081, 0.014999999999999, 0.0008069730266129, true, 2, null, 0);

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.01364526875011, 0.08738481720087, 0.01330055827532, true, 4, null, 0);

Count = selMgr.GetSelectedObjectCount();

// Face Set 1
fbFaceObj1[0] = (Face2)selMgr.GetSelectedObject6(1, 1);

// Face Set 2
fbFaceObj2[0] = (Face2)selMgr.GetSelectedObject6(1, 2);

//fbFaceObj2[0] = selMgr.GetSelectedObject6(1, 4)

v1 = fbFaceObj1;
v2 = fbFaceObj2;

Part.ClearSelection2(true);

int[] edges = new int[1];

Array edgeArray = default(Array);

edges[0] = 0;

```

```
//edges(1) = 0
edgeArray = edges;
myFeature = fm.InsertFilletBeadFeature3(0, 0.003, 0.003, 2, 0.003, 0.006, 0, 0.003, 0.003, 2, 0.003, 0, 1, edgeArray, 0, null, v1, v2);
Part.ClearSelection2(true);
}
public SldWorks swApp;
}
}
```

Insert Weldment Features Example (VBA)

This example shows how to insert the following weldment features into the FeatureManager design tree:

- * End cap feature
- * Fillet bead feature
- * Gusset feature
- * Structural weldment
- * Sub weld folder
- * Weldment trim feature

```
'-----
' Preconditions:
' 1. Open public_documents\samples\tutorialweldments\weldment_box2.sldprt
' 2. Expand the Cut list folder in the FeatureManager design tree
'    and observe its contents.
' 3. Delete End cap1 from the FeatureManager design tree.
' 4. Change the path specified for profilePathName, if necessary.
'
' Postconditions: Observe the following in the FeatureManager design tree:
' * Gusset1 moves to Sub Folder1 in the Cut list folder
' * Trim/Extend8 feature appears at the bottom of the design tree
'   and at the end of the Cut list folder
' * Structural Member6 feature appears at the bottom of the design tree
'   and at the end of the Cut list folder
' * End cap5 feature appears at the bottom of the design tree
'   and in the Cut list folder
' * Gusset5 feature appears at the bottom of the design tree
'   and in the Cut list folder
' * Fillet Bead5 feature appears at the bottom of the design tree
'   and in the Cut list folder
'
' NOTE: Because this part is used elsewhere,
' do not save any changes when you close it.
'-----
```

```
Option Explicit
Dim swApp As Object
Dim fm As FeatureManager
Dim selMgr As SelectionMgr
Dim Part As ModelDoc2
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Sub main()
    Dim myFeature As Object
    Dim obj(0) As Object
    Dim v As Variant
    Set swApp = Application.SldWorks

    Set Part = swApp.ActiveDoc
    Dim myModelView As Object
    Set myModelView = Part.ActiveView
    myModelView.FrameState = swWindowState_e.swWindowMaximized

    Set selMgr = Part.SelectionManager

    'InsertSubWeldFolder2

    boolstatus = Part.Extension.SelectByID2("Gusset1", "SOLIDBODY", 0, 0, 0, False, 0, Nothing, 0)
    Set obj(0) = selMgr.GetSelectedObject6(1, -1)
    v = obj

    Set fm = Part.FeatureManager
    Set myFeature = fm.InsertSubWeldFolder2((v))

    Part.ClearSelection2 True

    'InsertWeldmentTrimFeature2

    Dim obj1(0) As Object
    Dim obj2(0) As Object

    Dim Options As Long
    Dim v1 As Variant
    Dim v2 As Variant

    boolstatus = Part.Extension.SelectByID2("Structural Member1[2]", "SOLIDBODY", 0, 0, 0, True, 2, Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Structural Member1[1]", "SOLIDBODY", 0, 0, 0, True, 1, Nothing, 0)
```

```

Set selMgr = Part.SelectionManager
Dim Count As Long
Count = selMgr.GetSelectedObjectCount
If Count = 2 Then
    Set obj1(0) = selMgr.GetSelectedObject2(1)
    v1 = obj1
    Set obj2(0) = selMgr.GetSelectedObject2(2)
    v2 = obj2
    Part.ClearSelection2 True

    Options = swWeldmentTrimExtendOption_AllowTrimmedExtensionTrim + swWeldmentTrimExtendOption_AllowTrimmingExtensionTrim + swWeldmentTrimExtendOption_CopedCut + swWeldmentTrimExtendOption_WeldGap

    Set myFeature = fm.InsertWeldmentTrimFeature2(1, Options, 0.01, (v1), (v2))
End If

Part.ClearSelection2 True

'InsertFilletBeadFeature3

Dim fbFaceObj1(0) As Object
Dim fbFaceObj2(1) As Object

Part.ClearSelection2 True
boolstatus = Part.Extension.SelectByID2("", "FACE", 0.0412896304482, 0.02548020566445, 0, True, 1, Nothing, 0)
boolstatus = Part.Extension.SelectByID2("", "FACE", 0.09804264728081, 0.014999999999999, 8.069730266129E-04, True, 2, Nothing, 0)
boolstatus = Part.Extension.SelectByID2("", "FACE", 0.01364526875011, 0.08738481720087, 0.01330055827532, True, 4, Nothing, 0)

Count = selMgr.GetSelectedObjectCount

' Face Set 1
Set fbFaceObj1(0) = selMgr.GetSelectedObject6(1, 1)
' Face Set 2
Set fbFaceObj2(0) = selMgr.GetSelectedObject6(1, 2)
Set fbFaceObj2(1) = selMgr.GetSelectedObject6(1, 4)

v1 = fbFaceObj1
v2 = fbFaceObj2

Dim fbEdges() As Long
ReDim fbEdges(0 To 0) As Long
fbEdges(0) = 0

Set myFeature = fm.InsertFilletBeadFeature3(0, 0.003, 0.003, 2, 0.003, 0.006, 0, 0.003, 0.003, 2, 0.003, 0, 1, fbEdges, 0, Nothing, v1, v2)

Part.ClearSelection2 True

'InsertStructuralWeldment3

Dim errors As Long
Dim warnings As Long
Dim sketchSegments As Variant
Dim profilePathName As String
Dim structuralMember As Feature
Dim group As StructuralMemberGroup
Dim GroupArray2(0) As Object
Dim vGroups2 As Variant

errors = 0
warnings = 0

Set Part = swApp.ActiveDoc

Part.ClearSelection2 True

Part.InsertSketch2 True
sketchSegments = Part.SketchManager.CreateCornerRectangle(0, 0, 0, 1#, 2#, 0)
Set group = fm.CreateStructuralMemberGroup
group.Segments = (sketchSegments)
Set GroupArray2(0) = group
vGroups2 = GroupArray2

```

Part.ViewZoomtofit2
Part.InsertSketch2 True

profilePathName = "C:\Program Files\SOLIDWORKS Corp\SOLIDWORKS\lang\english\weldment profiles\ansi inch\section5 X 10.SLDLFP"

Set structuralMember = fm.InsertStructuralWeldment3(profilePathName, swSolidworksWeldmentEndCondOptions_e.swEndConditionButt1, 0#, False, (vGroups2))

Part.ClearSelection2 True
'InsertEndCapFeature

Dim endCapFeature As Feature
Dim y As Variant
Dim faceECOObj1(0) As Object
Dim face As Face2

Set selMgr = Part.SelectionManager
boolstatus = Part.Extension.SelectByID2("", "FACE", 0.6023443450227, 0.6150000000001, -1.013201139555, True, 1, Nothing, 0)
Set face = selMgr.GetSelectedObject6(1, -1)
Set faceECOObj1(0) = face
y = faceECOObj1
Set endCapFeature = fm.InsertEndCapFeature2(0.005, False, True, 0.003, 0.5, 0.003, y)

Part.ClearSelection2 True

'InsertGussetFeature2

Dim faceGFObj(1) As Object
Dim z As Variant

boolstatus = Part.Extension.SelectByID2("", "FACE", 0.02539999999988, 1.94542628561, 0.00429028534694, True, 1, Nothing, 0)
boolstatus = Part.Extension.SelectByID2("", "FACE", 0.07780718114736, 1.9746, -0.001856983219, True, 2, Nothing, 0)

Count = selMgr.GetSelectedObjectCount
If Count = 2 Then
 Set faceGFObj(0) = selMgr.GetSelectedObject6(1, 1)
 Set faceGFObj(1) = selMgr.GetSelectedObject6(1, 2)
 z = faceGFObj
Part.ClearSelection2 True

Set myFeature = fm.InsertGussetFeature2(0.005, 0, 0, False, 0.025, 0.025, 0.015, 0.7853981633975, 0.015, True, 0.005, 0, False, False, False, z)

End If
End Sub

Insert Weldment Features Example (VB.NET)

This example shows how to insert the following weldment features into the FeatureManager design tree:

- * End cap feature
- * Fillet bead feature
- * Gusset feature
- * Structural weldment
- * Sub weld folder
- * Weldment trim feature

```

'-----
' Preconditions:
' 1. Open public_documents\samples\tutorialweldments\weldment_box2.sldprt
' 2. Expand the Cut list folder in the FeatureManager design tree and
'    observe its contents.
' 3. Delete End cap1 from the FeatureManager design tree.
' 4. Change the path specified for profilePathName, if necessary.
'
' Postconditions: Observe the following in the FeatureManager design tree:
' * Gusset1 moves to the sub weld folder, Sub Folder1, in the
'   Cut list folder
' * Trim/Extend8 feature appears at the bottom of the design tree
'   and at the end of the Cut list folder
' * Structural Member6 feature appears at the bottom of the design tree
'   and at the end of the Cut list folder
' * End cap5 feature appears at the bottom of the design tree
'   and in the Cut list folder
' * Gusset5 feature appears at the bottom of the design tree
'   and in the Cut list folder
' * Fillet Bead5 feature appears at the bottom of the design tree
'   and in the Cut list folder
'
' NOTE: Because this part is used elsewhere,
' do not save any changes when you close it.
'-----

```

Imports SolidWorks.Interop.sldworks

Imports SolidWorks.Interop.swconst

Imports System

Partial Class SolidWorksMacro

Dim fm As FeatureManager

Dim selMgr As SelectionMgr

Dim Part As ModelDoc2

Dim boolstatus As Boolean

Sub main()

Dim myFeature As Object

Dim obj(0) As Body2

Dim v As Array

Part = swApp.ActiveDoc

Dim myModelView As Object

myModelView = Part.ActiveView

myModelView.FrameState = swWindowState_e.swWindowMaximized

selMgr = Part.SelectionManager

'InsertSubWeldFolder2

boolstatus = Part.Extension.SelectByID2("Gusset1", "SOLIDBODY", 0, 0, 0, False, 0, Nothing, 0)

obj(0) = selMgr.GetSelectedObject6(1, -1)

v = obj

fm = Part.FeatureManager

myFeature = fm.InsertSubWeldFolder2(v)

Part.ClearSelection2(True)

'InsertWeldmentTrimFeature2

Dim obj1(0) As Body2

Dim obj2(0) As Body2

Dim Options As Long

Dim v1 As Array

Dim v2 As Array

boolstatus = Part.Extension.SelectByID2("Structural Member1[2]", "SOLIDBODY", 0, 0, 0, True, 2, Nothing, 0)

boolstatus = Part.Extension.SelectByID2("Structural Member1[1]", "SOLIDBODY", 0, 0, 0, True, 1, Nothing, 0)

selMgr = Part.SelectionManager

Dim Count As Long

Count = selMgr.GetSelectedObjectCount

If Count = 2 Then

obj1(0) = selMgr.GetSelectedObject2(1)

v1 = obj1

obj2(0) = selMgr.GetSelectedObject2(2)

v2 = obj2

Part.ClearSelection2(True)


```
'edges(1) = 0
edgeArray = edges
myFeature = fm.InsertFilletBeadFeature3(0, 0.003, 0.003, 2, 0.003, 0.006, 0, 0.003, 0.003, 2, 0.003, 0, 1, edgeArray, 0, Nothing, v1, v2)
Part.ClearSelection2(True)
End Sub
Public swApp As SldWorks
End Class
```

Insert Wrap Feature Example (VBA)

This example shows how to insert a wrap feature.

```
'-----
' Preconditions: Open public_documents\samples\tutorial\api\cylinder20.sldprt.
'
' Postconditions:
' 1. Selects the Front plane and creates a sketch of a circle.
' 2. Selects the sketch and nonplanar face.
' 3. Inserts a wrap feature on the selected nonplanar face.
' 4. Examine the graphics area and FeatatureManager design
'    tree.
'
' NOTE: Because the part is used elsewhere, do not save changes.
'-----
Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim boolstatus As Boolean
Dim swFeatMgr As SldWorks.FeatureManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim swSketchSegment As SldWorks.SketchSegment
Dim swSketchManager As SldWorks.SketchManager

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swModelDocExt = swModel.Extension
    Set swSketchManager = swModel.SketchManager

    'Create sketch for wrap feature
    swSketchManager.InsertSketch True
    boolstatus = swModelDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    Set swSketchSegment = swSketchManager.CreateCircle(-0.006377, 0.026536, 0#, 0.004153, 0.006985, 0#)
    swSketchManager.InsertSketch True

    swModel.ClearSelection2 True

    ' Mark the sketch to use for wrap feature as 4
    boolstatus = swModelDocExt.SelectByID2("Sketch2", "SKETCH", 0, 0, 0, True, 4, Nothing, swSelectOptionDefault)

    ' Mark the nonplanar face on which to place wrap feature as 1
    boolstatus = swModelDocExt.SelectByID2("", "FACE", -2.01300616999447E-02, 2.48040612188447E-02, 5.80282618765864E-02, True, 1, Nothing, 0)
    Set swFeatMgr = swModel.FeatureManager

    ' Create a wrap feature of type emboss
    swFeatMgr.InsertWrapFeature 0, 0.005, False

    swModel.ClearSelection2 True

End Sub
```